Table of Contents	
01	Title Page
02	Block Diagram
03	FS2530 FOR S32K566
04	S32K566 POWER
05	S32K566 B DOMAIN
06	S32K566 A DOMAIN
07	S32K566 A DOMAIN
08	S32K566 A DOMAIN
09	ENET 60PIN CON
10	ENET TJA1410A
11	ENET TJA1120A
12	CAN TJA1462A
13	CAN TJA1482 & TJA1483
14	S32K5 IO IND/CTRL
15	LED INDICATORS
16	OpenSDA FOR S32K5
17	LIN SJA1124
18	FS2530 FOR S32J100
19	S32J100 POWER & GND
20	S32J100 IO
21	S32J100 SERDES
22	S32J100 RESET & JTAG
23	PCIE CLOCK
24	PCIE
25	10GE SFP
26	ENET TJA1230A
27	ENET TJA1230A
28	ENET TJA1120B
29	ENET TJA1120B
30	ENET TJA1410B
31	ENET PHY RJ45
32	CAN TJA1462A
33	OPNET KD7251
34	OPNET KD7251
35	PF5123 & PWR JUMPER
36	POWER INDICATORS
37	Power Tree
38	SETTINGS

Revision Control

Rev	Date	Description	Approver
X1	7/13/26	Intial Draft	Edmund Zhang
A	8/25/26	Release	Edmund Zhang

X-RING-OPTIC-ETH

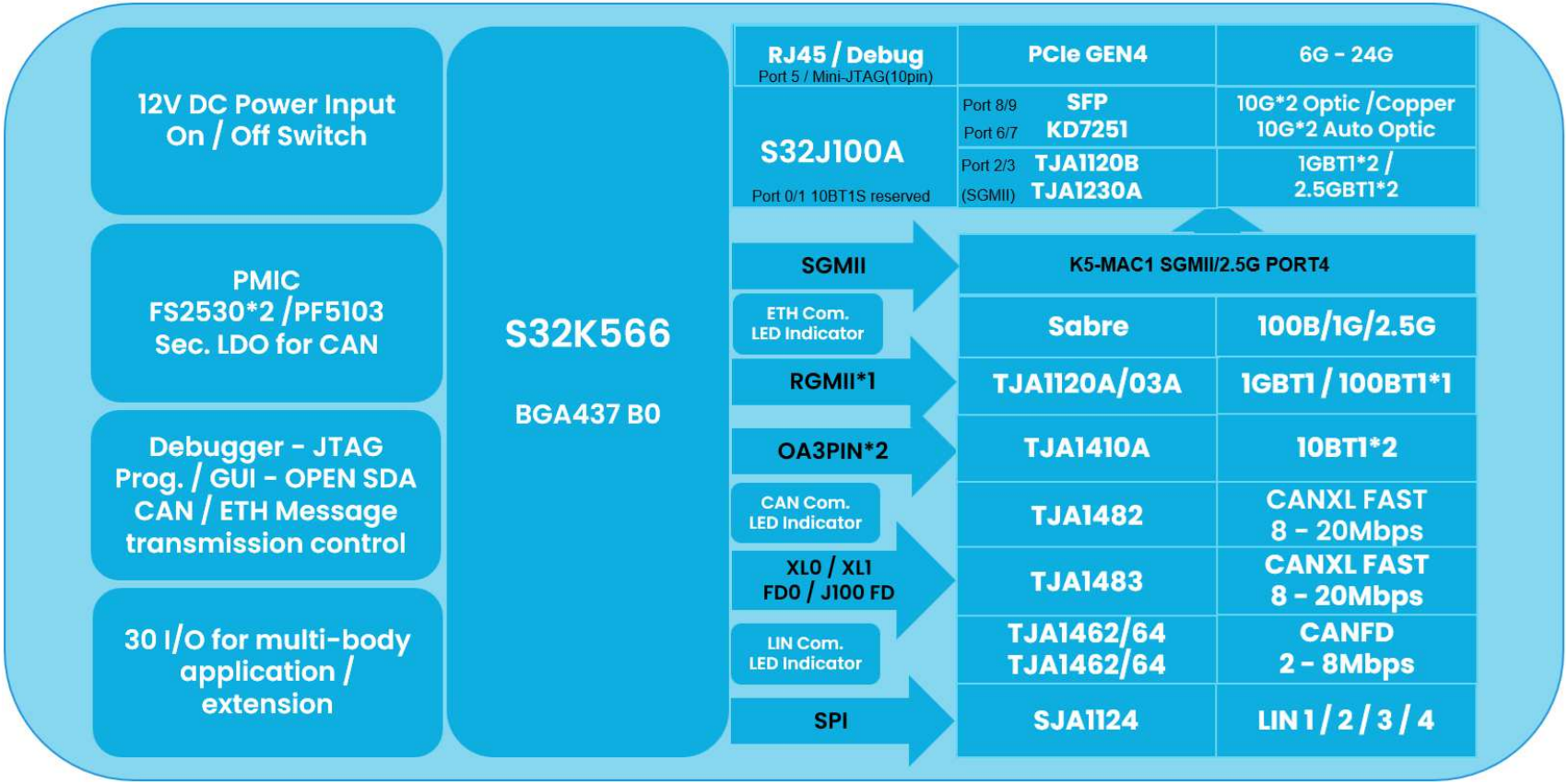
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Austin, TX 78735-8598

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Classification: Company Internal

Designer: Jun Qiao	Drawing Title: X-RING-OPTIC-ETH	
Drawn by: Jun Qiao	Page Title: 01 - Title Page	
Approved: Edmund Zhang	Size C	Document Number SCH-97122 / PDF: SPF-97122 — Rev A
	Date: Tuesday, August 25, 2026	Sheet 1 of 38

PROPOSED BLOCK DIAGRAM



Classification: Company Internal Use

Drawing Title: X-RING-OPTIC-ETH

Page Title: 02 - Block Diagram

Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

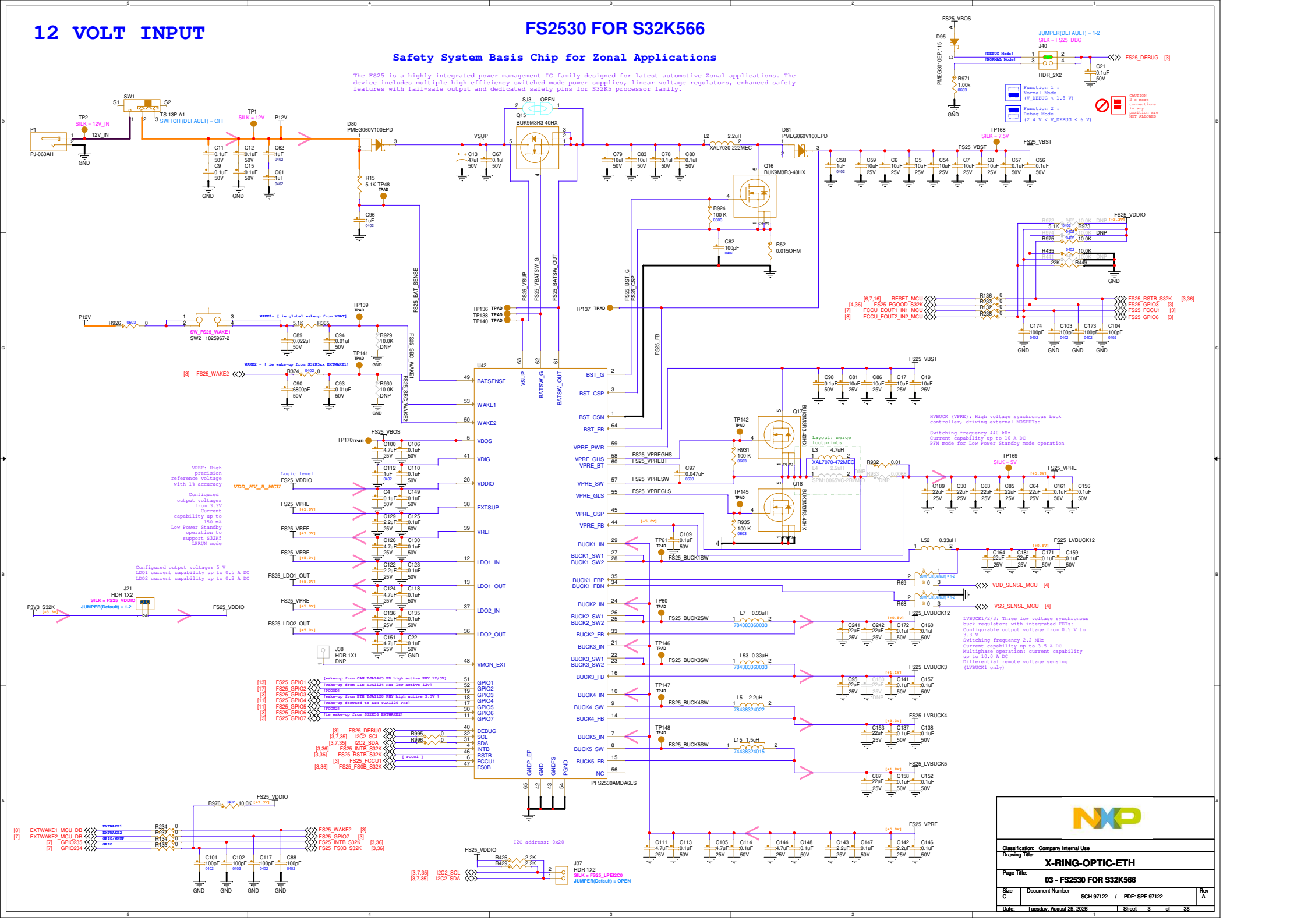
Date: Tuesday, August 25, 2026 | Sheet 2 of 38

12 VOLT INPUT

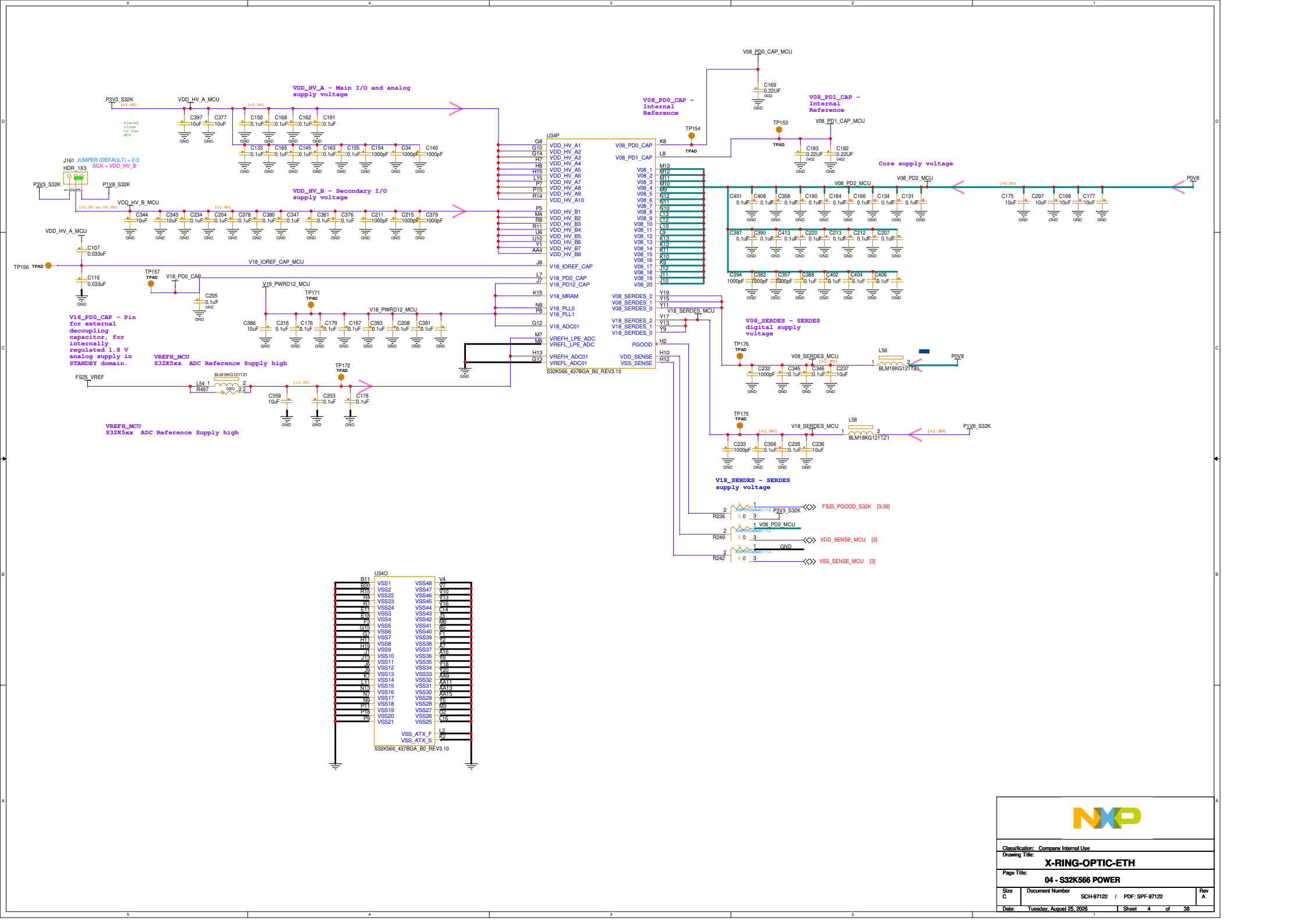
FS2530 FOR S32K566

Safety System Basis Chip for Zonal Applications

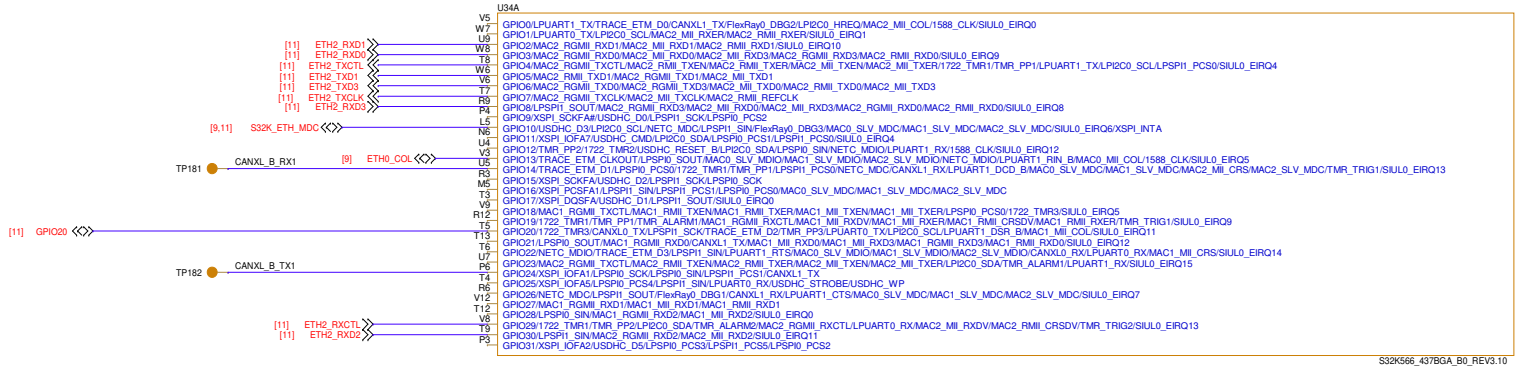
The FS25 is a highly integrated power management IC family designed for latest automotive Zonal applications. The device includes multiple high efficiency switched mode power supplies, linear voltage regulators, enhanced safety features with fail-safe output and dedicated safety pins for S32K5 processor family.

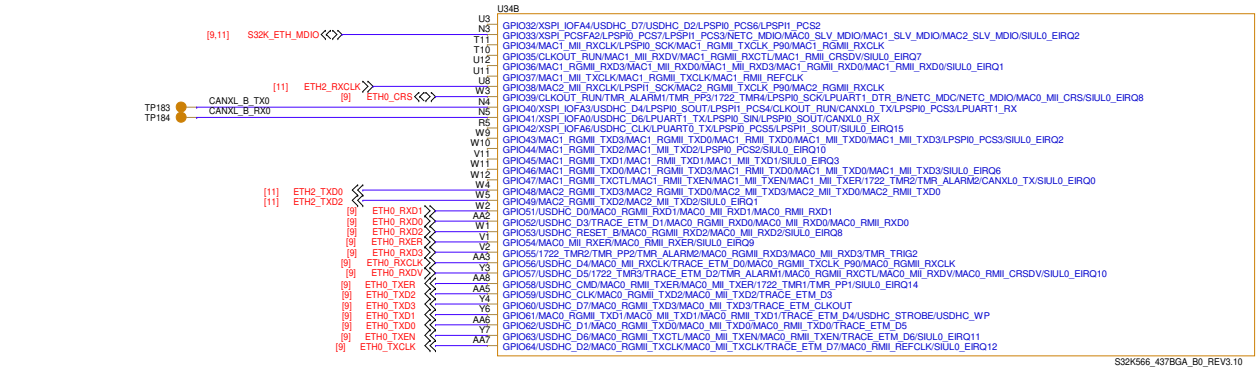


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Drawing Title:	X-RING-OPTIC-ETH
Page Title:	03 - FS2530 FOR S32K566
Size: C	Document Number: SCH-97122 / PDF: SPF-97122
Rev: A	
Date:	Tuesday, August 25, 2026
Sheet	3 of 38

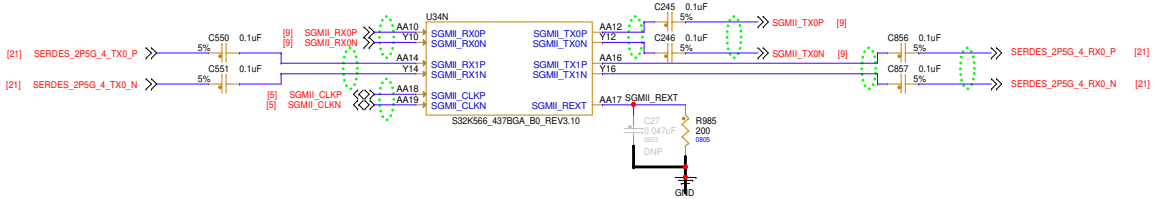

VDD_HV_A – Main I/O and analog supply voltage
VDD_HV_B – Secondary I/O supply voltage
V08_PD0_CAP – Internal Reference
V08_PD1_CAP – Internal Reference
Core supply voltage
V08_SERDES – SERDES digital supply voltage
V18_SERDES – SERDES supply voltage
V18_PD0_CAP – Pin for external decoupling capacitor, for internally regulated 1.8 V analog supply in STANDBY domain.
VREFH_MCU S32K5xx ADC Reference Supply high
U34P
U34O
S32K566_437BGA_B0_REV3.10
P3V3_S32K
P1V8_S32K
P0V8
FS25_VREF
FS25_PGOOD_S32K [3,36]
VDD_SENSE_MCU [3]
VSS_SENSE_MCU [3]
Classification: Company Internal Use
Drawing Title: X-RING-OPTIC-ETH
Page Title: 04 - S32K566 POWER
Size C
Document Number SCH-97122 / PDF: SPF-97122
Rev A
Date: Tuesday, August 25, 2026
Sheet 4 of 38

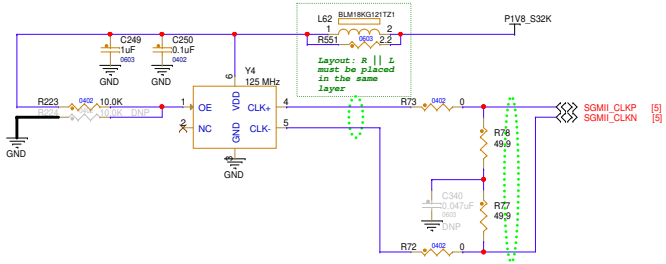
POWER DOMAIN - VDD_HV_B





SGMII

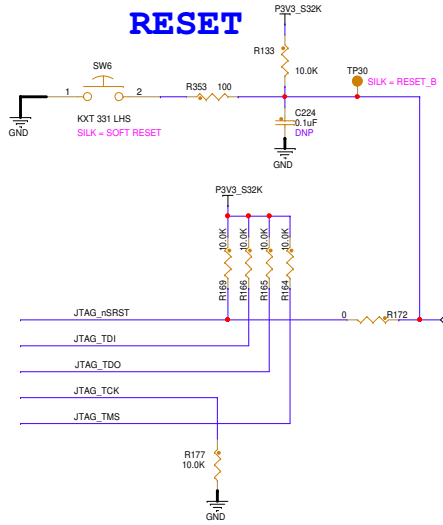


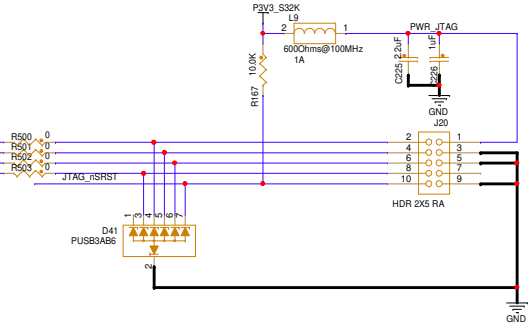


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Size C
Document Number SCH-97122 / PDF: SPF-97122
Rev A
Date: Tuesday, August 25, 2026
Sheet 5 of 38

POWER DOMAIN – VDD_HV_A

RESET





U34C

GPIO065/1722_TMR1/MAC2_TC14_ED/LPI2C0_SDAS/LPSPI0_PCS1/LPSPI1_PCS0/FlexRay0_DBG1/LPUART0_DCD_B/TMR_TRIG1/SIUL0_EIRQ4
GPIO066/MAC3_TC14_ED/1722_TMR2/TMR_PP2/LPUART0_RTS/LPSPI0_PCS4/LPSPI1_PCS2/LPI2C0_SDA/TMR_TRIG2/SIUL0_EIRQ5
GPIO067/LPI2C0_SCL/1722_TMR3/TMR_PP3/MAC1_TC14_RX/LPSPI0_SCK/FlexRay0_DBG3/CANXL0_RX/CANXL1_RX/LPUART0_RX/SIUL0_EIRQ3/WKPU20
GPIO068/MAC3_TC14_RX/1722_TMR4/LPSPI1_SCK/LPI2C0_SCL/LPUART0_RX/SIUL0_EIRQ6/WKPU18
GPIO069/MAC4_TC14_ED/1722_TMR1/TMR_PP1/LPSPI0_PCS3/LPI2C0_SDA/LPUART0_CTS/TMR_TRIG1/SIUL0_EIRQ9
GPIO070/MAC2_TC14_TX/1722_TMR2/TMR_PP2/LPSPI0_PCS2/LPUART1_TX/CANXL1_TX/LPUART0_DSR_B/TMR_TRIG2
GPIO071/1722_TMR3/TMR_PP3/MAC2_TC14_RX/LPUART0_DTR_B/LPI2C0_SCLS/LPSPI0_PCS0/LPSPI1_PCS1/FlexRay0_DBG2/CANXL1_RX/LPUART1_RX/SIUL0_EIRQ2/WKPU21
GPIO072/LPSPI0_SIN/1722_TMR4/TMR_ALARM2/MAC1_TC14_TX/LPUART0_TX/CANXL1_TX/LPI2C0_HREQ
GPIO073/LPI2C0_SDA/1722_TMR1/TMR_PP1/TMR_ALARM1/MAC1_TC14_ED/LPSPI0_SOUT/FlexRay0_DBG0/LPUART0_RIN_B/TMR_TRIG1/SIUL0_EIRQ7
GPIO074/MAC4_TC14_TX/1722_TMR2/TMR_PP2/TMR_ALARM2/LPSPI0_PCS7/LPSPI1_PCS5/LPUART1_TX/TMR_TRIG2
GPIO075/MAC4_TC14_RX/1722_TMR3/TMR_PP3/TMR_ALARM1/LPI2C0_SCL/LPSPI0_PCS6/LPSPI1_PCS4/LPUART1_RX/SIUL0_EIRQ12/WKPU22
GPIO076/MAC3_TC14_TX/1722_TMR4/LPUART0_TX/LPSPI0_PCS5/LPSPI1_PCS3
GPIO077/LPSPI1_SOUT/LPI2C0_SCL/NETC_MDC/CANXL0_RX/LPUART0_RX/LPUART1_RX/1588_CLK/SIUL0_EIRQ15/WKPU19
GPIO078/CANXL0_TX/LPSPI1_SIN/LPI2C0_SDA/LPUART0_TX/LPUART1_TX/NETC_MDIO/1588_CLK
GPIO080/SAI0_SYNC/FXIO0_D31/FXIO0_D15/LPSPI2_SOUT/FCCU_EOUT1/CAN1_RX/LPUART2_RX/SIUL1_EIRQ0/WKPU34/FCCU_ERR_IN1
GPIO081/LPUART3_TX/CAN0_TX/FXIO0_D22/LPSPI2_PCS3/FCCU_EOUT2/FXIO0_D6/SAI0_D1/LPUART13_TX/CAN1_TX/SENT1_CH[5]/SIUL1_EIRQ1/FCCU_ERR_IN2
GPIO082/SAI0_D0/CAN3_TX/LPSPI2_PCS0/FXIO0_D7/LPUART7_TX/FXIO0_D23/LPUART2_RTS/LPUART12_TX/SIUL1_EIRQ11/WKPU39
GPIO083/LPUART5_TX/FXIO0_D25/CAN7_TX/LPSPI2_PCS2/FXIO0_D9/LPUART2_TX/LPUART2_CTS/SIUL1_EIRQ3/WKPU25
GPIO084/SAI0_D2/FXIO0_D26/FXIO0_D10/LPUART2_TX/LPSPI2_SCK/CAN1_RX/LPUART12_RX/SENT1_CH[6]/SIUL1_EIRQ3
GPIO085/CAN1_TX/LPUART2_TX/LPSPI2_SIN/SAI0_D3/FXIO0_D12/FXIO0_D28/CAN0_RX/LPUART7_RX/LPUART9_RX/SIUL1_EIRQ5/WKPU42
GPIO086/FXIO0_D13/LPUART9_TX/LPSPI2_PCS3/FXIO0_D29/CAN4_RX/LPUART2_RX/SIUL1_EIRQ2/WKPU36
GPIO087/LPUART2_RTS/FXIO0_D14/FXIO0_D30/CAN4_TX/LPUART6_TX/CAN0_TX/LPUART5_RX/SIUL1_EIRQ12/WKPU37
GPIO088/FXIO0_D11/FXIO0_D27/LPUART6_TX/LPSPI2_PCS0/CAN0_RX/CAN4_RX/LPUART11_RX/SENT0_CH[0]/SIUL1_EIRQ9
GPIO089/LPSPI2_PCS1/FXIO0_D0/FXIO0_D16/CAN7_RX/LPUART2_RX/LPUART6_RX/SIUL1_EIRQ9/WKPU24
GPIO090/FXIO0_D16/FXIO0_D0/CAN6_TX/LPUART8_RX/SIUL1_EIRQ10/WKPU26
GPIO091/LPUART13_TX/FXIO0_D26/FXIO0_D10/CAN5_RX/SIUL1_EIRQ4/WKPU27
GPIO092/FXIO0_D8/FXIO0_D24/CAN4_TX/CAN0_TX/LPUART5_TX/LPUART6_RX/SENT0_CH[1]/SIUL1_EIRQ10
GPIO093/FXIO0_D25/FXIO0_D9/LPUART8_TX/CAN6_RX/SIUL1_EIRQ7/WKPU38
GPIO094/FXIO0_D20/LPUART4_TX/LPSPI2_PCS3/CAN0_TX/FXIO0_D4/SENT1_CH[1]
GPIO095/FXIO0_D6/LPSPI2_SIN/FXIO0_D22/CAN2_RX/LPUART3_RX/SIUL1_EIRQ8/WKPU29

S32K566_437BGA_B0_REV3.10

Layout Note:
CANXL_TX0/1, CANXL_RX0/1, CANFD_TX2, CANFD_RX2
should be routed in high priority by short traces with less vias

U34D

GPIO096/CAN2_TX/LPUART3_TX/FXIO0_D23/LPSPI2_SCK/FXIO0_D7/SIUL1_EIRQ9
GPIO097/SAI1_D1/FXIO0_D17/FXIO0_D1/CAN7_RX/LPUART11_RX/LPUART9_RX/SIUL1_EIRQ0/WKPU23
GPIO098/FXIO0_D18/LPUART9_TX/CAN7_TX/FXIO0_D2/LPUART11_TX/SAI1_D0/SIUL1_EIRQ10
GPIO099/FXIO0_D21/SAI1_MCLK/FXIO0_D5/LPUART11_RX/LPUART5_RX/SENT1_CH[0]/SIUL1_EIRQ1/WKPU28
GPIO100/FXIO0_D12/LPSPI2_PCS0/SAI0_D2/CAN5_TX/FXIO0_D28/CAN0_RX/LPUART4_RX/SENT1_CH[2]/SIUL1_EIRQ13/WKPU33
GPIO101/FXIO0_D4/LPSPI2_PCS1/CAN3_TX/SAI0_D3/FXIO0_D20/CAN5_RX/LPUART4_RX/SIUL1_EIRQ14/WKPU43
GPIO102/FXIO0_D5/LPSPI2_SOUT/LPUART4_TX/FXIO0_D21/CAN3_RX/LPUART7_RX/SIUL1_EIRQ15/WKPU31
GPIO103/FXIO0_D3/SAI1_BCLK/LPUART4_TX/CAN6_TX/FXIO0_D19
GPIO104/SAI1_D1/LPUART7_TX/FXIO0_D24/SAI0_D1/FXIO0_D8/LPUART5_TX/LPUART11_TX/SIUL1_EIRQ5
GPIO105/SAI0_D0/FXIO0_D17/FXIO0_D1/CAN4_RX/HSE2_LPUART_CTS/LPUART4_RX/LPUART7_RX/SIUL1_EIRQ2/WKPU32
GPIO106/SAI0_SYNC/FXIO0_D3/SAI1_SYNC/FXIO0_D19/LPSPI2_PCS1/HSE2_LPUART_RTS/CAN2_RX/LPUART10_RX/LPUART8_RX/SENT1_CH[3]/SIUL1_EIRQ4/WKPU40
GPIO107/LPUART8_TX/CAN2_TX/FXIO0_D2/FXIO0_D18/LPUART10_TX/LPSPI2_SIN/HSE2_LPUART_RX
GPIO108/FXIO0_D13/LPUART12_TX/CAN1_TX/LPSPI2_SOUT/FXIO0_D29/HSE2_LPUART_TX
GPIO109/LPUART5_TX/SAI1_SYNC/CAN6_RX/SIUL1_EIRQ6/WKPU41
GPIO110/CAN4_TX/FXIO0_D15/SAI1_MCLK/LPSPI2_PCS1/SAI0_MCLK/FXIO0_D31/LPUART6_RX/SIUL1_EIRQ7/WKPU44
GPIO111/SAI0_BCLK/SAI1_BCLK/FXIO0_D30/FXIO0_D14/LPSPI2_PCS2/CAN0_RX/LPUART13_RX/LPUART2_CTS/LPUART3_RX/SENT1_CH[4]/SIUL1_EIRQ6/WKPU35
GPIO112/LPSPI2_SCK/FXIO0_D11/FXIO0_D27/CAN1_RX/CAN3_RX/LPUART12_RX/SIUL1_EIRQ8/WKPU30
GPIO113/FXIO0_D13/LPUART2_TX/LPUART12_RTS/LPUART11_TX/LPSPI2_SIN/CAN1_RX/SENT0_CH[2]
GPIO114/FXIO0_D12/CAN1_TX/LPUART10_TX/LPSPI2_SCK/LPUART12_RX/LPUART3_RX/SENT0_CH[6]
GPIO115/FXIO0_D31/CAN4_TX/LPSPI2_PCS2/SENT1_CH[7]
GPIO116/FXIO0_D0/FXIO0_D16/CAN6_TX/LPUART13_RX/SIUL1_EIRQ15
GPIO126/FXIO0_D0/FXIO0_D16/LPSPI2_SIN/LPUART10_TX/CAN2_RX/SENT0_CH[5]/SIUL1_EIRQ11
GPIO127/LPSPI2_SCK/FXIO0_D1/FXIO0_D17/CAN3_RX/LPUART10_RX/LPUART8_RX/SENT0_CH[4]/SIUL1_EIRQ12/WKPU65

S32K566_437BGA_B0_REV3.10

U34E

GPIO128/CAN3_TX/LPSPI2_SOUT/LPUART8_TX/FXIO0_D2/FXIO0_D18/LPUART11_RX/SENT0_CH[3]/SIUL1_EIRQ13/WKPU64
GPIO129/FXIO0_D3/LPUART11_TX/LPUART7_TX/FXIO0_D19/LPUART12_TX/LPSPI2_PCS2/SIUL1_EIRQ14
GPIO130/CAN4_TX/FXIO0_D4/FXIO0_D20/LPSPI2_SIN/LPUART12_RX/LPUART7_RX/SIUL1_EIRQ0
GPIO131/FXIO0_D5/FXIO0_D21/LPUART11_TX/LPSPI2_SOUT/CAN4_RX/LPUART5_RX/SIUL1_EIRQ1
GPIO132/CAN5_TX/LPUART5_TX/FXIO0_D6/FXIO0_D22/LPSPI2_SCK/LPUART11_RX/SIUL1_EIRQ2
GPIO133/FXIO0_D7/FXIO0_D23/LPSPI2_PCS0/LPUART12_TX/CAN5_RX/LPUART5_RX/SIUL1_EIRQ3
GPIO134/CAN2_TX/FXIO0_D8/FXIO0_D24/LPSPI2_PCS1/CAN7_TX/LPUART12_RX/LPUART9_RX/SIUL1_EIRQ4/WKPU67
GPIO135/CAN6_TX/LPUART12_TX/FXIO0_D9/FXIO0_D25/LPUART9_TX/LPUART13_TX/LPSPI2_PCS3/CAN7_RX/SIUL1_EIRQ5
GPIO136/FXIO0_D10/FXIO0_D26/CAN6_RX/LPUART10_RX/LPUART13_RX/SENT0_CH[7]/SIUL1_EIRQ6/WKPU66
GPIO137/CAN1_TX/LPUART3_TX/FXIO0_D11/FXIO0_D27/LPUART13_TX/SIUL1_EIRQ7
GPIO138/FXIO0_D12/FXIO0_D28/CAN1_RX/LPUART3_RX/SIUL1_EIRQ8

S32K566_437BGA_B0_REV3.10

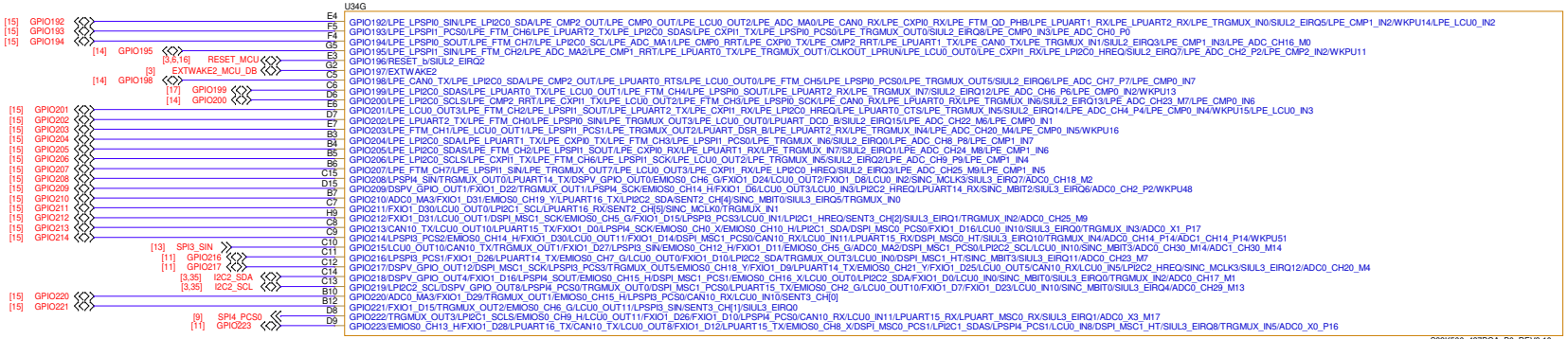
U34F

GPIO148/LPE_FTM_CH6/LPE_LPSPI0_SIN/LPE_FTM_FLT0
GPIO156/LPE_ADC_MA3/LPE_LPSPI0_SOUT/LPE_FTM_FLT0/LPE_TRGMUX_IN3
GPIO157/LPE_FTM_CH7/LPE_FTM_FLT1/LPE_FTM_QD_PHA/LPE_TRGMUX_IN4
GPIO158/LPE_ADC_MA3/LPE_FTM_FLT2/LPE_FTM_QD_PHB/LPE_TRGMUX_IN5
GPIO159/LPE_FTM_CH5/LPE_LPSPI0_SCK/LPE_FTM_FLT0
GPIO160/LPE_LPSPI1_SCK/LPE_CXPI0_TX/LPE_LPUART0_RTS/LPE_LPI2C0_SCLS/LPE_FTM_CH3/LPE_CMP1_OUT/LPE_LCU0_OUT2/LPE_TRGMUX_OUT7/SIUL2_EIRQ0/LPE_ADC_CH18_M2/WKPU10/LPE_CMP2_IN1/LPE_LCU0_IN2
GPIO161/LPE_LPSPI1_SCK/LPE_FTM_CH0/LPE_TRGMUX_OUT6/LPE_LPUART0_CTS/SIUL2_EIRQ1/LPE_ADC_CH3_P3/LPE_CMP2_IN0
GPIO162/LPE_LCU0_OUT3/LPE_LPSPI1_PCS1/LPE_FTM_CH1/LPE_CMP1_OUT/LPE_TRGMUX_OUT0/LPE_ADC_MA0/LPE_LPI2C0_SCLS/LPE_CXPI0_RX/LPE_LPUART0_RX/SIUL2_EIRQ10/LPE_ADC_CH19_M3/LPE_CMP2_IN3/NMI_B/WKPU17
GPIO163/LPE_LPSPI1_SOUT/LPE_LPI2C0_SDAS/LPE_FTM_CH4/LPE_CMP1_RRT/LPE_LPUART0_TX/LPE_TRGMUX_OUT1/JTAG_TDO/TRACEnoETM_SWO/LPE_CAN0_TX/CLKOUT_LPRUN/LPE_CMP1_OUT/LPE_LCU0_OUT1/LPE_LPI2C0_SCL/LPE_FTM_QD_PHA/SIUL2_EIRQ4/LPE_ADC_CH5_P5
GPIO164/LPE_FTM_CH6/LPE_LPSPI1_SCK/LPE_LCU0_OUT0/LPE_CMP1_RRT/LPE_TRGMUX_OUT5/LPE_ADC_MA1/LPE_CAN0_RX/LPE_LPI2C0_HREQ/LPE_LPUART0_RX/SIUL2_EIRQ6/WKPU8/LPE_ADC_CH21_M5/LPE_LCU0_IN0/JTAG_TCK/SWD_CLK
GPIO165/LPE_LPI2C0_SDAS/LPE_CMP0_OUT/LPE_FTM_CH7/LPE_CAN0_TX/JTAG_TMS/SWD_DIO/LPE_LPSPI1_PCS0/LPE_LPUART0_CTS/LPE_TRGMUX_IN0/SIUL2_EIRQ13/LPE_ADC_CH1_P1
GPIO166/LPE_LPSPI1_SIN/LPE_LPI2C0_SDA/LPE_CMP0_OUT/LPE_CMP0_RRT/LPE_LPUART0_RTS/LPE_FTM_CH5/JTAG_TDI/LPE_CAN0_RX/LPE_FTM_QD_PHB/LPE_TRGMUX_IN1/SIUL2_EIRQ12/LPE_ADC_CH17_M1
GPIO167/LPE_LCU0_OUT3/LPE_LPSPI0_PCS0/LPE_LPI2C0_SCL/LPE_FTM_CH4/LPE_CMP0_OUT/LPE_TRGMUX_OUT4/CLKOUT_STANDBY/LPE_CXPI0_TX/LPE_CAN0_TX/LPUART_RIN_B/SIUL2_EIRQ11/LPE_ADC_X0_P16/LPE_CMP1_IN1/LPE_LCU0_IN3
GPIO168/LPE_LPSPI0_PCS2/LPE_LCU0_OUT1/LPE_ADC_MA2/LPE_CMP0_RRT/CLKOUT_LPRUN/LPE_CXPI0_RX/LPE_FTM_FLT3/LPE_LPUART1_RX/LPE_TRGMUX_IN2/SIUL2_EIRQ9/LPE_ADC_X1_P17/WKPU9/LPE_CMP1_IN0/LPE_LCU0_IN1
GPIO169/LPE_LPSPI0_PCS1/LPE_LPI2C0_SCL/LPE_CMP1_OUT/LPUART_DTR_B/LPE_LCU0_OUT0/LPE_CAN0_RX/LPE_FTM_FLT3/LPE_FTM_QD_PHA/LPE_TRGMUX_IN3/SIUL2_EIRQ14/WKPU12/LPE_ADC_X2_M16/LPE_CMP0_IN0/LPE_LCU0_IN0
GPIO170/LCU0_OUT1/LPE_FTM_CH5/LPE_LPSPI0_SCK/LPE_FTM_FLT0/LPE_LPUART1_TX/LPE_CMP2_OUT/LPE_CMP2_RRT/CLKOUT_STANDBY/LPE_TRGMUX_IN4/LPE_ADC_X3_M17/LPE_LCU0_IN1
GPIO171/LPE_CXPI1_TX/LPE_LPUART1_TX/LPE_FTM_FLT3/SIUL2_EIRQ4/LPE_ADC_CH27_M11
GPIO172/LPE_CXPI1_RX/LPE_FTM_QD_PHB/LPE_LPUART1_RX/LPE_TRGMUX_IN0/SIUL2_EIRQ5/LPE_ADC_CH11_P11
GPIO173/LPE_LCU0_OUT1/LPE_FTM_CH0/LPE_LPUART2_RX/LPE_TRGMUX_IN1/SIUL2_EIRQ6/LPE_ADC_CH28_M12/LPE_CMP2_IN6
GPIO174/LPE_LPUART2_TX/LCU0_OUT0/LPE_FTM_QD_PHA/LPE_TRGMUX_IN7/SIUL2_EIRQ7/LPE_ADC_CH12_P12/LPE_CMP2_IN4
GPIO175/LPE_LPUART2_TX/LPE_FTM_CH2/LPE_LPSPI0_PCS1/LPE_LCU0_OUT3/LPE_TRGMUX_IN6/SIUL2_EIRQ8/LPE_ADC_CH13_P13/LPE_CMP2_IN5
GPIO176/LPE_LCU0_OUT2/LPE_FTM_CH1/LPE_LPSPI0_PCS3/LPE_LPUART2_RX/LPE_TRGMUX_IN2/SIUL2_EIRQ9/LPE_ADC_CH29_M13/LPE_CMP2_IN7
GPIO177/LPE_FTM_CH4/LPE_LPSPI0_PCS2/LPE_LPUART1_TX/LPE_LCU0_OUT3/LPE_CMP2_RRT/LPE_CAN0_RX/LPE_TRGMUX_IN3/SIUL2_EIRQ10/LPE_ADC_CH30_M14
GPIO178/LPE_LPSPI0_PCS3/LPE_CMP2_OUT/LPE_LCU0_OUT2/LPE_LPUART0_RTS/LPE_CAN0_TX/LPE_LPUART1_RX/SIUL2_EIRQ11/LPE_ADC_CH14_P14
GPIO179/LPE_LPSPI0_SIN/LPE_CMP1_RRT/LPE_FTM_CH6/LPE_LPUART0_CTS/SIUL2_EIRQ12/LPE_ADC_CH15_P15
GPIO180/LPE_CMP1_OUT/LPE_CXPI0_TX/LPE_LPSPI0_SCK/CLKOUT_LPRUN/LPE_FTM_CH5/LPE_LPUART0_RX/SIUL2_EIRQ13/LPE_ADC_CH31_M15
GPIO181/LPE_LPUART0_TX/LPE_FTM_CH7/LPE_LPSPI0_SOUT/LPE_ADC_MA2/LPE_CMP0_OUT/LPE_CXPI0_RX/SIUL2_EIRQ14
GPIO182/LPE_ADC_MA0/LPE_CMP0_RRT/LPE_FTM_CH5/LPE_LPUART2_RX/SIUL2_EIRQ15
GPIO183/LPE_ADC_MA1/LPE_FTM_CH4/LPE_LPSPI1_SOUT/LPE_LPI2C0_SDAS/LPE_TRGMUX_OUT1/LPE_LPUART0_RX/SIUL2_EIRQ0
GPIO184/LPE_FTM_CH1/LPE_TRGMUX_OUT2/LPE_LPI2C0_SCL/LPE_LPSPI1_PCS0/LPE_CXPI1_TX/LPE_LPUART0_TX/SIUL2_EIRQ1
GPIO185/LPE_LPSPI1_PCS1/LPE_TRGMUX_OUT6/LPE_CXPI1_RX/LPE_FTM_FLT3/LPE_LPI2C0_HREQ/LPUART_DCD_B/SIUL2_EIRQ2/LPE_ADC_CH26_M10
GPIO186/LPE_LPSPI1_SCK/LPE_FTM_CH5/LPE_LPUART2_TX/LPE_LPI2C0_SCLS/LPE_ADC_MA2/LPE_LPUART0_RTS/LPE_TRGMUX_OUT0/SIUL2_EIRQ3
GPIO187/LPE_FTM_CH0/LPE_LPI2C0_SDA/LPE_LPSPI1_SIN/LPE_TRGMUX_OUT3/LPE_LPUART0_CTS/SIUL2_EIRQ4/LPE_ADC_CH10_P10
GPIO188/LPE_FTM_CH3/LPE_ADC_MA0/LPE_TRGMUX_OUT7/LPE_CAN0_RX/LPUART_DSR_B/SIUL2_EIRQ5
GPIO189/LPE_LPUART1_TX/LPE_FTM_CH2/LPE_CAN0_TX/LPUART_DTR_B/LPE_ADC_MA1/SIUL2_EIRQ5
GPIO190/LPE_TRGMUX_OUT4/LPE_FTM_CH7/LPE_ADC_MA2/LPUART_RIN_B/SIUL2_EIRQ6
GPIO191/LPE_CMP0_RRT/LPE_TRGMUX_OUT5/LPE_FTM_CH6/LPE_LPUART1_RX/SIUL2_EIRQ7

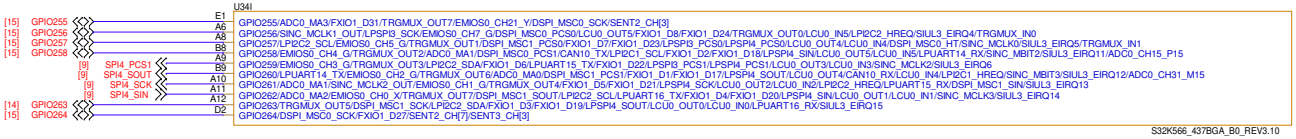
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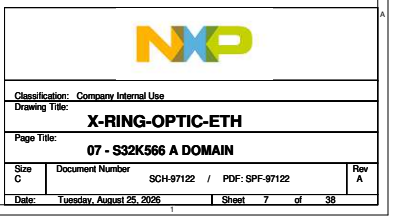
Classification:	Company Internal Use
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Page Title:	06 - S32K566 A DOMAIN
Size: C	Document Number: SCH-97122 / PDF: SPF-97122 — Rev: A
Date:	Tuesday, August 25, 2026 — Sheet 6 of 38

POWER DOMAIN - VDD_HV_A









POWER DOMAIN – VDD_HV_A

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[15]	GPIO304	H16	GPIO304/LPUART17_TX/LCU1_OUT11/LPSPI5_PCS0/EMIOS1_CH10_H/EMIOS2_CH10_H/ADC1_MA2/LCU1_IN11/SIUL4_EIRQ5/ADC1_CH28_M12
[15]	GPIO305	H17	GPIO305/LCU1_OUT5/LPSPI5_PCS4/LPUART17_TX/EMIOS1_CH11_H/EMIOS2_CH11_H/ADC1_MA0/CAN14_RX/LCU1_IN5/SIUL4_EIRQ4/ADC1_CH23_M7/WKPU59
[15]	GPIO306	H18	GPIO306/LPSPI5_PCS5/LCU1_OUT4/EMIOS1_CH0_X/ADC1_MA1/EMIOS1_CH1_H/CAN14_TX/LCU1_IN4/LPUART17_RX/SIUL4_EIRQ0/ADC1_CH7_P7
[15]	GPIO307	C16	GPIO307/LPSPI5_PCS1/DSPV_GPIO_OUT28/LCU1_OUT7/EMIOS2_CH2_H/EMIOS2_CH8_X/CAN15_RX/LCU1_IN7/LPUART19_RX/SIUL4_EIRQ14/ADC1_CH2_P2/WKPU60
[15]	GPIO308	J16	GPIO308/EMIOS2_CH9_H/LPSPI5_SOUT/LCU1_OUT1/EMIOS1_CH9_H/LCU1_IN1/LPUART18_RX/LPUART19_RX/SIUL4_EIRQ10/ADC1_CH20_M4
[15]	GPIO309	C18	GPIO309/DSPV_GPIO_OUT21/LPI2C3_SCL/EMIOS1_CH5_H/EMIOS1_CH23_X/LPSPI5_SCK/LCU1_OUT0/CAN16_RX/LCU1_IN0/ADC1_CH3_P3/WKPU61
[15]	GPIO310	C19	GPIO310/LPSPI5_SIN/EMIOS1_CH15_H/EMIOS2_CH15_H/CAN16_TX/DSPV_GPIO_OUT16/LCU1_OUT5/LCU1_IN5/LPI2C3_HREQ/LPUART20_RX/SIUL4_EIRQ11/ADC1_CH25_M9
[15]	GPIO311	J17	GPIO311/LPSPI5_PCS7/LPUART18_TX/LCU1_OUT0/EMIOS2_CH14_H/EMIOS1_CH14_H/LPUART19_TX/LCU1_IN0/SIUL4_EIRQ6/ADC1_CH4_P4
[15]	GPIO312	J18	GPIO312/LPSPI5_PCS1/LCU1_OUT3/EMIOS1_CH7_H/EMIOS2_CH7_H/LPI2C3_SCL/LCU1_IN3/SIUL4_EIRQ7/ADC1_CH26_M10
[15]	GPIO313	J19	GPIO313/LCU1_OUT2/EMIOS1_CH6_H/EMIOS2_CH0_X/LPI2C3_SDA/LCU1_IN2/SIUL4_EIRQ8/ADC1_CH10_P10
[15]	GPIO314	D16	GPIO314/LPUART19_TX/DSPV_GPIO_OUT29/LCU1_OUT6/ADC1_MA1/EMIOS1_CH4_H/EMIOS2_CH4_H/CAN15_RX/LCU1_IN6/SIUL4_EIRQ1/ADC1_CH18_M2
[15]	GPIO315	D17	GPIO315/CAN15_TX/LPI2C3_SDA/EMIOS1_CH16_X/ADC1_MA0/EMIOS1_CH3_H/LPSPI5_PCS0/DSPV_GPIO_OUT25/LCU1_OUT2/LCU1_IN2/SIUL4_EIRQ9/ADC1_CH0_P0
[17]	SPI5_PCS0	D19	GPIO316/DSPV_GPIO_OUT17/LPUART20_TX/LPSPI5_PCS0/EMIOS2_CH3_H/LPI2C3_SCL/EMIOS2_CH16_X/LCU1_OUT4/HSE2_BBSM_OUTPUT/CAN16_RX/LCU1_IN4/SIUL4_EIRQ12/ADC1_CH9_P9/HSE2_BBSM_INPUT
[17]	SPI5_SOUT	D18	GPIO317/LCU1_OUT1/EMIOS1_CH22_X/LPSPI5_SOUT/EMIOS1_CH4_H/CAN16_TX/DSPV_GPIO_OUT22/HSE2_BBSM_OUTPUT/LCU1_IN1/ADC1_CH19_M3/HSE2_BBSM_INPUT
[15]	GPIO318	K14	GPIO318/LCU1_OUT9/EMIOS1_CH6_H/EMIOS2_CH6_H/CAN11_RX/LCU1_IN9/LPUART17_RX/LPUART19_RX/SIUL4_EIRQ5/ADC1_X1_P17/WKPU57
[15]	GPIO319	K16	GPIO319/LPSPI5_PCS6/LCU1_OUT11/EMIOS2_CH22_X/EMIOS2_CH4_H/CAN11_RX/LCU1_IN11/LPUART20_RX/ADC1_CH22_M6/WKPUS8

S32K566_437BGA_B0_REV3.10

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[17]	GPIO320	E16	GPIO320/DSPV_GPIO_OUT30/LCU1_OUT3/EMIOS2_CH8_X/EMIOS2_CH2_H/CAN15_TX/LCU1_IN3/ADC1_CH16_M0
[17]	GPIO321	F16	GPIO321/DSPV_GPIO_OUT31/LCU1_OUT8/EMIOS1_CH8_X/EMIOS1_CH2_H/CAN12_RX/LCU1_IN8/LPUART20_RX/ADC1_CH1_P1/WKPU54
[15]	GPIO322	E19	GPIO322/LPUART20_TX/LCU1_OUT10/EMIOS2_CH18_Y/EMIOS1_CH18_Y/DSPV_GPIO_OUT18/LPSPI5_SOUT/LCU1_IN10/LPI2C3_HREQ/SIUL4_EIRQ13/ADC1_X0_P16
[17]	SPI5_SCK	K17	GPIO323/LPUART20_TX/EMIOS2_CH17_Y/EMIOS1_CH17_Y/LCU1_OUT10/LPSPI5_SCK/HSE2_BBSM_OUTPUT/CAN11_TX/LCU1_IN10/ADC1_CH6_P6/HSE2_BBSM_INPUT
[17]	SPI5_SIN	K18	GPIO324/LPUART18_TX/LCU1_OUT9/EMIOS2_CH23_X/LPSPI5_SIN/EMIOS2_CH6_H/ADC1_MA0/HSE2_BBSM_OUTPUT/LCU1_IN9/LPI2C3_HREQ/ADC1_CH24_M8/HSE2_BBSM_INPUT
[15]	GPIO325	K19	GPIO325/LCU1_OUT8/LPSPI5_PCS3/EMIOS2_CH16_X/EMIOS2_CH3_H/ADC1_MA1/LCU1_IN8/LPUART18_RX/ADC1_CH8_P8
[15]	GPIO326	F17	GPIO326/DSPV_GPIO_OUT26/EMIOS1_CH8_X/CAN12_TX/LCU1_OUT9/EMIOS1_CH2_H/LCU1_IN9/SIUL4_EIRQ0/ADC1_CH17_M1
[15]	GPIO327	F18	GPIO327/DSPV_GPIO_OUT23/EMIOS1_CH21_Y/LPSPI5_PCS2/LCU1_OUT11/EMIOS2_CH21_Y/CAN13_RX/LCU1_IN11/SIUL4_EIRQ15/ADC1_X2_M16
[15]	GPIO328	F19	GPIO328/LPI2C3_SDA/FCCU_EOUT1/LPUART18_TX/EMIOS1_CH19_Y/LCU1_OUT3/LPSPI5_SIN/EMIOS2_CH19_Y/DSPV_GPIO_OUT19/CAN14_TX/CAN12_RX/LCU1_IN3/SIUL4_EIRQ2/ADC1_CH21_M5/WKPU62/FCCU_ERR_IN1
[15]	GPIO329	H14	GPIO329/LCU1_OUT6/EMIOS1_CH5_H/EMIOS2_CH5_H/CAN13_RX/LCU1_IN6/SIUL4_EIRQ1/ADC1_CH29_M13/WKPU56
[16]	LPUART18_RX	G16	GPIO330/LCU1_OUT10/EMIOS2_CH13_H/ADC1_MA1/EMIOS1_CH13_H/CAN13_RX/LCU1_IN10/LPUART17_RX/LPUART18_RX/SIUL4_EIRQ2/ADC1_CH12_P12/WKPU55
[16]	LPUART18_TX	G17	GPIO331/LCU1_OUT6/EMIOS2_CH12_H/ADC1_MA0/EMIOS1_CH12_H/CAN13_TX/DSPV_GPIO_OUT27/LPUART18_TX/LCU1_IN6/SIUL4_EIRQ3/ADC1_CH11_P11
[3]	FCCU_EOUT2_IN2_MCU	G18	GPIO332/FCCU_EOUT2/EMIOS1_CH0_X/LCU1_OUT7/CAN13_TX/LPI2C3_SDAS/EMIOS1_CH1_H/DSPV_GPIO_OUT24/LPSPI5_PCS3/LCU1_IN7/SIUL4_EIRQ3/ADC1_CH27_M11/FCCU_ERR_IN2
[15]	GPIO333	G19	GPIO333/EMIOS1_CH20_Y/ADC1_MA2/CAN12_TX/LCU1_OUT2/EMIOS2_CH20_Y/DSPV_GPIO_OUT20/LPSPI5_SCK/LPI2C3_SCLS/CAN14_RX/LCU1_IN2/LPUART18_RX/SIUL4_EIRQ4/ADC1_CH5_P5/WKPU63
[15]	GPIO334	J14	GPIO334/LPUART17_TX/LPUART19_TX/EMIOS2_CH0_X/CAN11_TX/LCU1_OUT8/EMIOS2_CH1_H/LCU1_IN8/ADC1_X3_M17
[15]	GPIO335	J15	GPIO335/EMIOS1_CH3_H/LPSPI5_PCS2/EMIOS1_CH16_X/CAN13_TX/LCU1_OUT7/LCU1_IN7/ADC1_CH13_P13
[15]	GPIO336	B14	GPIO336/EMIOS2_CH0_X/ADC1_MA0/LCU1_OUT6/CAN11_RX/LCU1_IN6/SIUL4_EIRQ6
[15]	GPIO337	B15	GPIO337/EMIOS2_CH8_X/ADC1_MA1/LCU1_OUT5/CAN11_TX/LPUART19_TX/LCU1_IN5/SIUL4_EIRQ7
[15]	GPIO338	B13	GPIO338/EMIOS2_CH16_X/ADC1_MA2/LCU1_OUT7/CAN11_TX/LCU1_IN7/SIUL4_EIRQ8
[15]	GPIO339	A13	GPIO339/EMIOS2_CH22_X/LPI2C3_SDA/LPSPI5_SOUT/SIUL4_EIRQ0
[15]	GPIO340	A14	GPIO340/EMIOS2_CH23_X/LPI2C3_SCL/LPSPI5_SIN/CAN13_RX/SIUL4_EIRQ1
[15]	GPIO341	A15	GPIO341/EMIOS1_CH0_X/CAN13_TX/LPSPI5_SCK/LPI2C3_HREQ/LPUART19_RX/SIUL4_EIRQ2
[15]	GPIO342	A17	GPIO342/EMIOS1_CH8_X/CAN16_TX/LPSPI5_PCS0/LPUART19_RX/SIUL4_EIRQ3
[15]	GPIO343	A18	GPIO343/EMIOS1_CH22_X/LPUART19_TX/LPSPI5_PCS2/LCU1_OUT0/CAN16_RX/LCU1_IN0/LPUART17_RX/SIUL4_EIRQ4
[15]	GPIO344	A19	GPIO344/LPUART17_TX/EMIOS2_CH3_H/LPSPI5_PCS3/LCU1_OUT1/LCU1_IN1/SIUL4_EIRQ5
[14]	GPIO345	B21	GPIO345/EMIOS2_CH20_Y/EMIOS1_CH20_Y/LPUART17_TX/LPSPI5_PCS3/LCU1_OUT3/CAN15_TX/LCU1_IN2/SIUL4_EIRQ6
[15]	GPIO346	D20	GPIO346/EMIOS2_CH18_Y/CAN15_TX/EMIOS1_CH18_Y/LCU1_OUT2/LPUART17_TX/TRACE_ETM_D1/LCU1_IN2/SIUL4_EIRQ9
[14]	GPIO347	D21	GPIO347/EMIOS2_CH15_H/LPSPI5_PCS1/LPUART20_TX/EMIOS1_CH15_H/LCU1_OUT4/TRACE_ETM_D8/CAN15_RX/LCU1_IN4/SIUL4_EIRQ8
[14]	GPIO348	B17	GPIO348/EMIOS1_CH16_X/LPSPI5_PCS2/LCU1_OUT4/TRACE_ETM_CLKOUT/ADC1_MA0/LPUART17_TX/CAN14_RX/LCU1_IN4/LPUART19_RX/SIUL4_EIRQ10
[14]	GPIO349	B18	GPIO349/EMIOS1_CH23_X/EMIOS2_CH2_H/CAN14_TX/ADC1_MA1/TRACE_ETM_D0/LPSPI5_PCS3/LCU1_OUT3/LCU1_IN3/LPUART17_RX/SIUL4_EIRQ11
[15]	GPIO350	C21	GPIO350/ADC1_MA2/EMIOS2_CH21_Y/EMIOS1_CH21_Y/LPSPI5_PCS2/LCU1_OUT3/CAN15_RX/LCU1_IN3/LPUART17_RX/SIUL4_EIRQ7
[14]	GPIO351	F20	GPIO351/EMIOS1_CH19_Y/EMIOS2_CH19_Y/LCU1_OUT0/TRACE_ETM_D3/LCU1_IN0/SIUL4_EIRQ12/ADC1_CH31_M15

S32K566_437BGA_B0_REV3.10

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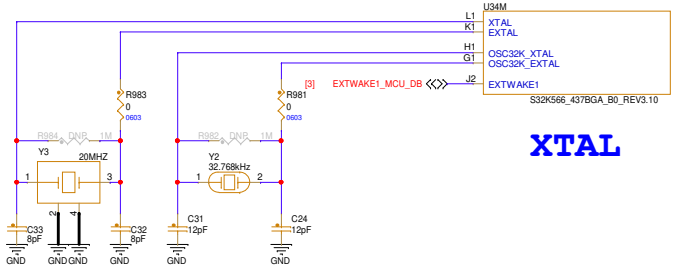
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[14]	GPIO353	E20	GPIO353/EMIOS2_CH14_H/LCU1_OUT1/EMIOS1_CH14_H/TRACE_ETM_D2/CAN15_RX/LCU1_IN1/SIUL4_EIRQ13/ADC1_CH15_P15
[15]	GPIO354	E21	GPIO354/EMIOS2_CH13_H/TRACE_ETM_D9/EMIOS1_CH13_H/LPSPI5_PCS4/LCU1_OUT5/CAN15_TX/LCU1_IN5/LPUART20_RX/SIUL4_EIRQ9
[15]	GPIO355	H20	GPIO355/EMIOS2_CH12_H/LPI2C3_SDA/LPUART20_TX/EMIOS1_CH12_H/TRACE_ETM_D4/LPSPI5_PCS0/LCU1_OUT10/CAN16_RX/LCU1_IN10/LPUART18_RX/SIUL4_EIRQ14
[15]	GPIO356	H21	GPIO356/EMIOS2_CH10_H/EMIOS1_CH10_H/CAN12_TX/LPUART19_TX/TRACE_ETM_D11/LPSPI5_SIN/LCU1_OUT7/LCU1_IN7/SIUL4_EIRQ11
[15]	GPIO357	J20	GPIO357/EMIOS2_CH9_H/LPI2C3_SCL/CAN16_TX/EMIOS1_CH9_H/TRACE_ETM_D5/LCU1_OUT8/LCU1_IN8/LPUART20_RX/SIUL4_EIRQ15/ADC1_CH30_M14
[15]	GPIO358	K21	GPIO358/EMIOS2_CH7_H/EMIOS1_CH7_H/CAN14_TX/LPUART17_TX/TRACE_ETM_D13/LPSPI5_PCS5/LCU1_OUT8/LCU1_IN8/SIUL4_EIRQ13
[15]	GPIO359	H20	GPIO359/EMIOS2_CH6_H/CAN12_TX/EMIOS1_CH23_X/TRACE_ETM_D6/LCU1_OUT9/CAN16_RX/LCU1_IN9/LPI2C3_HREQ/LPUART18_RX/ADC1_CH14_P14
[15]	GPIO360	L20	GPIO360/EMIOS2_CH5_H/LPUART18_TX/EMIOS1_CH22_X/LCU1_OUT11/TRACE_ETM_D7/CAN12_RX/LCU1_IN11
[15]	GPIO361	L21	GPIO361/LPSPI5_PCS6/EMIOS2_CH4_H/CAN11_TX/LCU1_OUT10/TRACE_ETM_D14/CAN14_TX/LCU1_IN10/SIUL4_EIRQ14
[15]	GPIO362	M21	GPIO362/EMIOS2_CH1_H/LPSPI5_PCS7/LCU1_OUT11/TRACE_ETM_D15/CAN11_RX/CAN14_RX/LCU1_IN11/SIUL4_EIRQ15
[15]	GPIO363	J21	GPIO363/EMIOS1_CH11_H/EMIOS2_CH11_H/LPI2C3_SDA/TRACE_ETM_D12/LPSPI5_SCK/LCU1_OUT9/CAN14_RX/LCU1_IN9/LPUART17_RX/SIUL4_EIRQ12
[15]	GPIO364	B16	GPIO364/EVTO_0/ADC1_MA3/LCU1_OUT0/EMIOS1_CH10_H/LPI2C3_SCLS/CAN11_RX/CAN16_RX/LCU1_IN0
[15]	GPIO365	C17	GPIO365/ADC1_MA3/CAN11_TX/CAN16_TX/EMIOS1_CH11_H/LPI2C3_SDAS/LPUART19_TX/LCU1_IN1/EVTI_0
[15]	GPIO366	E17	GPIO366/LCU1_OUT4/ADC1_MA2/EMIOS2_CH10_H/LCU1_IN4/EVTI_1
[15]	GPIO367	A20	GPIO367/EVTO_1/LCU1_OUT5/EMIOS1_CH3_H/LPSPI5_PCS5/CAN13_RX/CAN15_RX
[15]	GPIO368	B19	GPIO368/CAN13_TX/CAN14_TX/ADC1_MA2/EMIOS1_CH4_H/EVTI_1
[15]	GPIO369	C20	GPIO369/EVTO_1/ADC1_MA3/CAN15_TX/EMIOS1_CH5_H/LPSPI5_PCS1/CAN14_RX/LPUART17_RX
[15]	GPIO370	C20	GPIO370/EVTO_0/CAN12_TX/CAN16_TX/LCU1_OUT1/EMIOS1_CH2_H/LPUART18_TX/LPSPI5_PCS7/LPUART20_RX
[15]	GPIO371	F21	GPIO371/ADC1_MA3/EMIOS1_CH1_H/LPUART20_TX/LPSPI5_PCS6/CAN12_RX/LCU1_IN5/EVTI_0
[13]	CANXL3V3_RX1	P10	GPIO432/LPSPI0_SIN/CANXL1_RX/FlexRay0_A_RX
[13]	CANXL3V3_TX1	R10	GPIO433/FlexRay0_A_TX/LPSPI0_SOUT/CANXL1_TX
[15]	GPIO434	P12	GPIO434/FlexRay0_B_TX/LPSPI1_SCK
[15]	GPIO435	P13	GPIO435/LPSPI1_SIN/FlexRay0_B_RX
[15]	GPIO436	R13	GPIO436/FlexRay0_B_TX_EN_b/LPSPI1_SOUT
[15]	GPIO437	U13	GPIO437/FlexRay0_A_TX_EN_b/FlexRay0_DBG0/LPSPI1_PCS0

S32K566_437BGA_B0_REV3.10

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L1	XTAL	XTAL
K1	EXTAL	EXTAL
H1	OSC32K_XTAL	OSC32K_XTAL
G1	OSC32K_EXTAL	OSC32K_EXTAL
[3] EXTWAKE1_MCU_DB	J2	EXTWAKE1

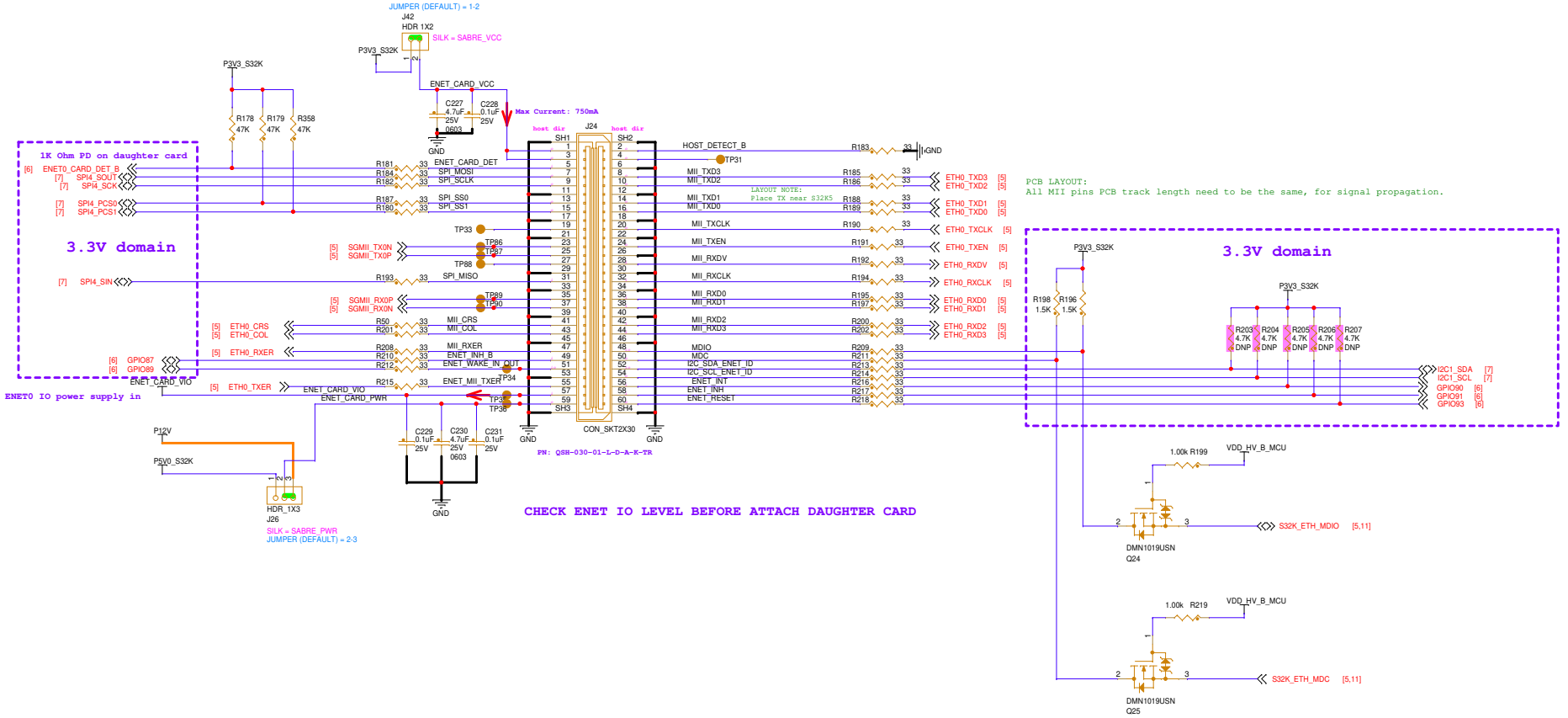
S32K566_437BGA_B0_REV3.10



XTAL

Classification: Company Internal Use			
Drawing Title: X-RING-OPTIC-ETH			
Page Title: 08 - S32K566 A DOMAIN			
Size: C	Document Number: SCH-97122 / PDF: SPF-97122		Rev: A
Date: Tuesday, August 25, 2026	Sheet 8 of 38		

ENET DAUGHTER CARD 60PIN CONNECTOR

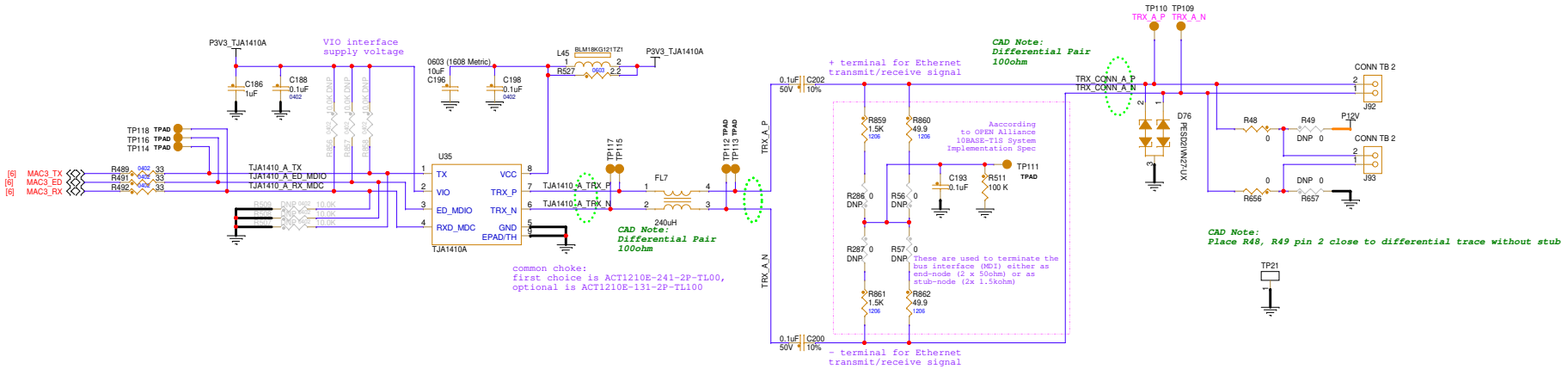


CHECK ENET IO LEVEL BEFORE ATTACH DAUGHTER CARD

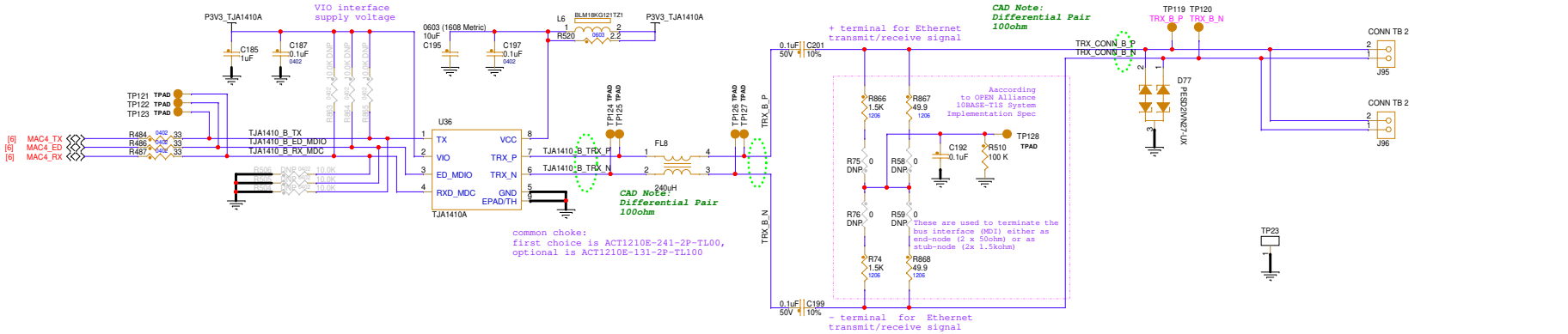
Classification: Company Internal Use		
Drawing Title: X-RING-OPTIC-ETH		
Page Title: 09 - ENET 60PIN CON		
Size C	Document Number SCH-97122 / PDF: SPF-97122	Rev A
Date: Tuesday, August 25, 2026	Sheet 9 of 38	

ETHERNET 10BASE-T1S

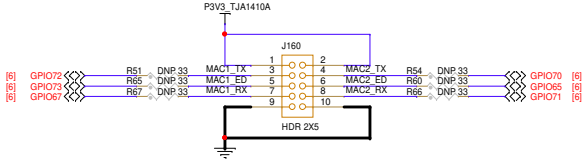
TJA1410 ASIL B Compliant 10BASE-T1S PMD Transceiver



TJA1410 ASIL B Compliant 10BASE-T1S PMD Transceiver



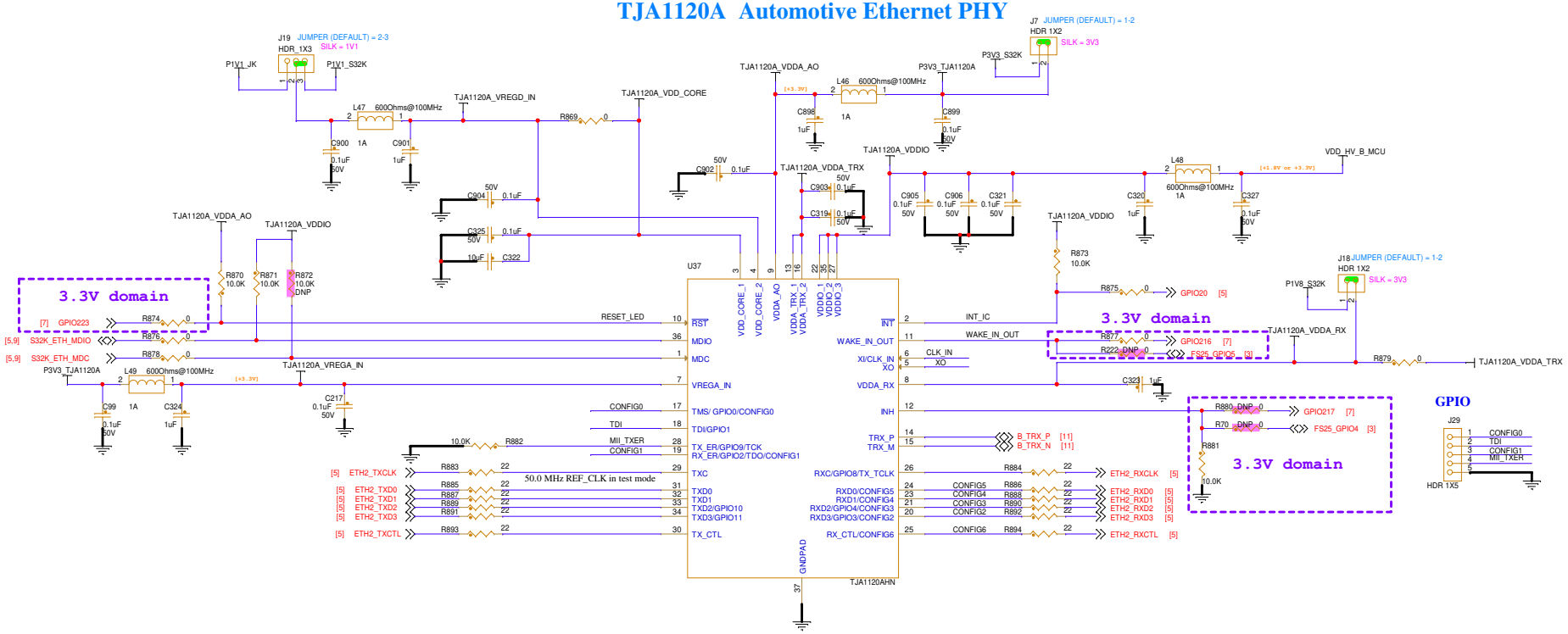


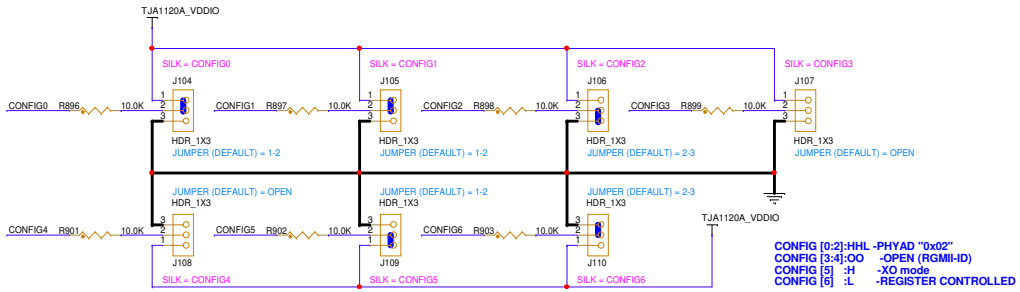


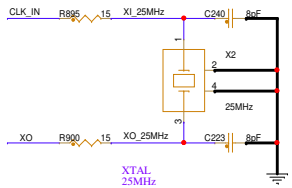
S32K5 silicon REV B
GPIO71: CANXL3V3_RX1
GPIO72: CANXL3V3_TX1

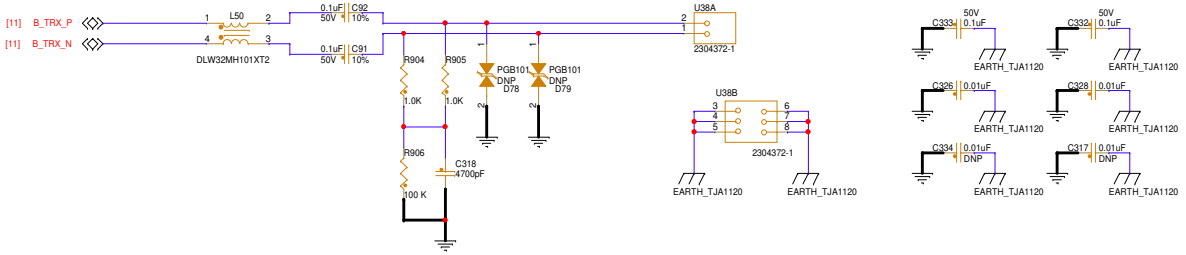
Classification: Company Internal Use			
Drawing Title: X-RING-OPTIC-ETH			
Page Title: 10 - ENET TJA1410A			
Size C	Document Number SCH-97122 / PDF: SPF-97122		Rev A
Date: Tuesday, August 25, 2026		Sheet 10 of 38	

TJA1120A Automotive Ethernet PHY



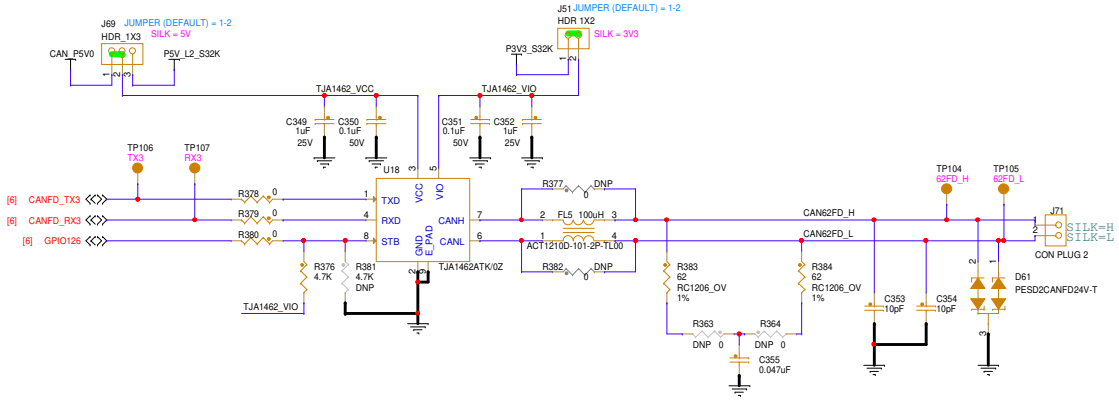






Classification:	<Classification>
Drawing Title:	X-RING-OPTIC-ETH
Page Title:	11 - ENET TJA1120A
Size: C	Document Number: SCH-97122 / PDF: SPF-97122 — Rev: A
Date: Tuesday, August 25, 2026	Sheet 11 of 38

TJA1462ATK or TJA1464ATK– CAN FD



Classification: Company Internal Use

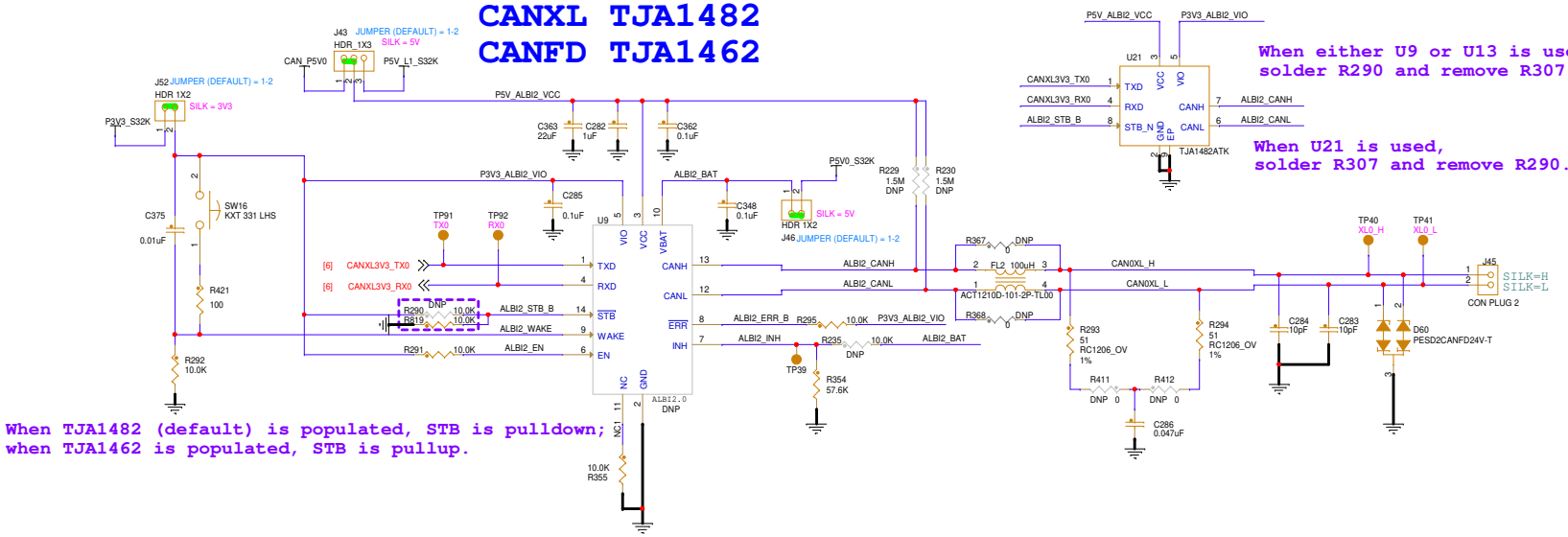
Drawing Title: X-RING-OPTIC-ETH

Page Title: 12 - CAN TJA1462A

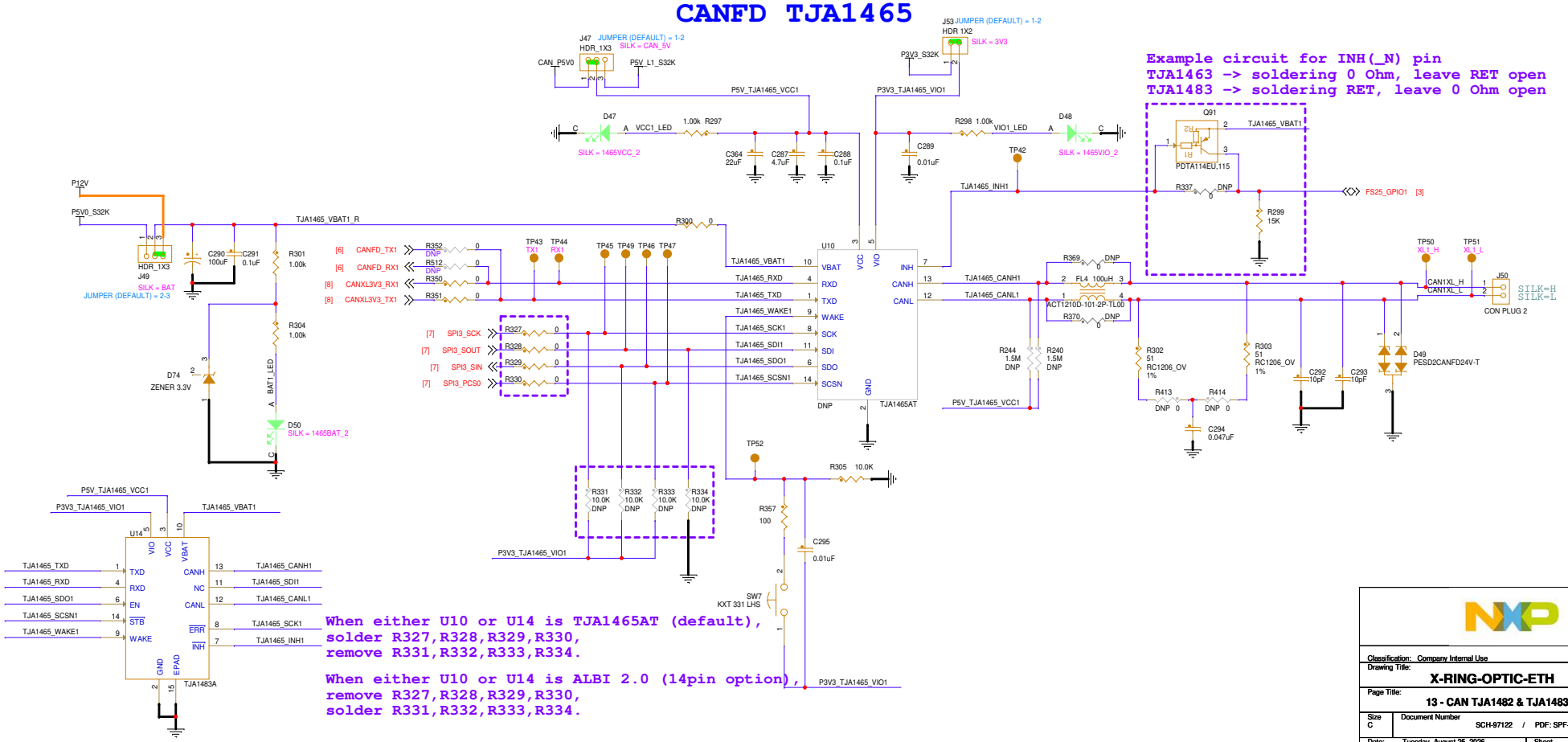
Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 Sheet 12 of 38

CANXL TJA1482
CANFD TJA1462



CANXL TJA1483
CANFD TJA1465



Classification: Company Internal Use

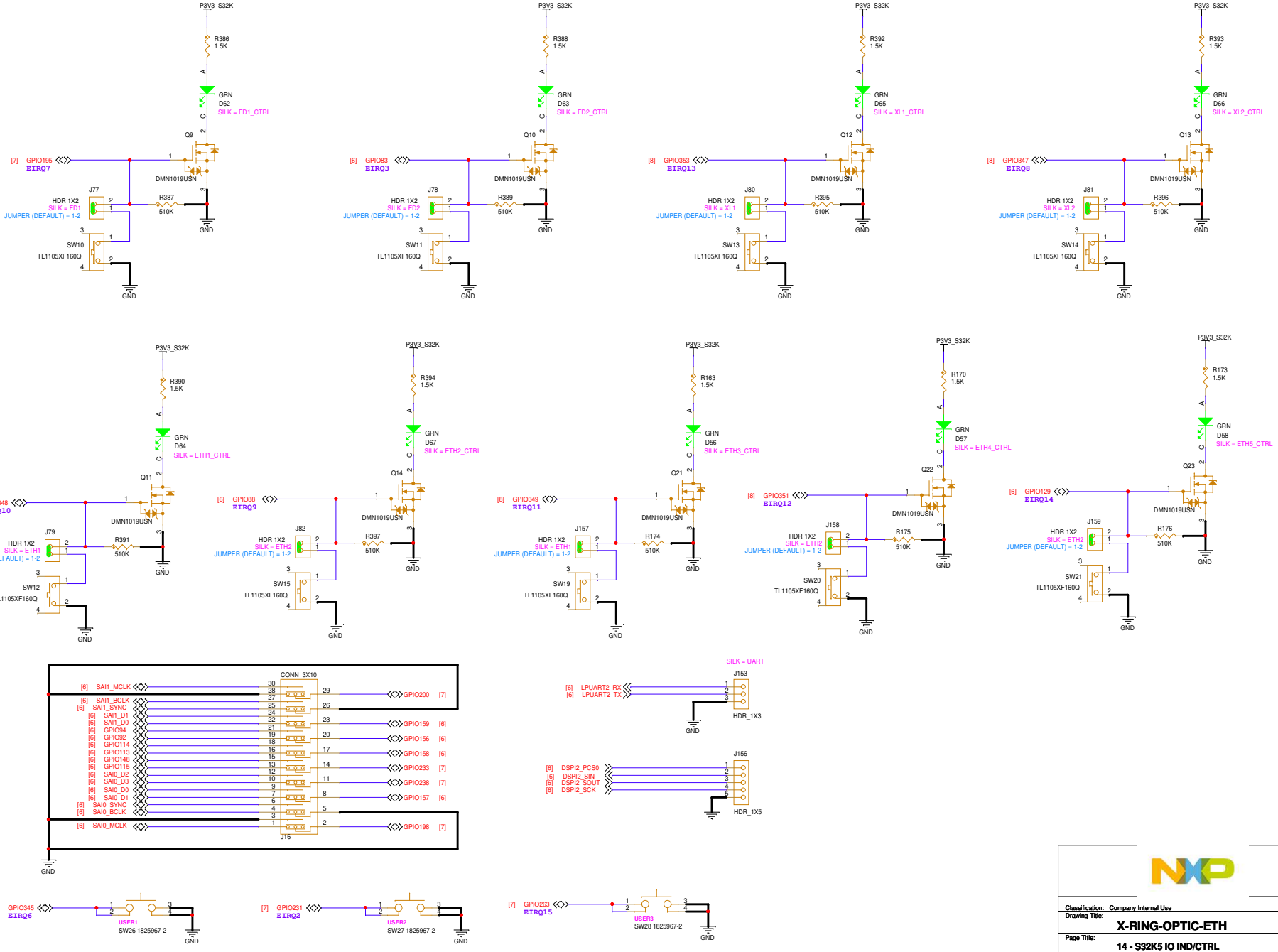
Drawing Title: X-RING-OPTIC-ETH

Page Title: 13 - CAN TJA1482 & TJA1483

Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 | Sheet 13 of 38

I/O pin control, LED indication



Classification: Company Internal Use

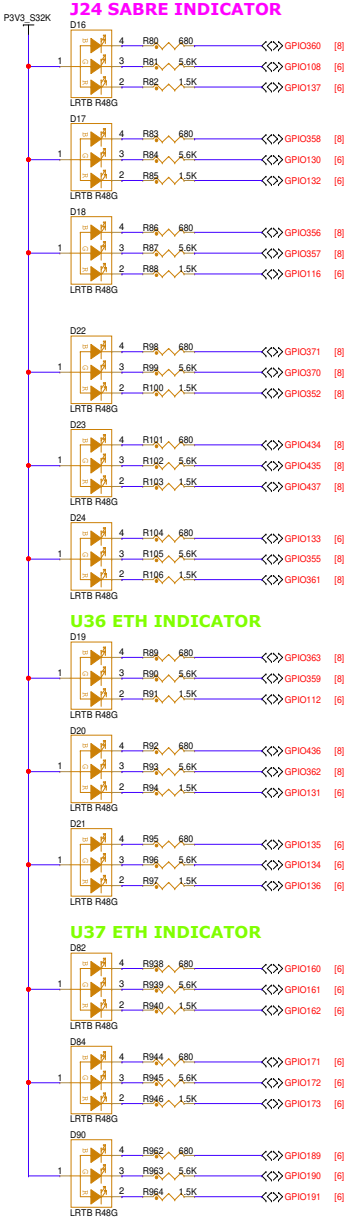
Drawing Title: **X-RING-OPTIC-ETH**

Page Title: **14 - S32K5 IO IND/CTRL**

Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 | Sheet 14 of 38

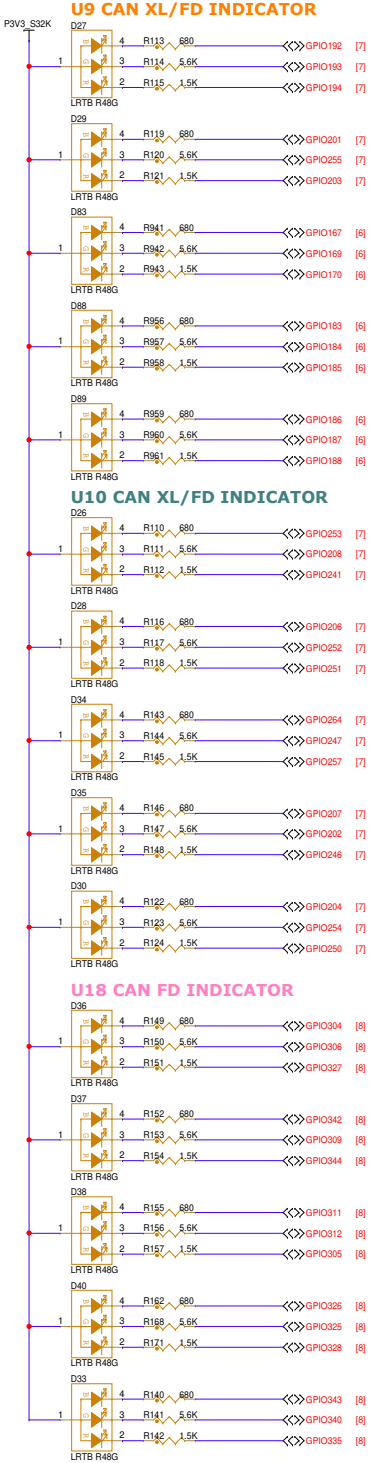
J24 SABRE INDICATOR



U36 ETH INDICATOR

U37 ETH INDICATOR

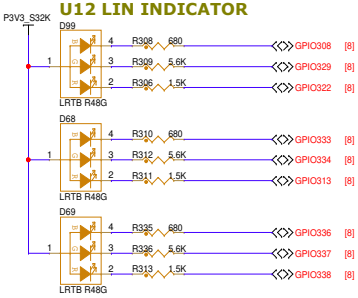
U9 CAN XL/FD INDICATOR



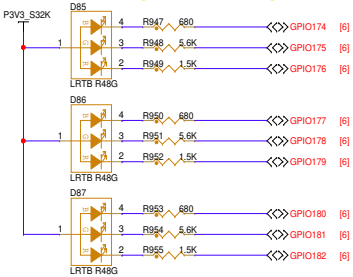
U10 CAN XL/FD INDICATOR

U18 CAN FD INDICATOR

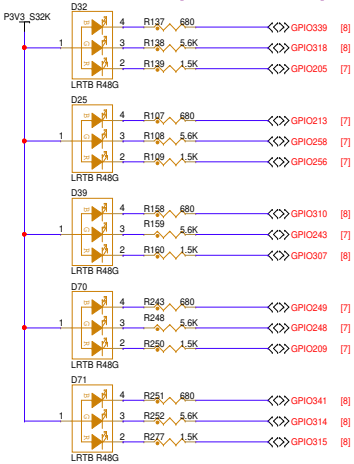
U12 LIN INDICATOR

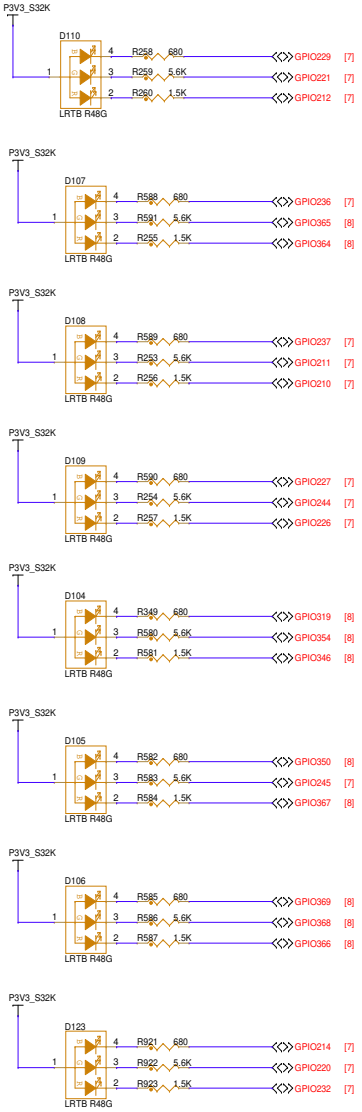


ETH IND (USED FOR U7)



CAN FD IND (USED FOR U11)





Classification: <Classification>

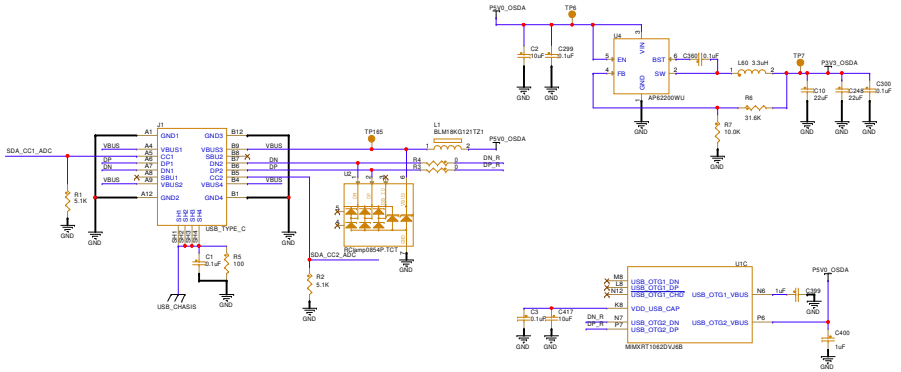
Drawing Title: **X-RING-OPTIC-ETH**

Page Title: **15 - LED INDICATORS**

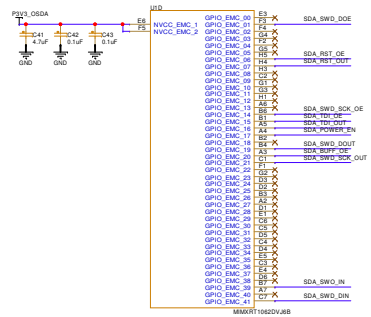
Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

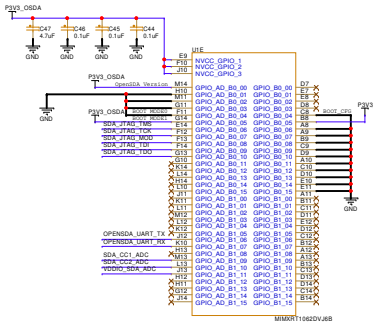
Date: Tuesday, August 25, 2026 | Sheet 15 of 38

USB Type C

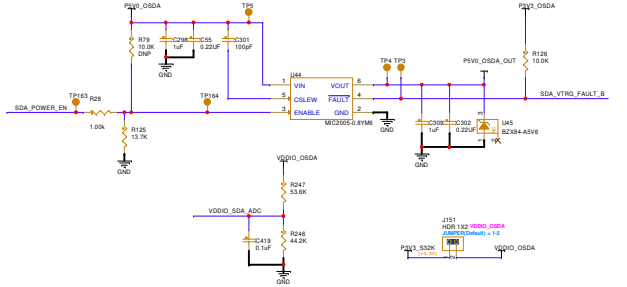


OpenSDA – Periph

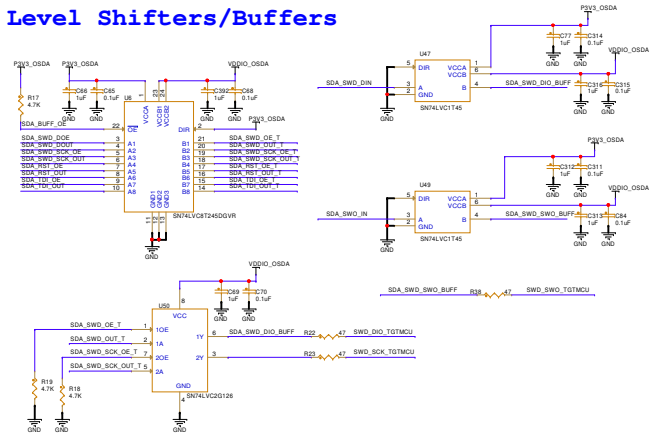




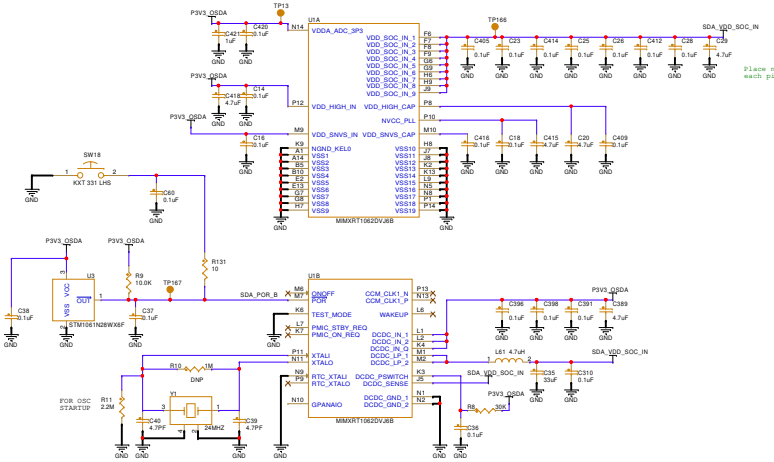
PWR Switch



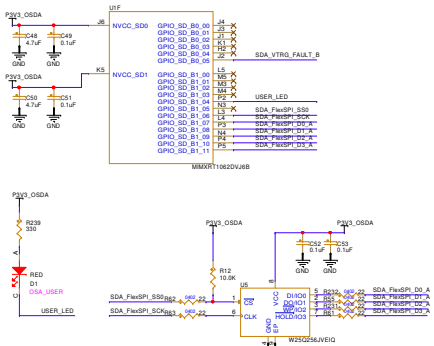
Level Shifters/Buffers



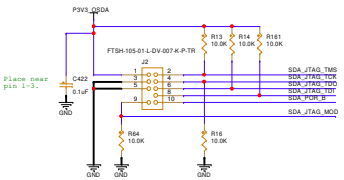
OpenSDA – PWR



OpenSDA – Ext Memory



OpenSDA – JTAG Connector



External signals

- OSDA_TX0 — LPUART1ti_TX [8]
- OSDA_RX0 — LPUART1ti_RX [8]
- RESET_TGTMCU — RESET_MCU [3,6,7]
- SWD_DIO_TGTMCU — JTAG_TMS/SWD_DIO [6]
- TDI_TGTMCU — JTAG_TDI [6]
- SWD_SWO_TGTMCU — JTAG_TDO [6]
- SWD_SCK_TGTMCU — JTAG_TCLK/SWD_CLK [6]

NXP

Classification: Company Internal Use

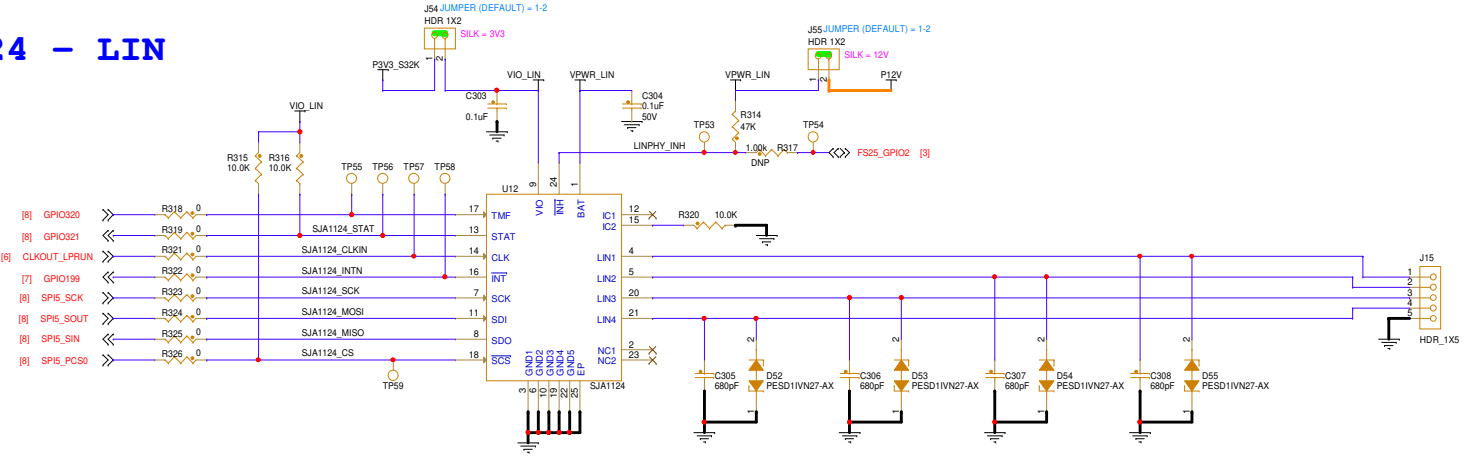
Drawing Title: **X-RING-OPTIC-ETH**

Page Title: **16 - OpenSDA FOR S32K5**

Size	Document Number	Rev
D	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 | Sheet 16 of 38

SJA1124 - LIN



Classification: <Classification>

Drawing Title: X-RING-OPTIC-ETH

Page Title: 17 - LIN SJA1124

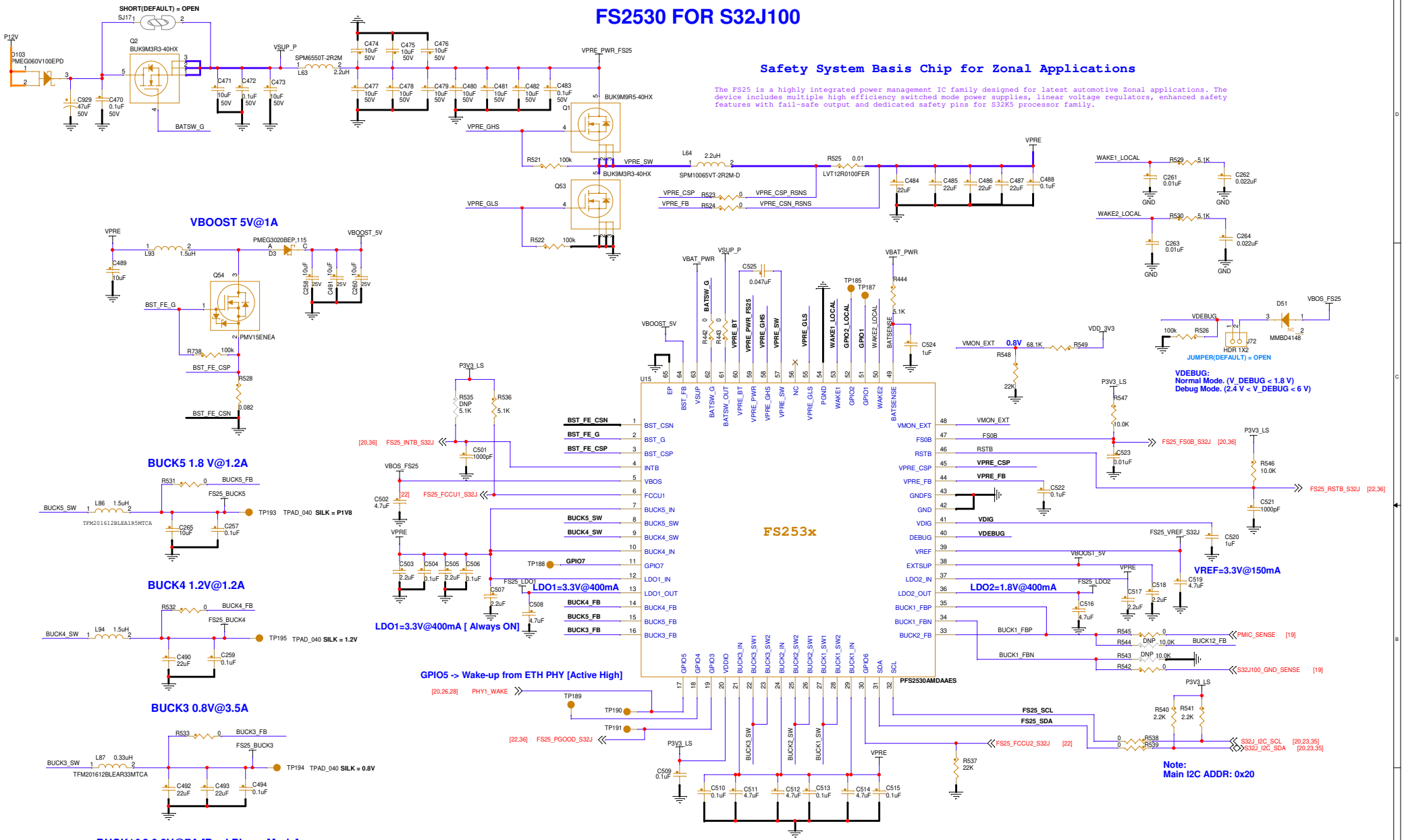
Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 | Sheet 17 of 38

FS2530 FOR S32J100

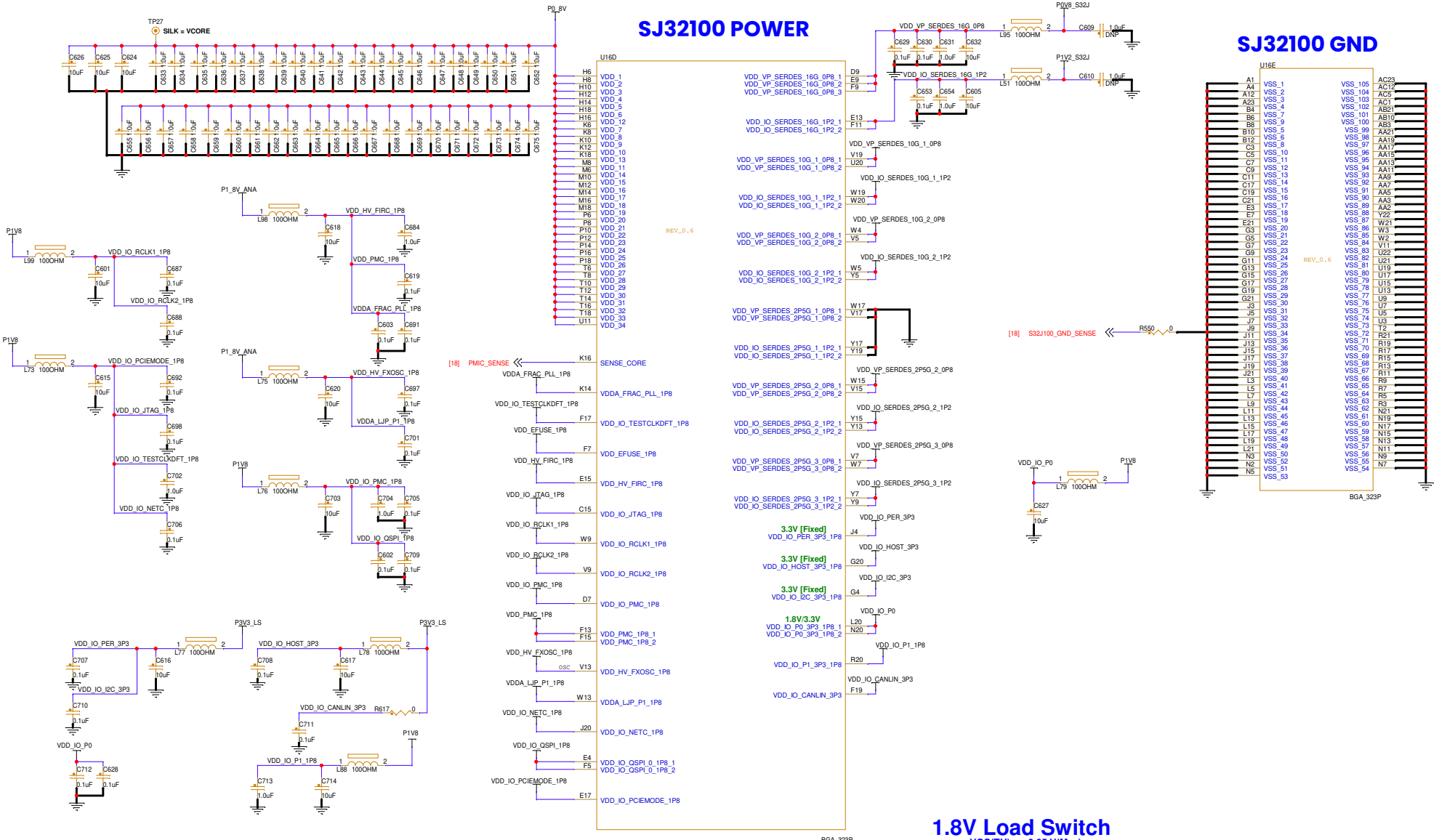
Safety System Basis Chip for Zonal Applications

The FS25 is a highly integrated power management IC family designed for latest automotive Zonal applications. The device includes multiple high efficiency switched mode power supplies, linear voltage regulators, enhanced safety features with fail-safe output and dedicated safety pins for S32K5 processor family.



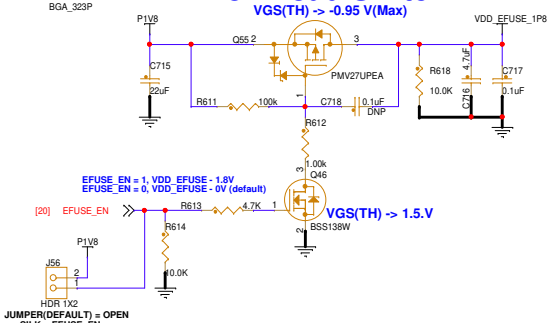
Classification:	<Classification>		
Drawing Title:	X-RING-OPTIC-ETH		
Page Title:	18 - FS2530 FOR S32J100		
Size: C	Document Number: SCH-97122 / PDF: SPF-97122		Rev: A
Date: Tuesday, August 25, 2026		Sheet 18 of 38	

SJ32100 POWER



SJ32100 GND

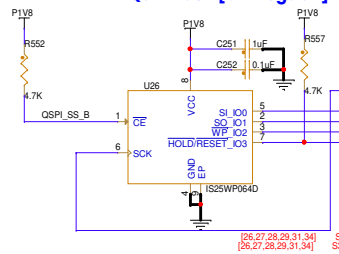
1.8V Load Switch



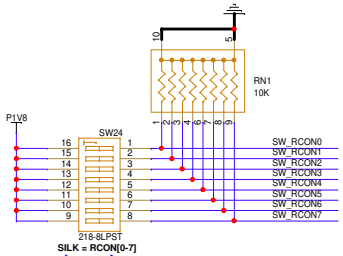
Classification: <Classification>			
Drawing Title: X-RING-OPTIC-ETH			
Page Title: 19 - S32J100 POWER & GND			
Size: C	Document Number: SCH-97122 / PDF: SPF-97122		Rev: A
Date: Tuesday, August 25, 2026	Sheet 19 of 38		

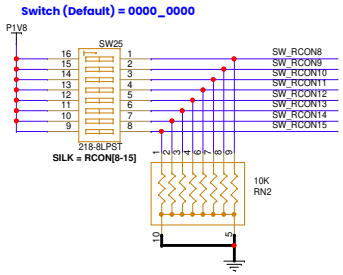
S32J100 (Ports, SPI, SMI, Config)

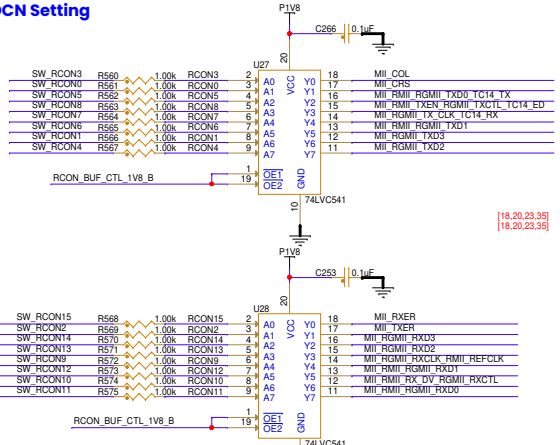
QSPI Flash [64 Megabit]

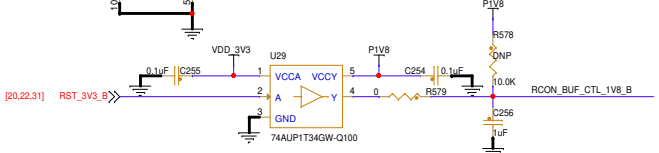


ROCN Setting

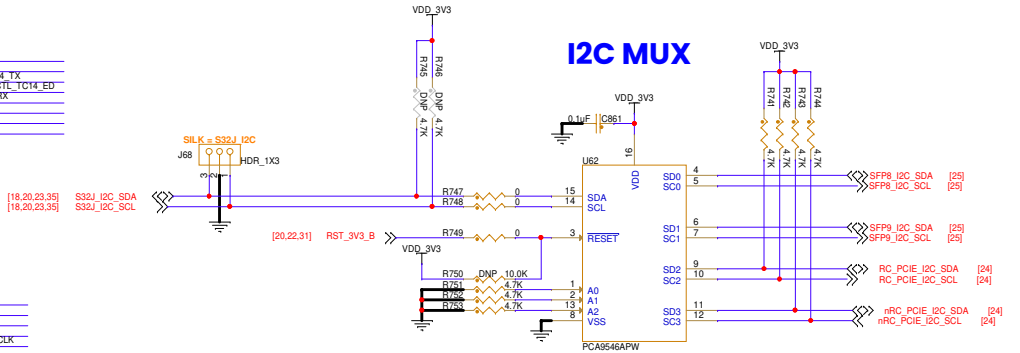








I2C MUX



U16A

BGA_323P

D2	PAD_013/BOOTMOD_0/TAMPER_IN/GPIO[13]	
E2	PAD_014/BOOTMOD_1/GPIO[14]	
A2	PAD_016/QSPI_0_CK_A/GPIO[16]	
B3	PAD_020/QSPI_0_CS_A0/GPIO[20]	
C2	PAD_015/QSPI_0_DATA_A_I[0]/QSPI_0_DATA_A_O[0]/GPIO[15]	
D1	PAD_017/QSPI_0_DATA_A_I[1]/QSPI_0_DATA_A_O[1]/GPIO[17]	
B2	PAD_018/QSPI_0_DATA_A_I[2]/QSPI_0_DATA_A_O[2]/GPIO[18]	
B1	PAD_019/QSPI_0_DATA_A_I[3]/QSPI_0_DATA_A_O[3]/GPIO[19]	
A18	PAD_021/EVTI_B_0/PCI_CLKREQ_N_I/CLKOUT_0/PCI_CLKREQ_N_O/GPIO[21]	
B17	PAD_022/PCI_WAKE_N_I/CLKOUT_3/EVTO_B_0/PCI_WAKE_N_O/GPIO[22]	
D17	PAD_023/PCI_PERESET_B_I/PCI_PERESET_B_O/GPIO[23]	
B23	PAD_034/TMR_1588_CLK_I/CLKOUT_1/TMR_1588_CLK_O/GPIO[34]	
C22	PAD_035/TMR_1588_TRIG2/TMR_1588_ALARM1/TMR_1588_PP2/GPIO[35]	
D23	PAD_037/ETH_MDC_I/ETH_MDC_O/GPIO[37]	
D22	PAD_036/ETH_MDIO_I/ETH_MDIO_O/GPIO[36]	
A19	PAD_024/CAN_24_RX/GPIO[24]	VDD_IO_CANLIN_3P3 [3.3V]
B20	PAD_026/CAN_24_TX/GPIO[26]	
D19	PAD_025/LIN_0_RX/GPIO[25]	
A21	PAD_027/LIN_0_TX/GPIO[27]	
B22	PAD_030/DSPI_0_SCK_I/DSPI_0_SCK_O/GPIO[30]	VDD_IO_HOST_3P3_1P8 [3.3V -Fixed]
B21	PAD_032/DSPI_0_SIN/GPIO[32]	
A22	PAD_033/DSPI_0_SOUT/GPIO[33]	
E19	PAD_029/DSPI_0_PCS0_I/DSPI_0_PCS0_O/GPIO[29]	
E20	PAD_028/DSPI_0_PCS1/GPIO[28]	
L2	PAD_003/DSPI_5_SCK_I/DSPI_5_SCK_O/GPIO[3]	VDD_IO_PER_3P3_1P8 [3.3V -Fixed]
L4	PAD_002/DSPI_5_SIN/GPIO[2]	
N4	PAD_001/DSPI_5_SOUT/GPIO[1]	
K2	PAD_006/DSPI_5_PCS0_I/DSPI_5_PCS0_O/GPIO[6]	
L1	PAD_004/DSPI_5_PCS1/GPIO[4]	
R4	PAD_000/EIRQ_0/ERR_2/GPIO[0]	
M2	PAD_005/EIRQ_1/ERR_3/GPIO[5]	
G2	PAD_009/I2C_2_SCL_I/I2C_2_SCL_O/GPIO[9]	VDD_IO_I2C_3P3_1P8 [3.3V -Fixed]
H2	PAD_007/I2C_2_SDA_I/I2C_2_SDA_O/GPIO[7]	
J2	PAD_008/EIRQ_2/GPIO[8]	
G1	PAD_010/EIRQ_3/GPIO[10]	
F2	PAD_011/EIRQ_4/GPIO[11]	
E1	PAD_012/EIRQ_5/CLKOUT_4/GPIO[12]	
J22	PAD_053/ETH_0_RMII_RXER/ETH_0_MII_RXER/RCON15/GPIO[53]	
G22	PAD_047/ETH_PHY_INT_6/ETH_0_RX_CLK/RCON9/ETH_0_TX_RGMII_LPBK_CLK/ETH_0_RGMII_TST_TXC/ETH_0_MII_RX_CLK/GPIO[47]	
E22	PAD_048/ETH_PHY_INT_7/ETH_0_MII_RXDV/ETH_0_RGMII_RXCTL/ETH_0_RMII_RXCRSDV/RCON10/XENO1_DEBUG0/ETH_0_RGMII_TST_TXCTL/GPIO[48]	
E23	PAD_049/ETH_PHY_INT_8/ETH_0_RGMII_RXD_0/ETH_0_RMII_RXD_0/ETH_0_MII_RXD_0/RCON11/XENO1_DEBUG1/ETH_0_RGMII_TST_TXD_0/GPIO[49]	
F22	PAD_050/ETH_PHY_INT_9/ETH_0_RGMII_RXD_1/ETH_0_RMII_RXD_1/ETH_0_MII_RXD_1/RCON12/DLL_OUT/XENO1_DEBUG2/ETH_0_RGMII_TST_TXD_1/GPIO[50]	
G23	PAD_051/DLL_RX_TEST_CLK/ETH_0_RGMII_RXD_2/ETH_0_MII_RXD_2/RCON13/XENO1_DEBUG3/ETH_0_RGMII_TST_TXD_2/GPIO[51]	
H23	PAD_052/ETH_0_RGMII_RXD_3/ETH_0_MII_RXD_3/RCON14/XENO1_DEBUG4/DLL_RX_SUBSAMPLE_DATA/ETH_0_RGMII_TST_TXD_3/GPIO[52]	
H22	PAD_040/ETH_PHY_INT_2/RCON2/XENO0_DEBUG2/ETH_0_RMII_TXER/ETH_0_MII_TXER/GPIO[40]	
N23	PAD_038/ETH_PHY_INT_0/ETH_0_MII_CRS/RCON0/XENO0_DEBUG0/GPIO[38]	
P23	PAD_041/ETH_PHY_INT_3/ETH_0_MII_COL/RCON3/XENO0_DEBUG3/GPIO[41]	
L22	PAD_045/XENO0_AFERX_I/ETH_0_TX_CLK/RCON7/XENO0_AFERX_O/ETH_0_RGMII_TXC/ETH_0_MII_TX_CLK/GPIO[45]	
M22	PAD_046/RCON8/ETH_0_RGMII_RXCTL/XENO0_AFEED_I/XENO0_AFEED_O/ETH_0_RGMII_TXCTL/ETH_0_RMII_TXEN/ETH_0_MII_TXEN/GPIO[46]	
N22	PAD_043/ETH_0_RGMII_RXD_0/RCON5/XENO0_AFETX/ETH_0_RGMII_TXD_0/ETH_0_RMII_TXD_0/ETH_0_MII_TXD_0/GPIO[43]	
L23	PAD_044/ETH_PHY_INT_5/ETH_0_TX_CLK_P0TXD1/ETH_0_RGMII_RXD_1/RCON6/ETH_0_RGMII_TXD_1/ETH_0_RMII_TXD_1/ETH_0_MII_TXD_1/GPIO[44]	
K22	PAD_042/DLL_TX_TEST_CLK/ETH_PHY_INT_4/ETH_0_RGMII_RXD_2/RCON4/XENO0_DEBUG4/ETH_0_RGMII_TXD_2/ETH_0_MII_TXD_2/GPIO[42]	
K23	PAD_039/ETH_PHY_INT_1/ETH_0_RGMII_RXD_3/RCON1/DLL_TX_SUBSAMPLE_DATA/XENO0_DEBUG1/ETH_0_RGMII_TXD_3/ETH_0_MII_TXD_3/GPIO[39]	
P22	PAD_054/EIRQ_6/GPIO[54]	
T22	PAD_057/XENO1_AFERX_I/XENO1_AFERX_O/GPIO[57]	
T23	PAD_056/XENO1_AFETX/GPIO[56]	
R22	PAD_058/XENO1_AFEED_I/XENO1_AFEED_O/GPIO[58]	
U23	PAD_055/EIRQ_7/GPIO[55]	

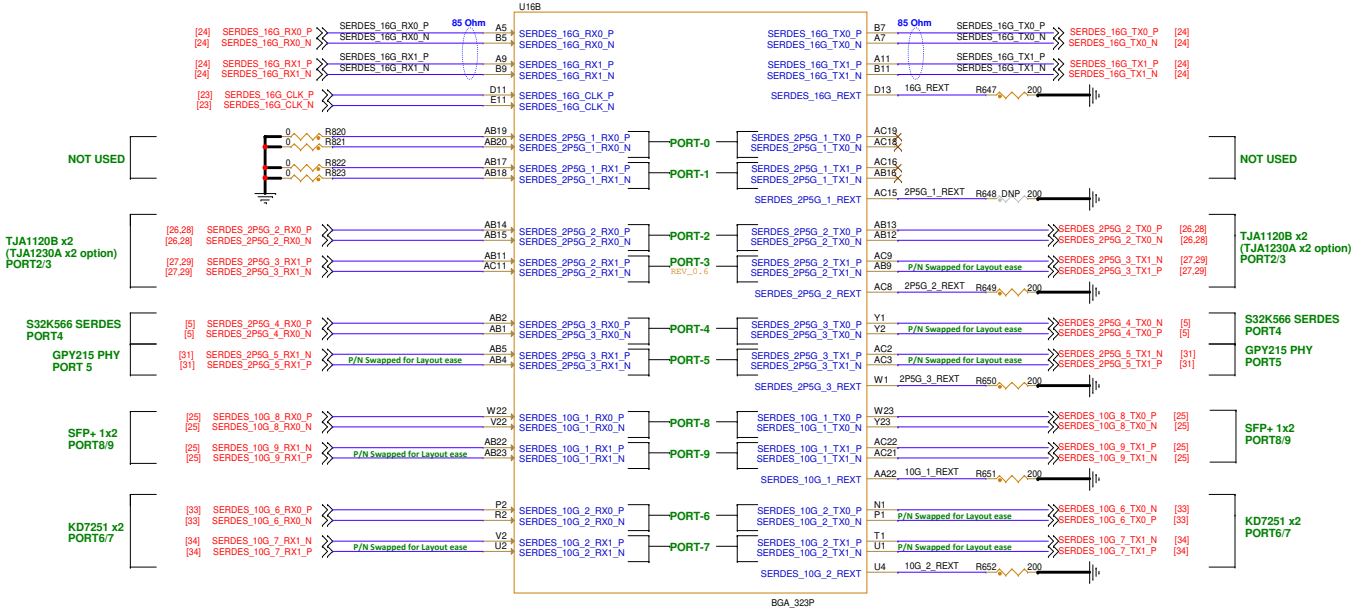
TJA1410B PORT0

TJA1410B PORT1

REV_0.6

Classification:	<Classification>
Drawing Title:	X-RING-OPTIC-ETH
Page Title:	20 - S32J100 IO
Size	C
Document Number	SCH-97122 / PDF: SPF-97122
Rev	A
Date:	Tuesday, August 25, 2026
Sheet	20 of 38

S32J100 SERDES INTERFACE



Classification: <Classification>
Drawing Title: X-RING-OPTIC-ETH
Page Title: 21 - S32J100 SERDES
Size C | Document Number SCH-97122 / PDF: SPF-97122 | Rev A
Date: Tuesday, August 25, 2026 | Sheet 21 of 38

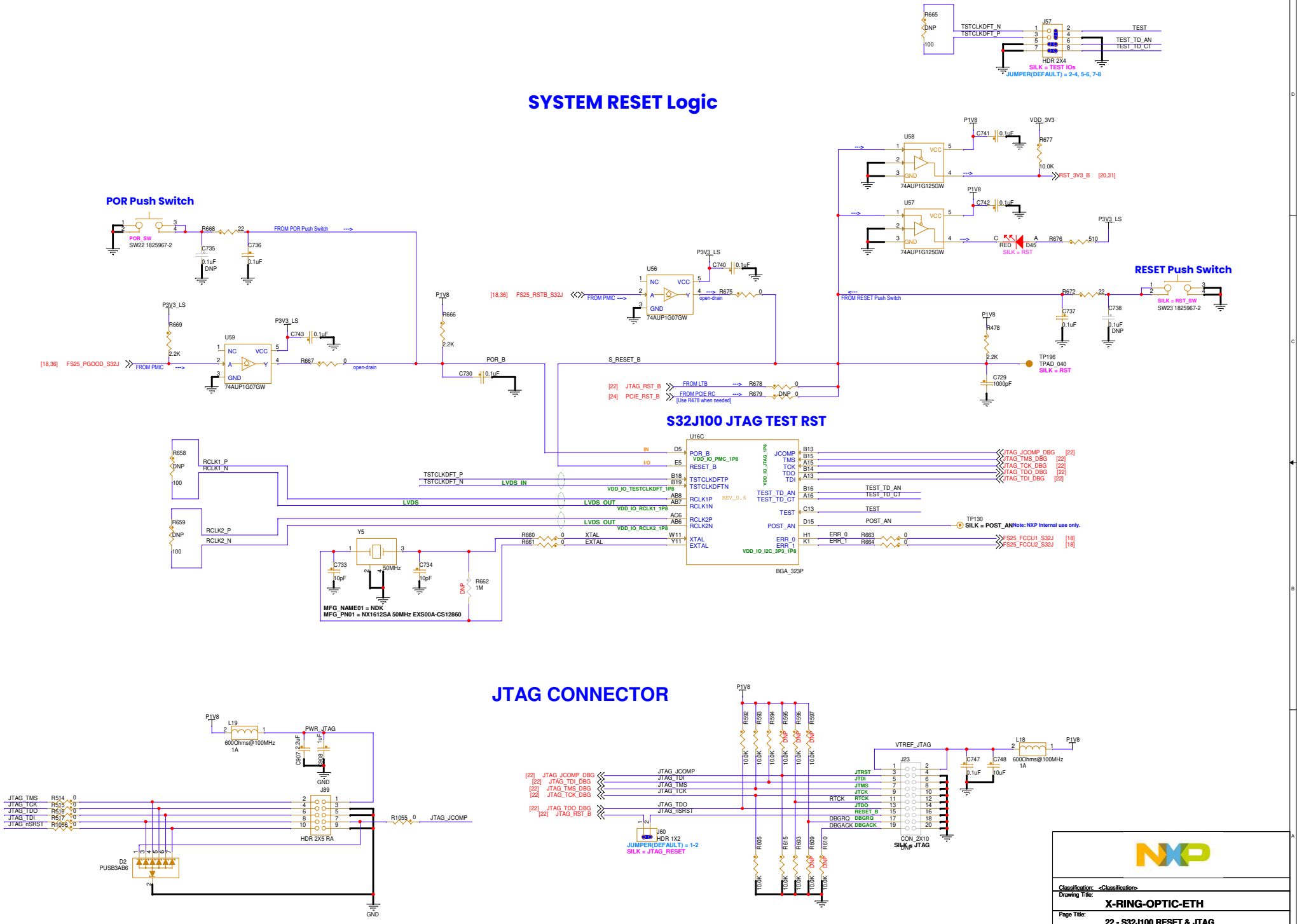
SYSTEM RESET Logic

POR Push Switch

RESET Push Switch

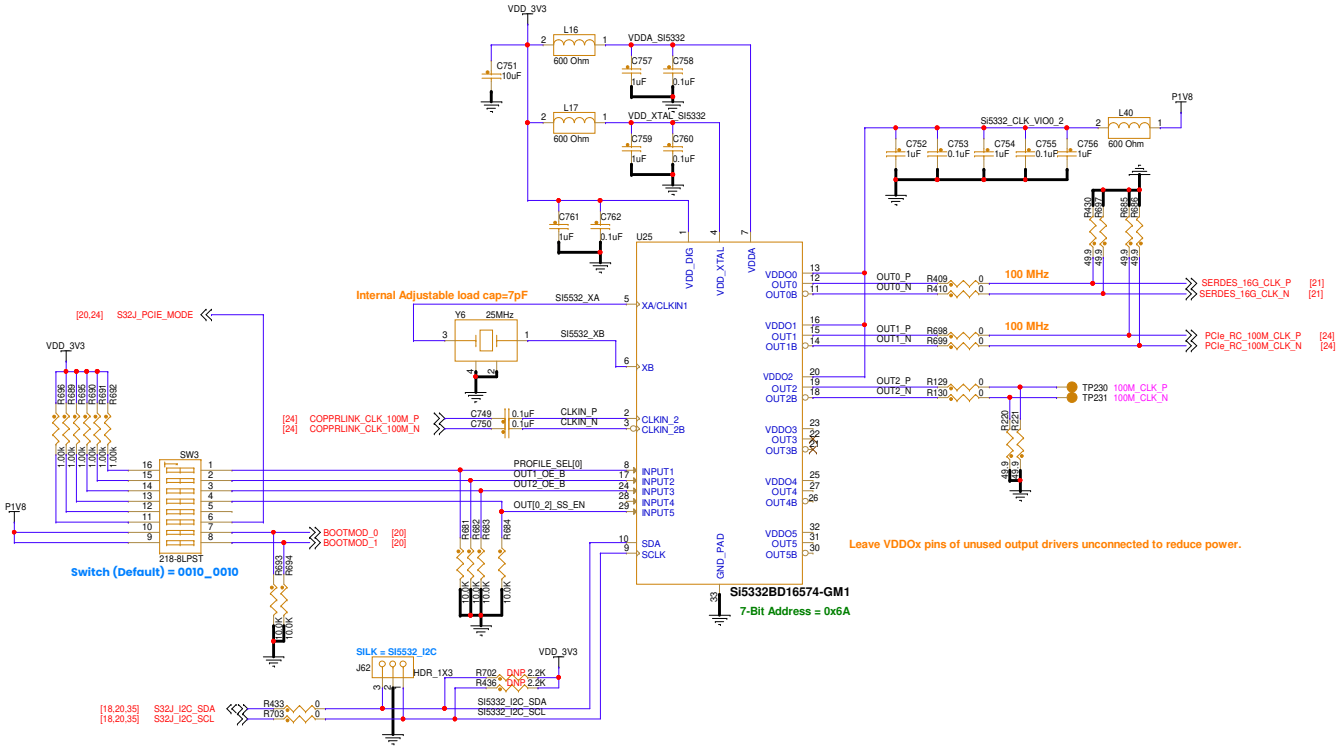
S32J100 JTAG TEST RST

JTAG CONNECTOR



Classification: <Classification>			
Drawing Title: X-RING-OPTIC-ETH			
Page Title: 22 - S32J100 RESET & JTAG			
Size C	Document Number SCH-97122 / PDF: SPF-97122		Rev A
Date: Tuesday, August 25, 2026	Sheet 22 of 38		

SYSTEM CLOCK GENERATOR
[OTP PROGRAMMED PART]



Design
======
Base I2C Address: 0x6A

Profile #1 - Enabled (Included in NVM),
PROFILE_SEL[0]: INPUT1 (P8)--- Low

Universal Hardware Input Pins:
- INPUT1 (P8) : PROFILE_SEL[0]
- INPUT2 (P17): Output Enable (Input: active LOW)
- INPUT3 (P24): Output Enable (Input: active LOW)
- INPUT4 (P28): None
- INPUT5 (P29): Spread Spectrum (Input: active HIGH)

Inputs:
- XAXB: 25 MHz
  - Crystal Mode
  - Adjusted Capacitance: 7 pF
- CLKIN2: Unused

Outputs:
- OUT0: 100 MHz HCSL 1.8 V 50 Ohms
- OUT1: 100 MHz HCSL 1.8 V 50 Ohms --OEB INPUT2 (P17)
- OUT2: 100 MHz HCSL 1.8 V 50 Ohms --OEB INPUT3 (P24)
- OUT[3:5]: Unused

Spread Spectrum:
- N0: OUT[0:2]
  - Direction - Center Spread
  - Amplitude - 0.5%

Profile #2 - Enabled (Included in NVM),
PROFILE_SEL[0]: INPUT1 (P8)--- High

Universal Hardware Input Pins:
- INPUT1 (P8) : PROFILE_SEL[0]
- INPUT2 (P17): None
- INPUT3 (P24): Output Enable (Input: active LOW)
- INPUT4 (P28): None
- INPUT5 (P29): None

Inputs:
- XAXB: Unused
- CLKIN2: 100 MHz Differential

Outputs:
- OUT0: 100 MHz (Buffer)Format: HCSL 1.8 V 50 Ohms - External
- OUT1: Unused
- OUT2: 100 MHz (Buffer)Format: HCSL 1.8 V 50 Ohms External-OEB Pin: INPUT3 (P24)
- OUT[3:5]: Unused

DIP Switch Setting:

##########
SW1: SI5532 Clock Mode
>>0= Clock From on board crystal
1= Clock From external MCIO connector

##########
SW2: SI5532 Output Clock "1" Enable/Disable
>>0= Enable [Applicable in RC mode only]
1= Disable [Applicable in RC mode only]

##########
SW3: SI5532 Output Clock "2" Enable/Disable
0= Enable
>>1= Disable

##########
SW4: PCIe Spread Spectrum Control
>>0= Disable
1= Enable
Direction - Center Spread
Amplitude - 0.5%

##########
SW5: S32J_PORT0_PWR_SEL
>> 0= 3.3V
1= 1.8V

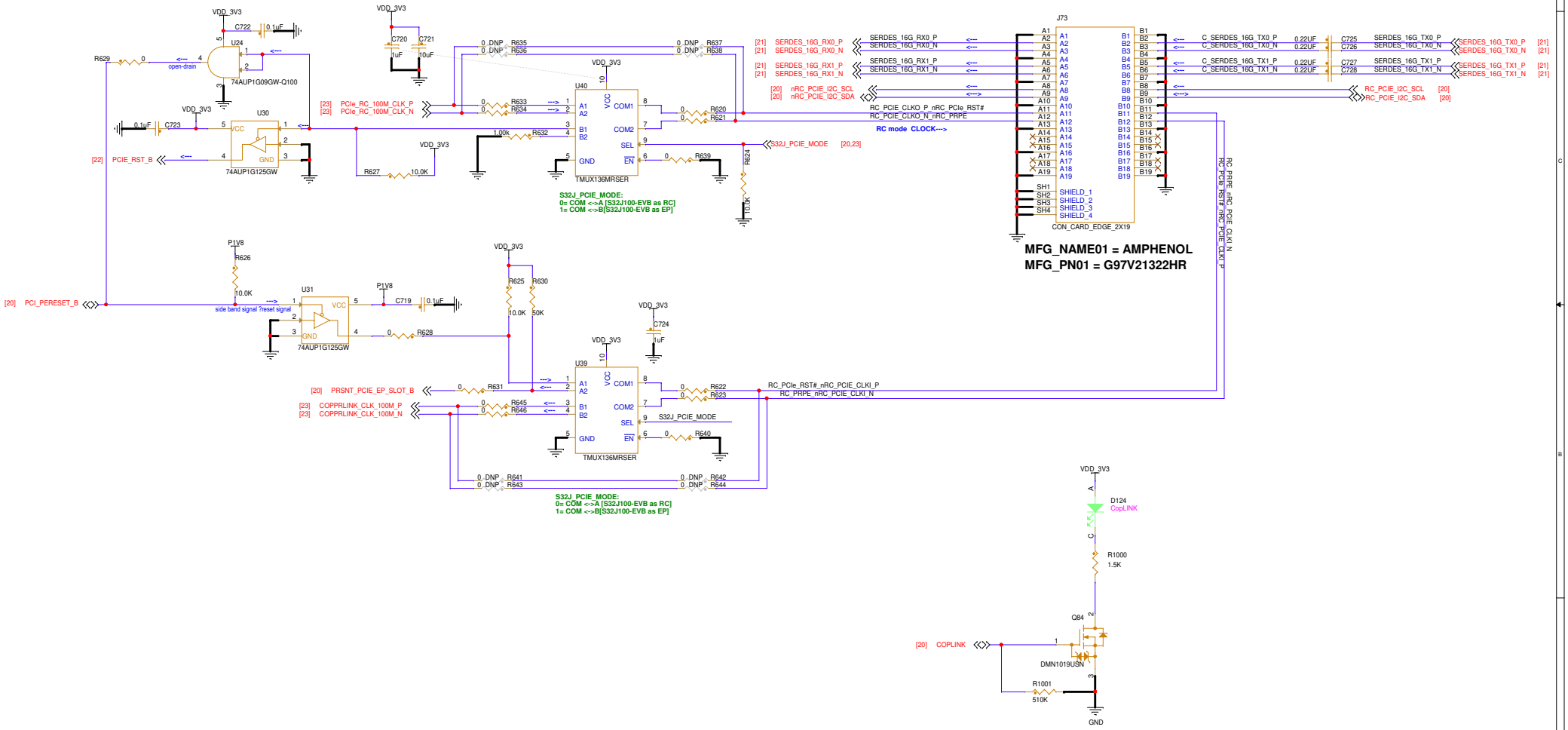
##########
SW6: S32J_PCIE_MODE
>>0= S32J100-EVB as RC
1= S32J100-EVB as EP

##########
SW[7:8]: BOOT_MODE_SEL

FUSE_SEL	BOOTMOD_0	BOOTMOD_1	Mode
0	0	1	Serial boot+XOSC in Crystal mode
0	1	0	Boot from RCON [Default]
1	0	0/1	Boot from fuses
1	1	0	Serial boot

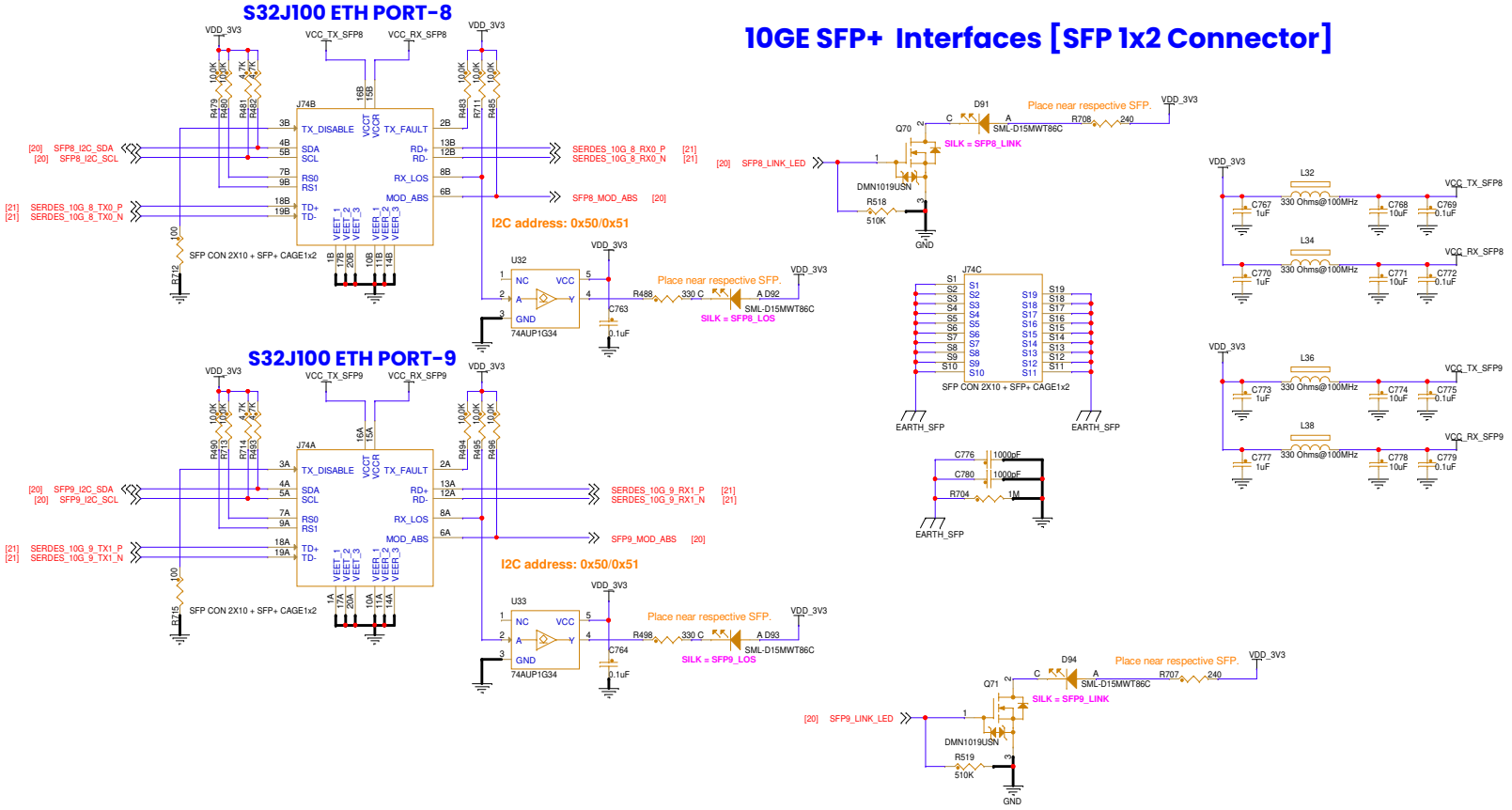
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Page Title: 23 - PCIE CLOCK
Size C | Document Number SCH-97122 / PDF: SPF-97122 | Rev A
Date: Tuesday, August 25, 2026 | Sheet 23 of 38

CopprLink MCIO Connector Interface [RC and nRC Mode]



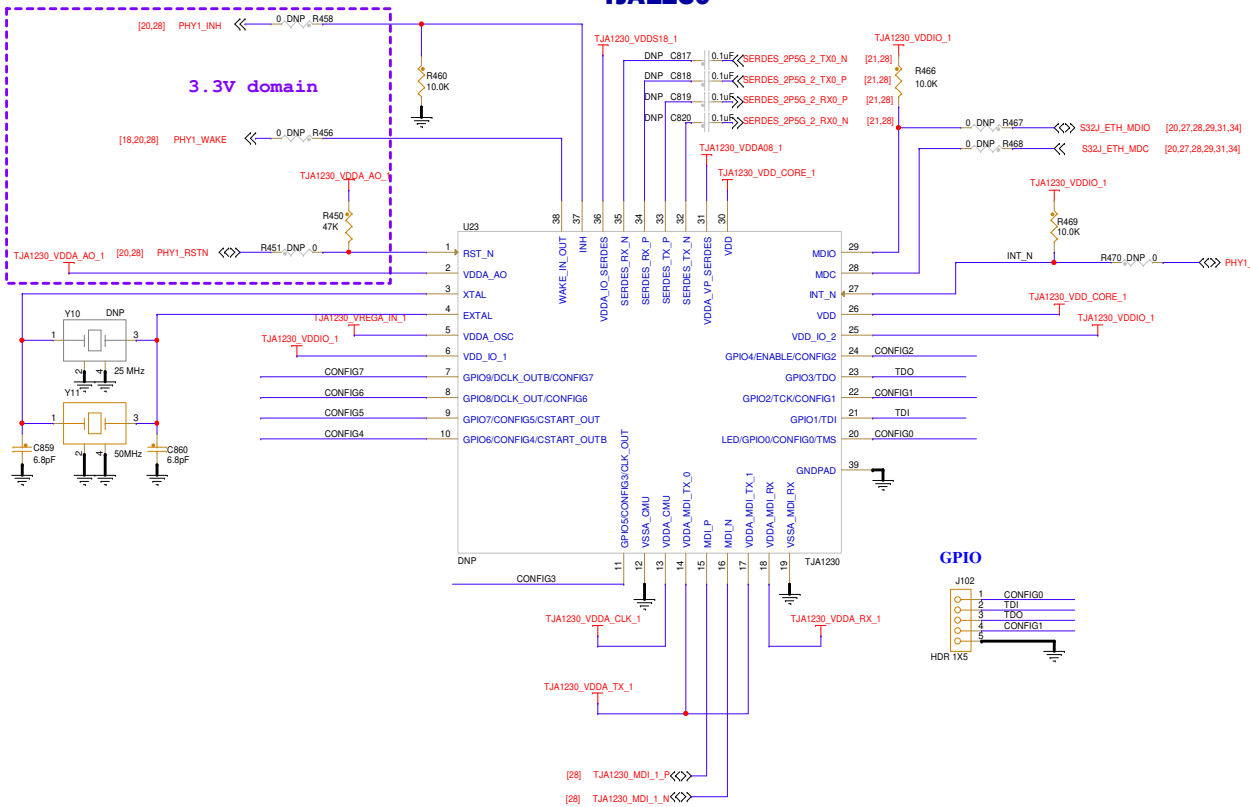
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Page Title:	24 - PCIE		
Size C	Document Number SCH-97122 / PDF: SPF-97122		Rev A
Date: Tuesday, August 25, 2026		Sheet 24 of 38	

10GE SFP+ Interfaces [SFP 1x2 Connector]

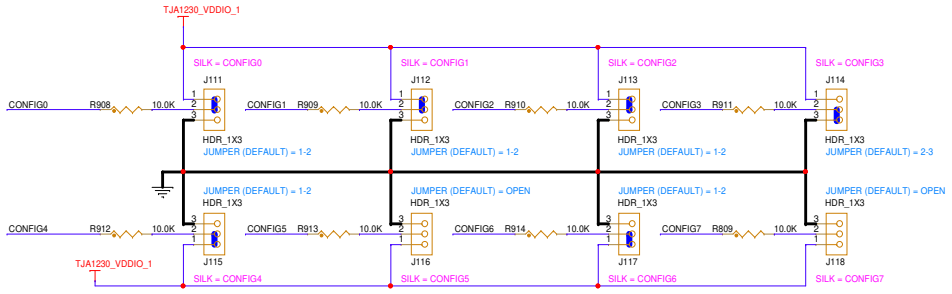


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Drawing Title: X-RING-OPTIC-ETH			
Page Title: 25 - 10GE SFP			
Size C	Document Number SCH-97122 / PDF: SPF-97122		Rev A
Date: Tuesday, August 25, 2026		Sheet 25 of 38	

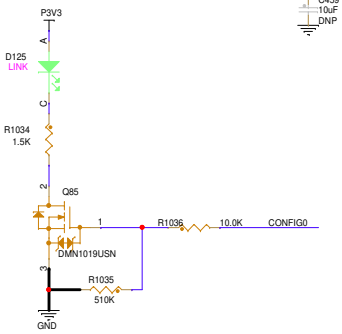
TJA1230



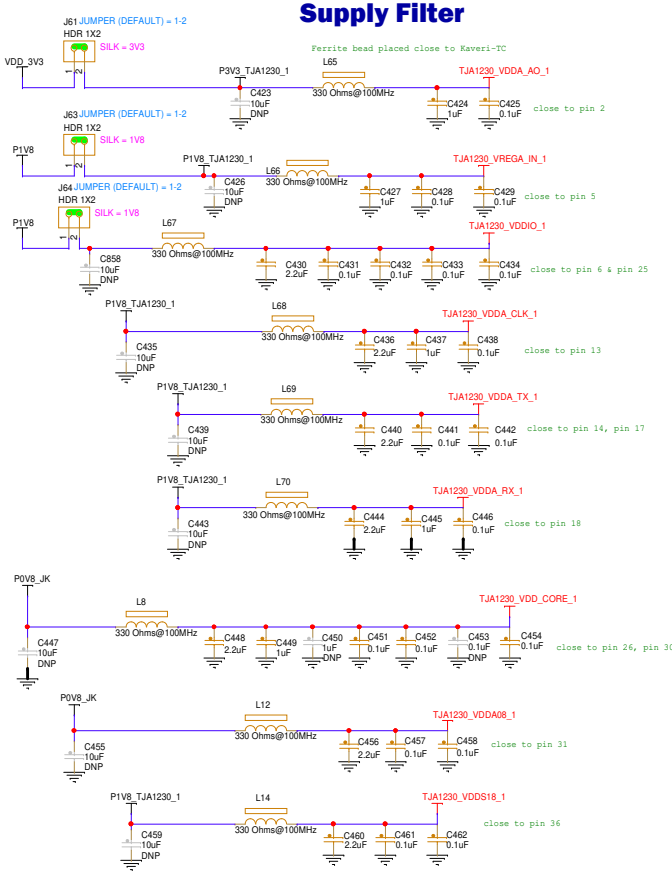
GPIO



CONFIG [0:2]:HHH - PHYAD "0x01"
CONFIG [3:4]:LH - preferred FOLLOWER
CONFIG [5] :O - OPEN (SGMII PHY)
CONFIG [6] :H - XO mode
CONFIG [7] :O - DISABLE CLKOUT&LED

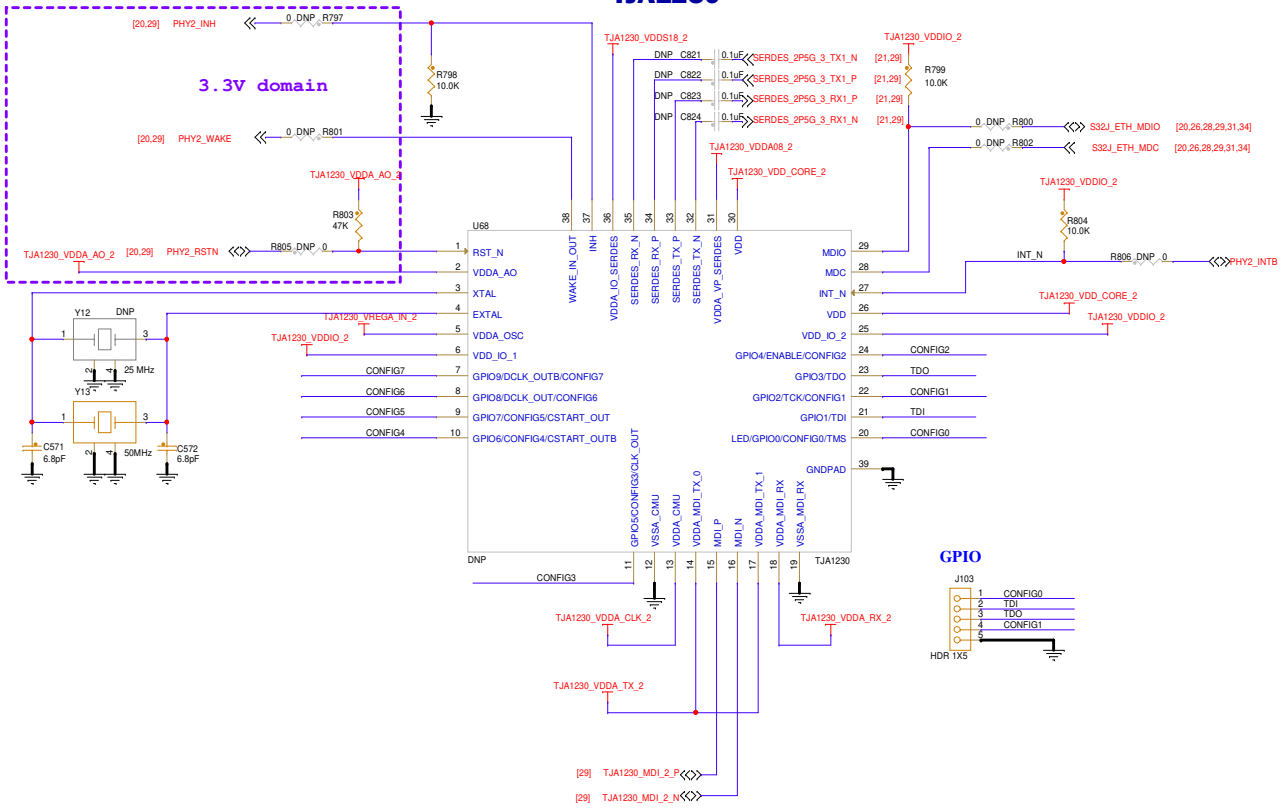


Supply Filter



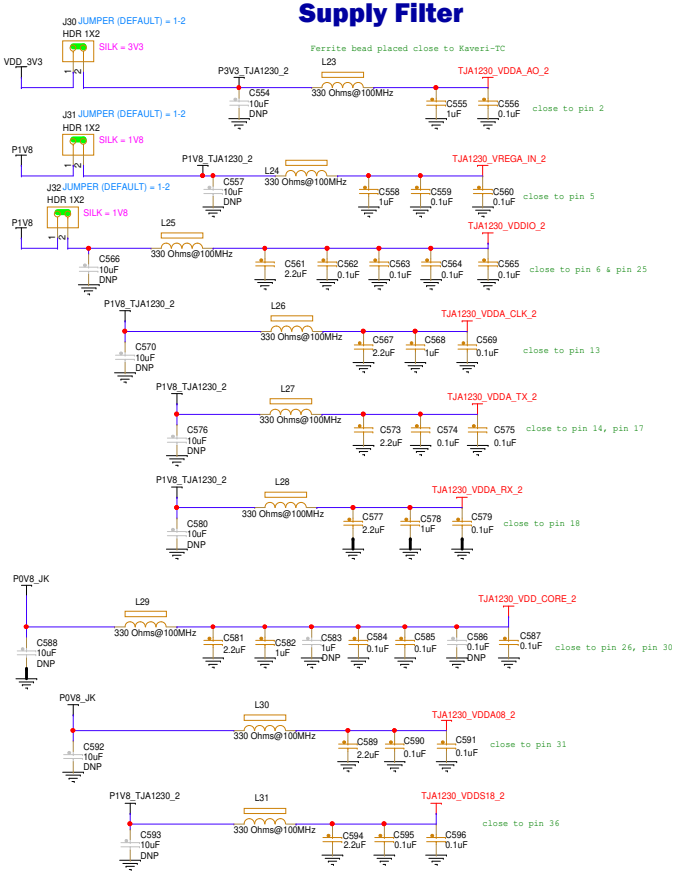
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Drawing Title:	X-RING-OPTIC-ETH		
Page Title:	26 - ENET TJA1230A		
Size: C	Document Number: SCH-97122 / PDF: SPF-97122		Rev: A
Date: Tuesday, August 25, 2026		Sheet 26 of 38	

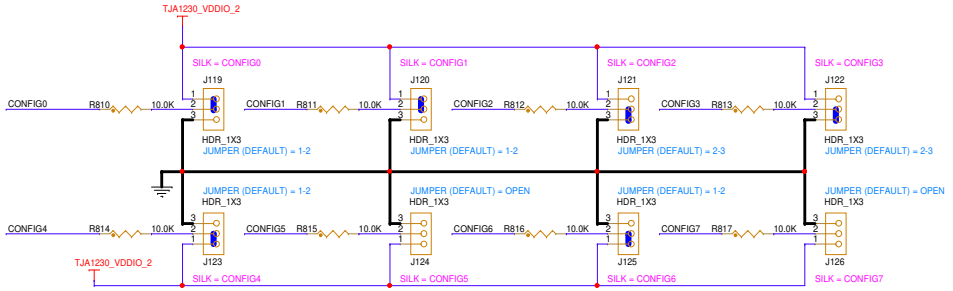
TJA1230

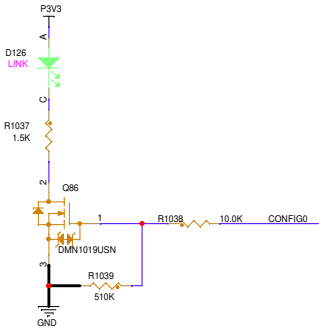


GPIO

Supply Filter







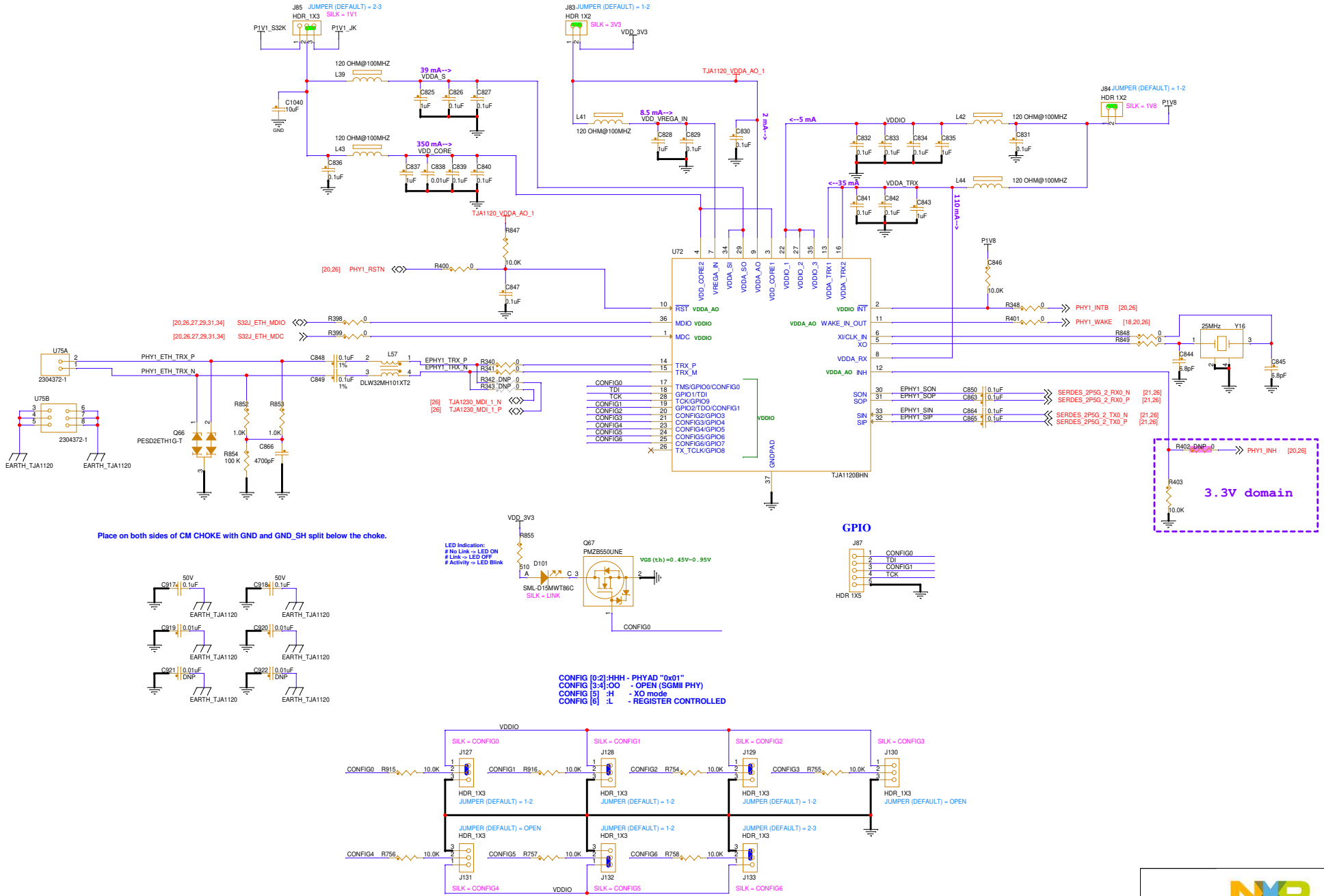
CONFIG [0:2]:HHL - PHYAD "0x02"
CONFIG [3:4]:LH - preferred FOLLOWER
CONFIG [5] :O - OPEN (SGMII PHY)
CONFIG [6] :H - XO mode
CONFIG [7] :O - DISABLE CLKOUT&LED

Classification: <Classification>
Drawing Title: **X-RING-OPTIC-ETH**
Page Title: **27 - ENET TJA1230A**

Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

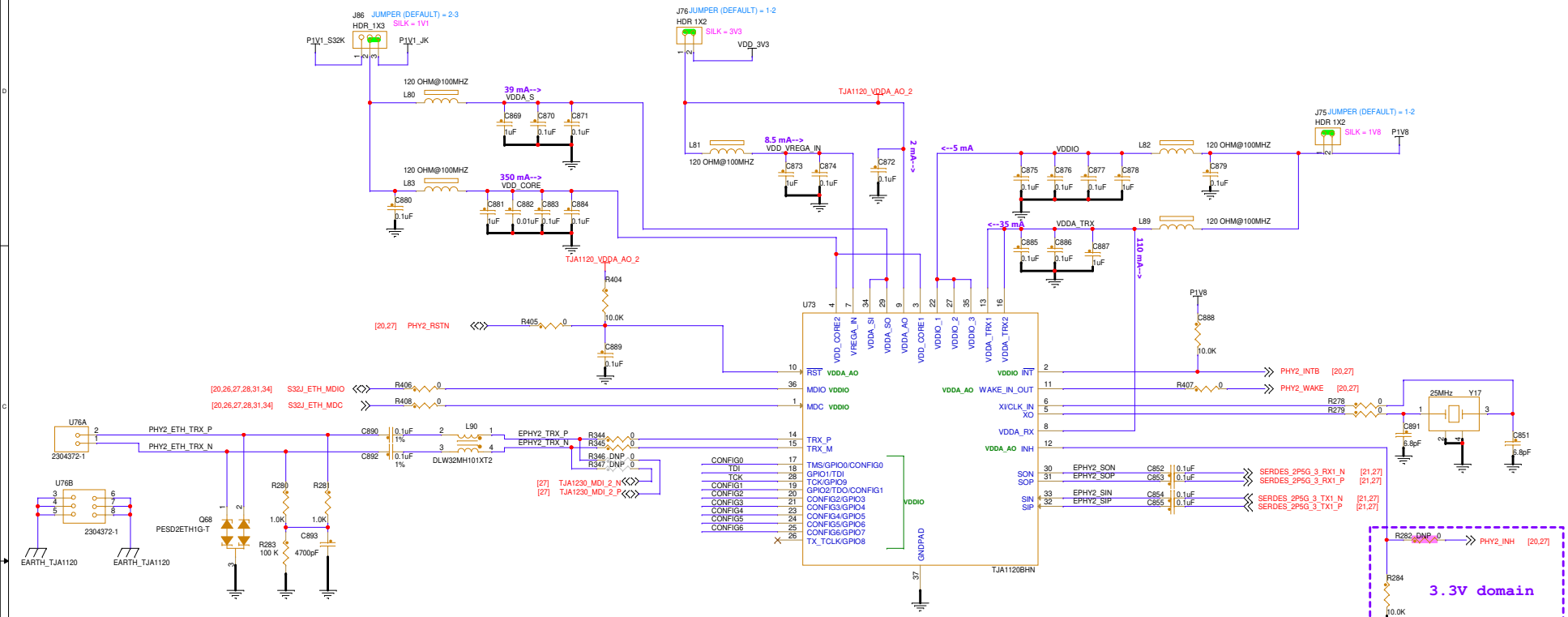
Date: Tuesday, August 25, 2026 | Sheet 27 of 38

TJA1120B

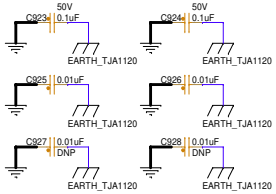


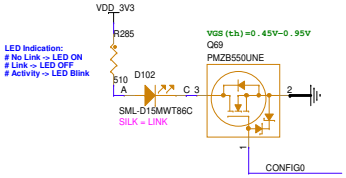
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Drawing Title: X-RING-OPTIC-ETH
Page Title: 28 - ENET TJA1120B
Size C | Document Number SCH-97122 / PDF: SPF-97122 | Rev A
Date: Tuesday, August 25, 2026 | Sheet 28 of 38

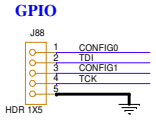
TJA1120B



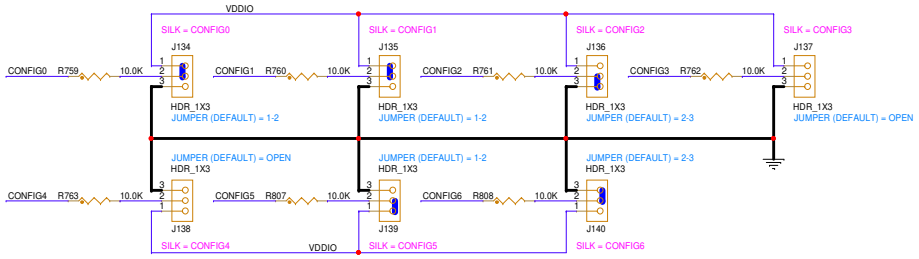
Place on both sides of CM CHOKE with GND and GND_SH split below the choke.





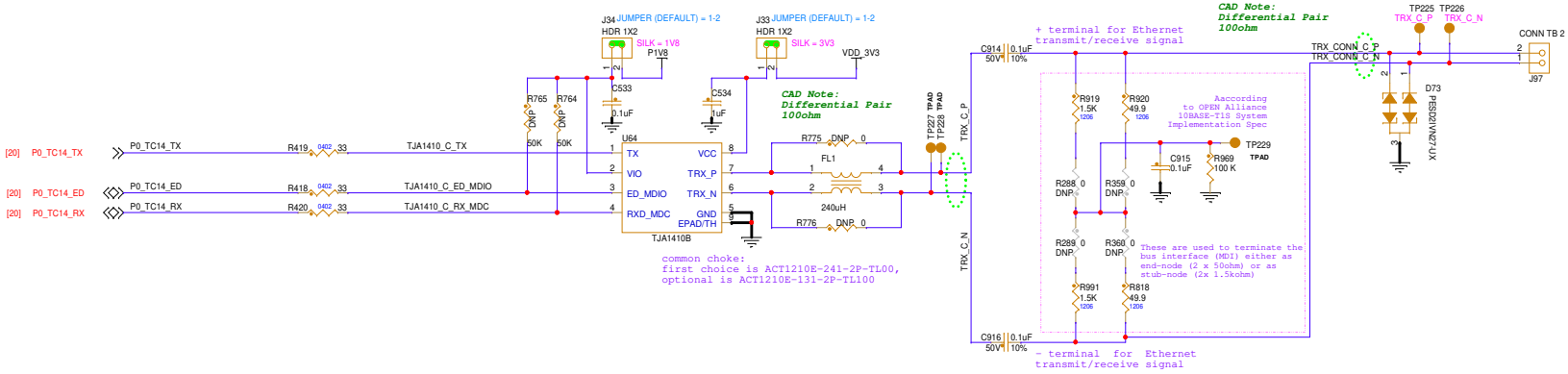


CONFIG [0:2]:HHL - PHYAD "0x02"
CONFIG [3:4]:OO - OPEN (SGMII PHY)
CONFIG [5] :H - XO mode
CONFIG [6] :L - REGISTER CONTROLLED

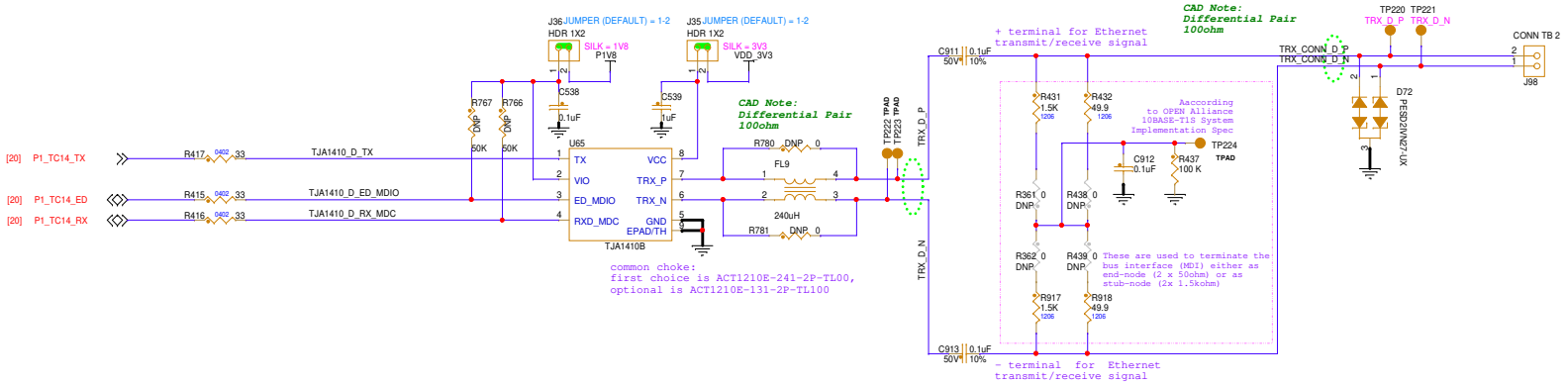


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Page Title: 29 - ENET TJA1120B		
Size C	Document Number SCH-97122 / PDF: SPF-97122	Rev A
Date: Tuesday, August 25, 2026	Sheet 29 of 38	

TJA1410 BluJay: 10BASE-T1S Ethernet PMD transceiver 1 [1.8V Fixed]

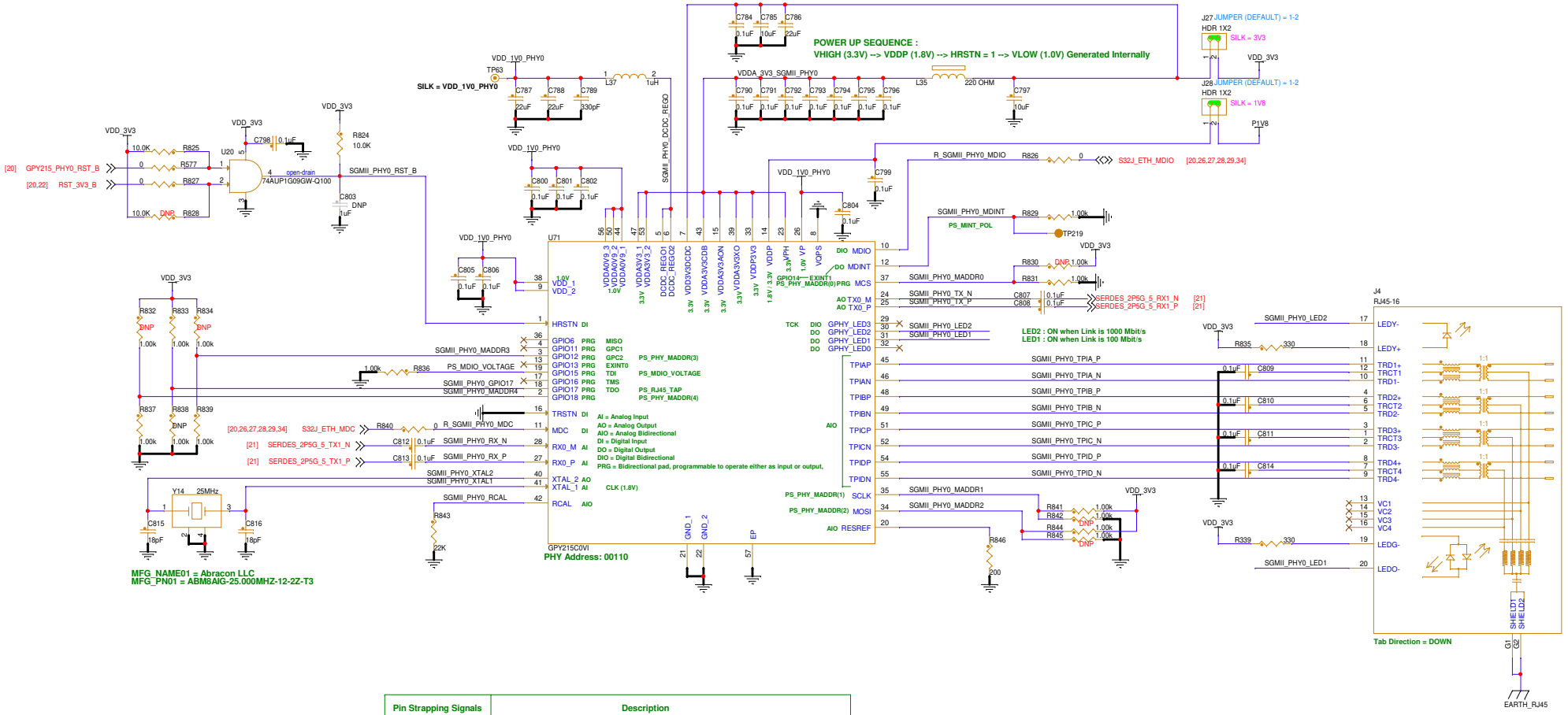


TJA1410 BluJay: 10BASE-T1S Ethernet PMD transceiver 2 [1.8V Fixed]



Classification: <Classification>
Drawing Title: X-RING-OPTIC-ETH
Page Title: 30 - ENET TJA1410B
Size C | Document Number SCH-97122 / PDF: SPF-97122 | Rev A
Date: Tuesday, August 25, 2026 | Sheet 30 of 38

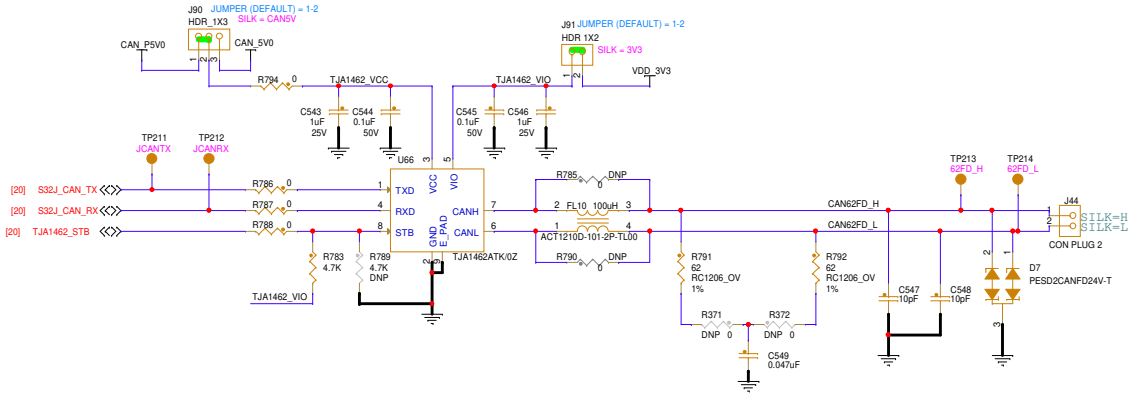
GPY215: 2.5GBASE-T SGMII PHY



Pin Strapping Signals	Description
PS_PHY_MADDR(4:0)	MDIO PHY Address: 00110
PS_MINT_POL	MDIO Interrupt (MDINT) Polarity >> 0 HIGH MDIO Interrupt (MDINT) is active high and configured in push-pull
PS_MDIO_VOLTAGE	MDIO Voltage This is to specify whether the maximum voltage level used by the MDIO signals is 3.3 V or 1.8 V (pin 10 to pin 13). >>0 LOW MDIO signals pads (pin 10 to pin 13) are supplied with 1.8 V. In this configuration the pin14 (VDDP) must be supplied with 1.8 V. 1 NORMAL MDIO signals pads (pin 10 to pin 13) are supplied with 3.3 V. In this configuration pin 14 (VDDP) must be supplied with 3.3 V.
PS_RJ45_TAP	RJ45 Pin Reversal >> 1 DOWN Tap down 0 UP Tap up

Classification: <Classification>		
Drawing Title: X-RING-OPTIC-ETH		
Page Title: 31 - ENET PHY RJ45		
Size C	Document Number SCH-97122 / PDF: SPF-97122	Rev A
Date: Tuesday, August 25, 2026	Sheet 31 of 38	

TJA1462ATK or TJA1464ATK - CANFD



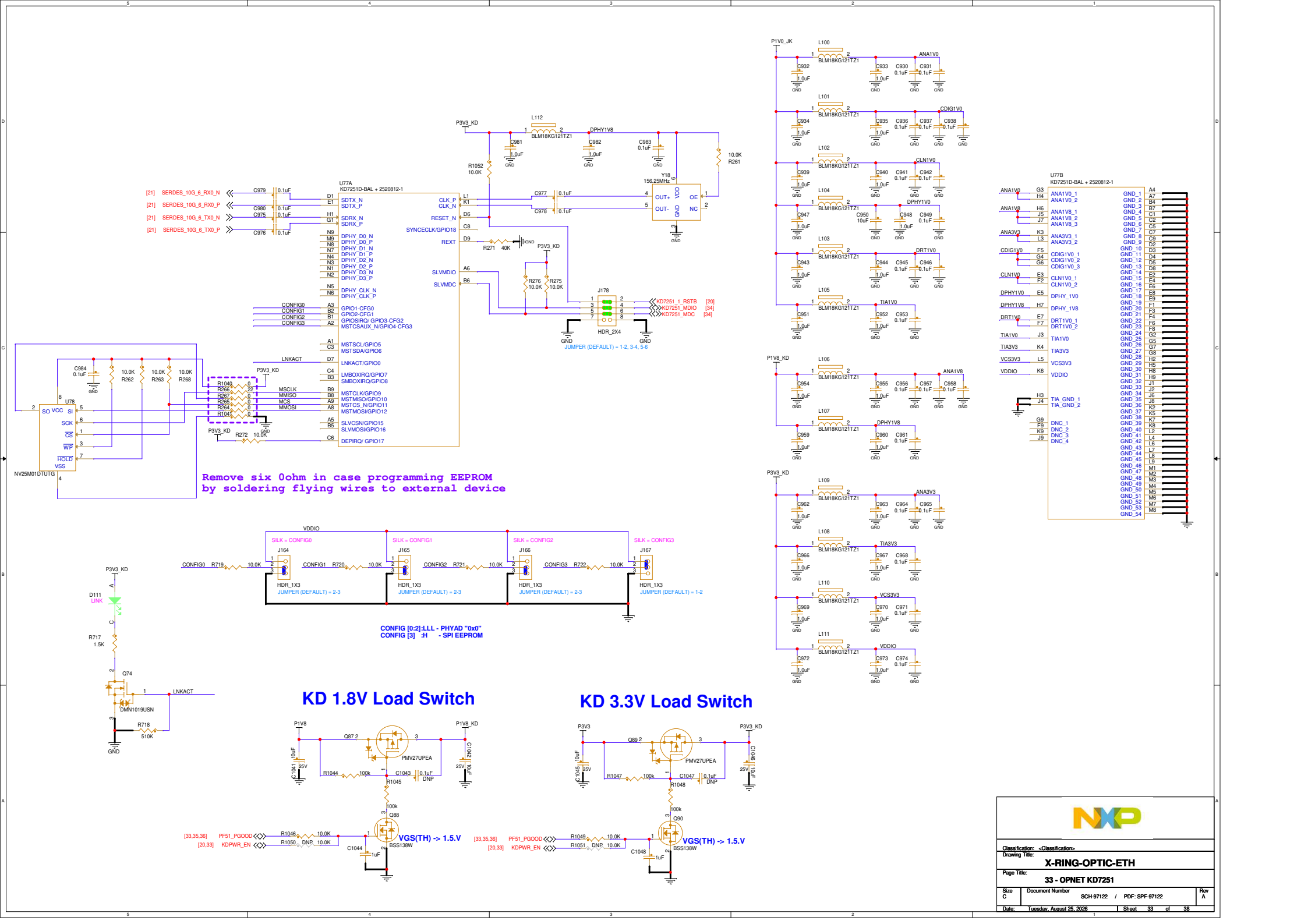
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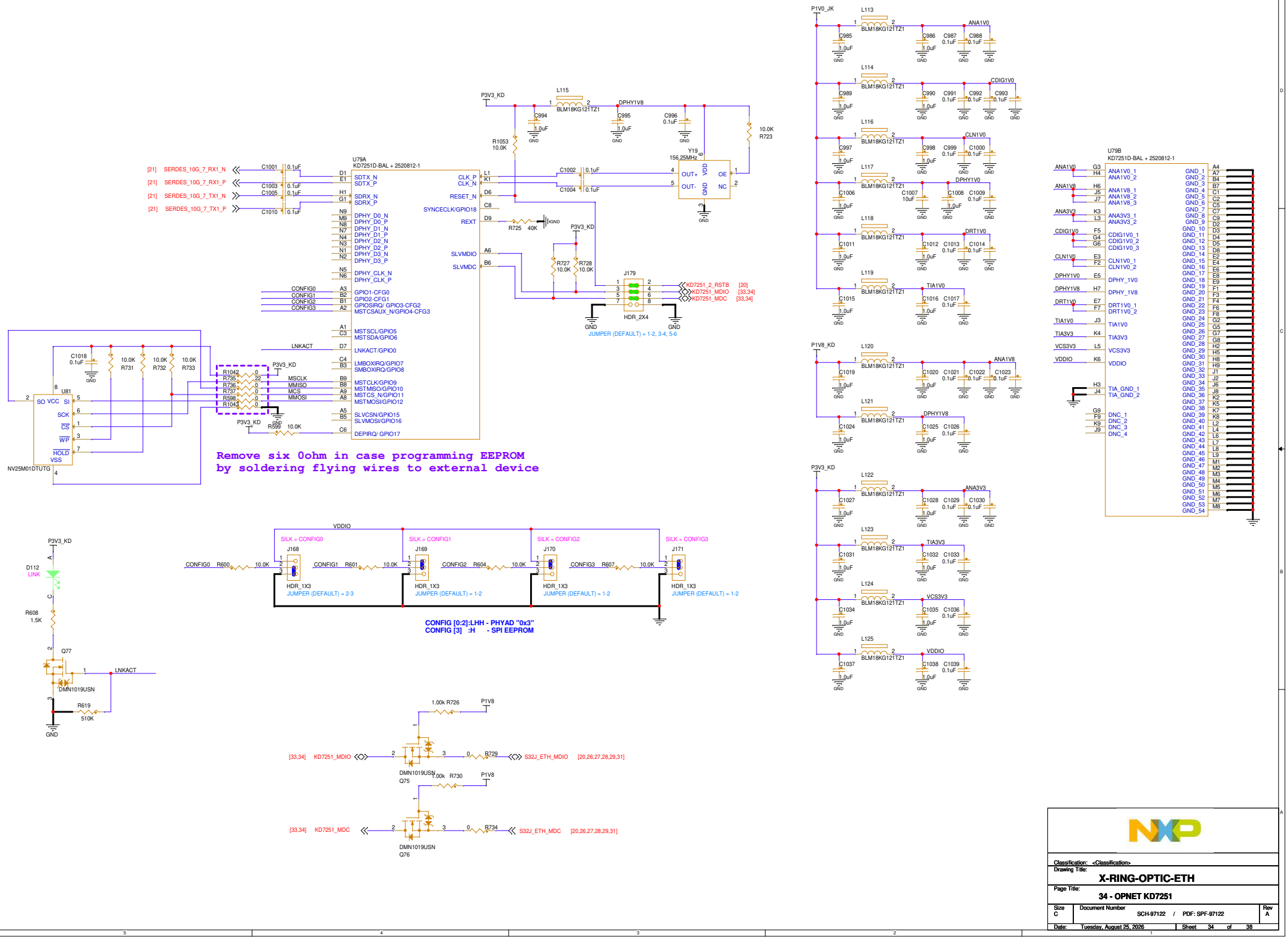
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Page Title: 32 - CAN TJA1462A

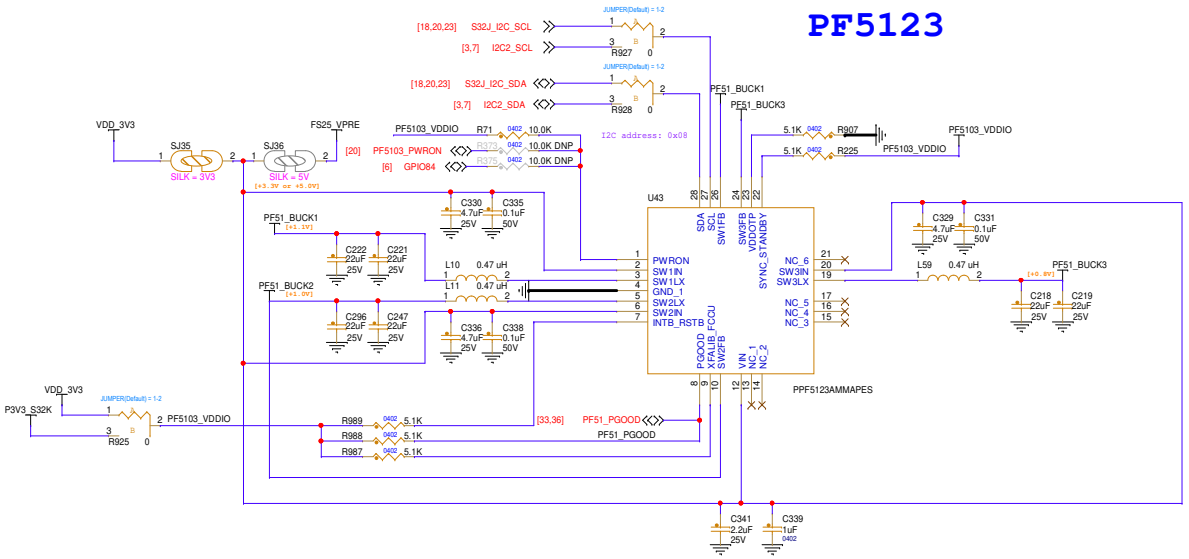
Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

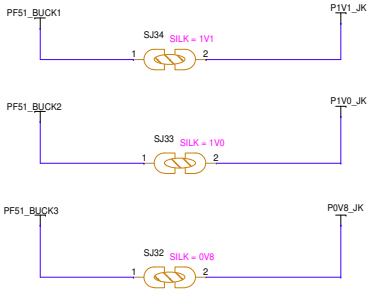
Date: Tuesday, August 25, 2026 Sheet 32 of 38


U77A
KD7251D-BAL + 2520812-1
SDTX_N
SDTX_P
SDRX_N
SDRX_P
DPHY_D0_N
DPHY_D0_P
DPHY_D1_N
DPHY_D1_P
DPHY_D2_N
DPHY_D2_P
DPHY_D3_N
DPHY_D3_P
DPHY_CLK_N
DPHY_CLK_P
CLK_P
CLK_N
RESET_N
SYNCECLK/GPIO18
REXT
SLVMDIO
SLVMDC
GPIO1-CFG0
GPIO2-CFG1
GPIOSIRQ/ GPIO3-CFG2
MSTCSAUX_N/GPIO4-CFG3
MSTSCL/GPIO5
MSTSDA/GPIO6
LNKACT/GPIO0
LMBOXIRQ/GPIO7
SMBOXIRQ/GPIO8
MSTCLK/GPIO9
MSTMISO/GPIO10
MSTCS_N/GPIO11
MSTMOSI/GPIO12
SLVCSN/GPIO15
SLVMOSI/GPIO16
DEPIRQ/ GPIO17
P3V3_KD
L112
BLM18KG121TZ1
DPHY1V8
Y18
156.25MHz
OUT+
OUT-
VDD
GND
OE
NC
10.0K
R261
R1052
10.0K
R271
40K
R276
10.0K
R275
10.0K
J178
HDR_2X4
KD7251_1_RSTB [20]
KD7251_MDIO [34]
KD7251_MDC [34]
JUMPER (DEFAULT) = 1-2, 3-4, 5-6
U78
NV25M01DTUTG
LNKACT
MSCLK
MMISO
MCS
MMOSI
R1040
R1041
R267
R265
R264
R1042
R272
10.0K
Remove six 0ohm in case programming EEPROM
by soldering flying wires to external device
VDDIO
SILK = CONFIG0
SILK = CONFIG1
SILK = CONFIG2
SILK = CONFIG3
J164
J165
J166
J167
HDR_1X3
JUMPER (DEFAULT) = 2-3
JUMPER (DEFAULT) = 1-2
CONFIG0 R719 10.0K
CONFIG1 R720 10.0K
CONFIG2 R721 10.0K
CONFIG3 R722 10.0K
CONFIG [0:2]:LLL - PHYAD "0x0"
CONFIG [3] :H - SPI EEPROM
D111
LINK
R717
1.5K
Q74
DMN1019USN
R718
510K
KD 1.8V Load Switch
P1V8
P1V8_KD
PMV27UPEA
BSS138W
R1044
100k
R1045
R1046
R1050
DNP
C1043
0.1uF
C1044
1uF
Q87
Q88
VGS(TH) -> 1.5V
[33,35,36] PF51_PGOOD
[20,33] KDPWR_EN
KD 3.3V Load Switch
P3V3
Q89
Q90
R1047
R1048
R1049
R1051
C1045
C1046
C1047
C1048
P1V0_JK
P1V8_KD
P3V3_KD
L100
L101
L102
L103
L104
L105
L106
L107
L108
L109
L110
L111
ANA1V0
CDIG1V0
CLN1V0
DPHY1V0
DRT1V0
TIA1V0
ANA1V8
ANA3V3
TIA3V3
VCS3V3
VDDIO
U77B
KD7251D-BAL + 2520812-1
ANA1V0_1
ANA1V0_2
ANA1V8_1
ANA1V8_2
ANA1V8_3
ANA3V3_1
ANA3V3_2
CDIG1V0_1
CDIG1V0_2
CDIG1V0_3
CLN1V0_1
CLN1V0_2
DPHY_1V0
DPHY_1V8
DRT1V0_1
DRT1V0_2
TIA_GND_1
TIA_GND_2
DNC_1
DNC_2
DNC_3
DNC_4
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
GND_11
GND_12
GND_13
GND_14
GND_15
GND_16
GND_17
GND_18
GND_19
GND_20
GND_21
GND_22
GND_23
GND_24
GND_25
GND_26
GND_27
GND_28
GND_29
GND_30
GND_31
GND_32
GND_33
GND_34
GND_35
GND_36
GND_37
GND_38
GND_39
GND_40
GND_41
GND_42
GND_43
GND_44
GND_45
GND_46
GND_47
GND_48
GND_49
GND_50
GND_51
GND_52
GND_53
GND_54
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Drawing Title:
X-RING-OPTIC-ETH
Page Title:
33 - OPNET KD7251
Size C
Document Number SCH-97122 / PDF: SPF-97122
Rev A
Date: Tuesday, August 25, 2026
Sheet 33 of 38

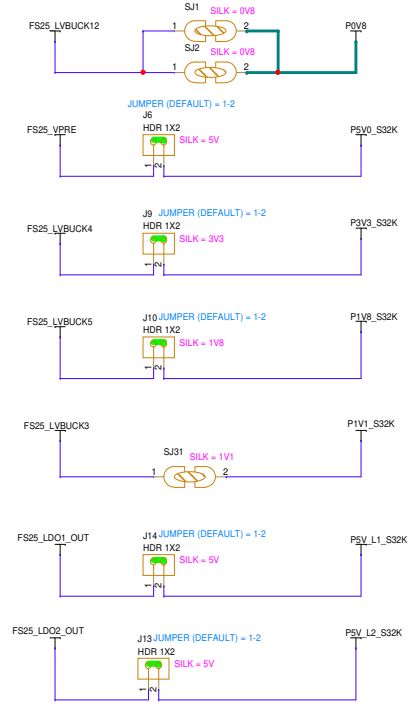

U79A
KD7251D-BAL + 2520812-1
SDTX_N
SDTX_P
SDRX_N
SDRX_P
DPHY_D0_N
DPHY_D0_P
DPHY_D1_N
DPHY_D1_P
DPHY_D2_N
DPHY_D2_P
DPHY_D3_N
DPHY_D3_P
DPHY_CLK_N
DPHY_CLK_P
GPIO1-CFG0
GPIO2-CFG1
GPIOSIRQ/ GPIO3-CFG2
MSTCSAUX_N/GPIO4-CFG3
MSTSCL/GPIO5
MSTSDA/GPIO6
LNKACT/GPIO0
LMBOXIRQ/GPIO7
SMBOXIRQ/GPIO8
MSTCLK/GPIO9
MSTMISO/GPIO10
MSTCS_N/GPIO11
MSTMOSI/GPIO12
SLVCSN/GPIO15
SLVMOSI/GPIO16
DEPIRQ/ GPIO17
CLK_P
CLK_N
RESET_N
SYNCECLK/GPIO18
REXT
SLVMDIO
SLVMDC
[21] SERDES_10G_7_RX1_N
[21] SERDES_10G_7_RX1_P
[21] SERDES_10G_7_TX1_N
[21] SERDES_10G_7_TX1_P
P3V3_KD
DPHY1V8
L115
BLM18KG121TZ1
Y19
156.25MHz
OUT+
OUT-
VDD
GND
OE
NC
R1053 10.0K
R723 10.0K
R725 40K
R727 10.0K
R728 10.0K
J179
HDR_2X4
KD7251_2_RSTB [20]
KD7251_MDIO [33,34]
KD7251_MDC [33,34]
JUMPER (DEFAULT) = 1-2, 3-4, 5-6
U81
NV25M01DTUTG
SO VCC SI
SCK
CS
WP
HOLD
VSS
R731 10.0K
R732 10.0K
R733 10.0K
R1042
R735
R736
R737
R598
R1043
MSCLK
MMISO
MCS
MMOSI
R599 10.0K
Remove six 0ohm in case programming EEPROM
by soldering flying wires to external device
VDDIO
J168
J169
J170
J171
HDR_1X3
SILK = CONFIG0
SILK = CONFIG1
SILK = CONFIG2
SILK = CONFIG3
CONFIG0 R600 10.0K
CONFIG1 R601 10.0K
CONFIG2 R604 10.0K
CONFIG3 R607 10.0K
JUMPER (DEFAULT) = 2-3
JUMPER (DEFAULT) = 1-2
JUMPER (DEFAULT) = 1-2
JUMPER (DEFAULT) = 1-2
CONFIG [0:2]:LHH - PHYAD "0x3"
CONFIG [3] :H - SPI EEPROM
D112
LINK
R608 1.5K
Q77
DMN1019USN
LNKACT
R619 510K
1.00k R726
P1V8
[33,34] KD7251_MDIO
R729
S32J_ETH_MDIO [20,26,27,28,29,31]
Q75
1.00k R730
[33,34] KD7251_MDC
R734
S32J_ETH_MDC [20,26,27,28,29,31]
Q76
P1V0_JK
P1V8_KD
P3V3_KD
L113
L114
L116
L117
L118
L119
L120
L121
L122
L123
L124
L125
BLM18KG121TZ1
ANA1V0
CDIG1V0
CLN1V0
DPHY1V0
DRT1V0
TIA1V0
ANA1V8
DPHY1V8
ANA3V3
TIA3V3
VCS3V3
VDDIO
U79B
KD7251D-BAL + 2520812-1
ANA1V0_1
ANA1V0_2
ANA1V8_1
ANA1V8_2
ANA1V8_3
ANA3V3_1
ANA3V3_2
CDIG1V0_1
CDIG1V0_2
CDIG1V0_3
CLN1V0_1
CLN1V0_2
DPHY_1V0
DPHY_1V8
DRT1V0_1
DRT1V0_2
TIA1V0
TIA3V3
VCS3V3
VDDIO
TIA_GND_1
TIA_GND_2
DNC_1
DNC_2
DNC_3
DNC_4
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
GND_11
GND_12
GND_13
GND_14
GND_15
GND_16
GND_17
GND_18
GND_19
GND_20
GND_21
GND_22
GND_23
GND_24
GND_25
GND_26
GND_27
GND_28
GND_29
GND_30
GND_31
GND_32
GND_33
GND_34
GND_35
GND_36
GND_37
GND_38
GND_39
GND_40
GND_41
GND_42
GND_43
GND_44
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GND_47
GND_48
GND_49
GND_50
GND_51
GND_52
GND_53
GND_54
Classification: <Classification>
Drawing Title:
X-RING-OPTIC-ETH
Page Title:
34 - OPNET KD7251
Size C
Document Number SCH-97122 / PDF: SPF-97122
Rev A
Date: Tuesday, August 25, 2026
Sheet 34 of 38

PF5123

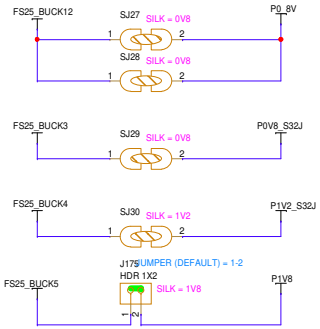


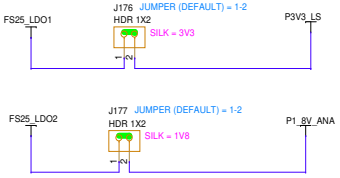


PWR SEL FOR S32K5

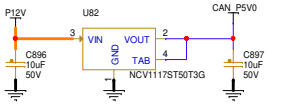


PWR SEL FOR S32J100



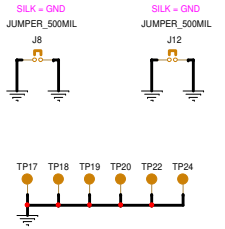


CAN 5V

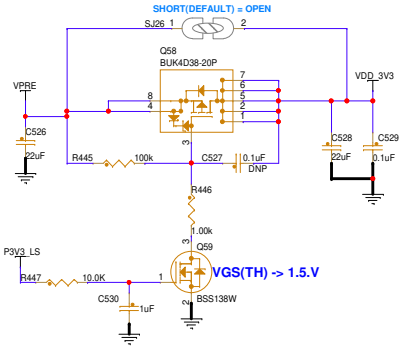


GROUND TEST POINTS

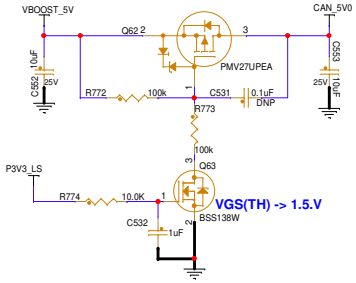
Layout Note:
Place close to BOARD EDGE



3.3V Load Switch

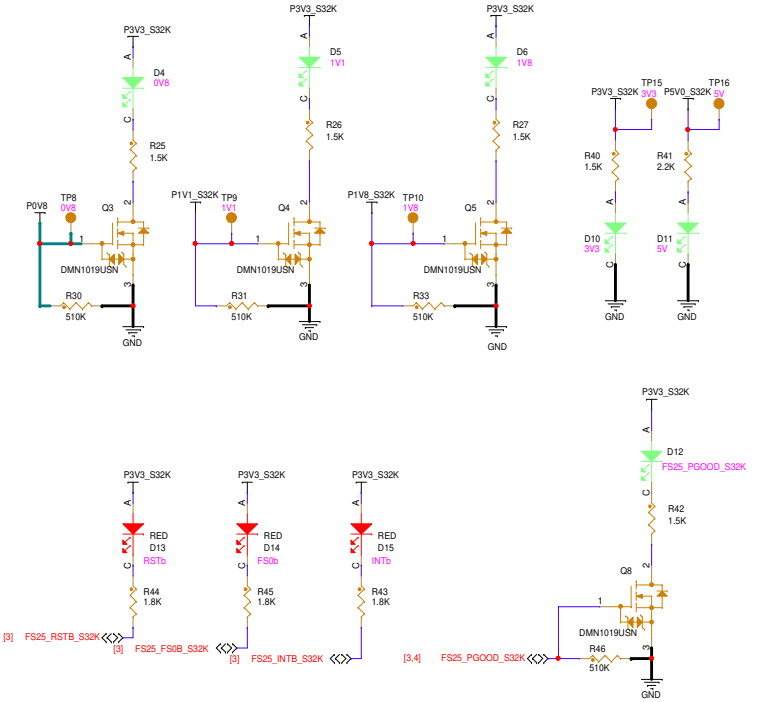


5V Load Switch

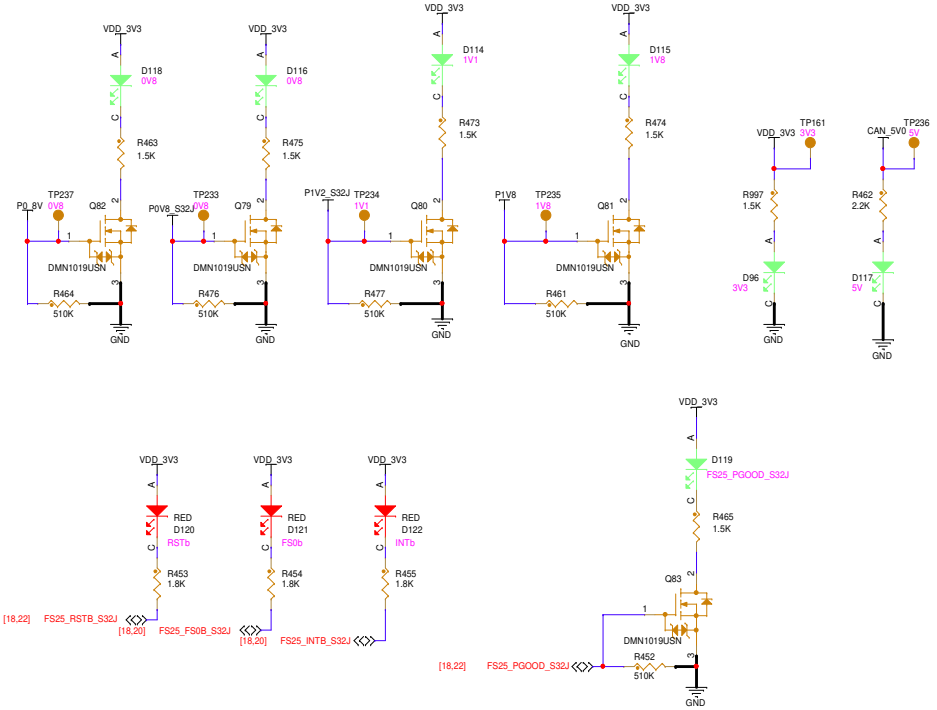


Classification: Company Internal Use		
Drawing Title: X-RING-OPTIC-ETH		
Page Title: 35 - PF5123 & PWR JUMPER		
Size C	Document Number SCH-97122 / PDF: SPF-97122	Rev A
Date: Tuesday, August 25, 2026	Sheet 35 of 38	

S32K FS25 LED

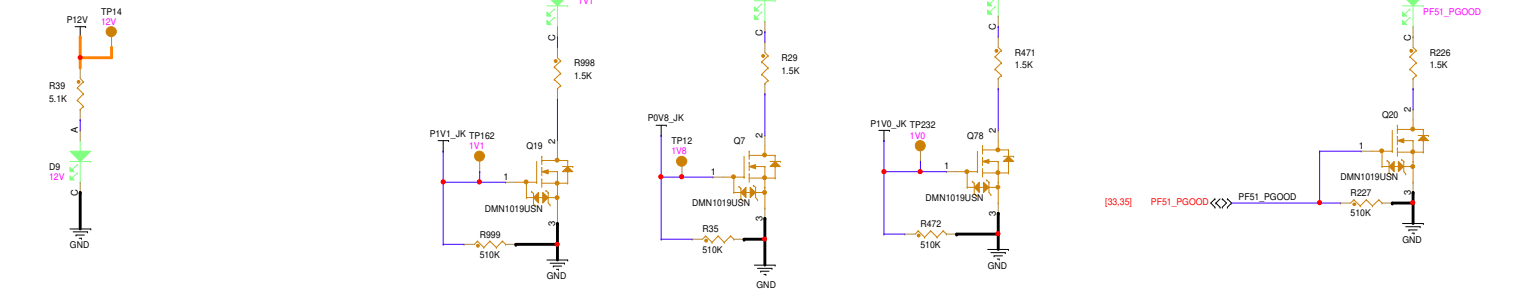


S32J FS25 LED

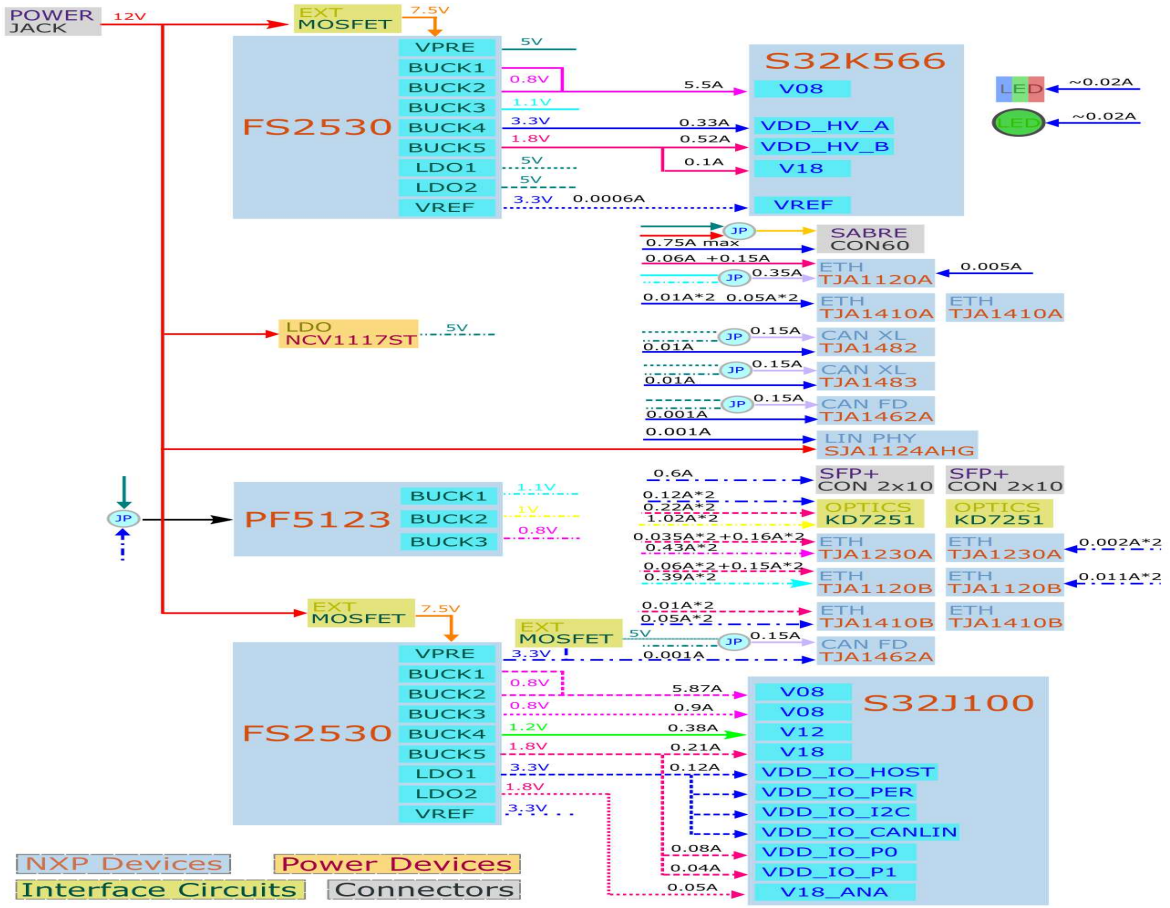


PF5103 LED

PWRIN LED



Classification: <Classification>			
Drawing Title: X-RING-OPTIC-ETH			
Page Title: 36 - POWER INDICATORS			
Size C	Document Number SCH-97122 / PDF: SPF-97122		Rev A
Date: Tuesday, August 25, 2026		Sheet 36 of 38	



Classification: <Classification>

Drawing Title: **X-RING-OPTIC-ETH**

Page Title: **37 - Power Tree**

Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 | Sheet 37 of 38

REF DES	JUMPER(DEFAULT)	PAGE NAME
J21,J40,R68,R69	1-2	03 - FS2530 FOR S32K566
J37	OPEN	03 - FS2530 FOR S32K566
R236,R242,R249	1-2	04 - S32K566 POWER
J161	2-3	04 - S32K566 POWER
J42	1-2	09 - ENET 60PIN CON
J26	2-3	09 - ENET 60PIN CON
J17	1-2	10 - ENET TJA1410A
J7,J18,J104,J105,J109	1-2	11 - ENET TJA1120A
J19,J106,J110	2-3	11 - ENET TJA1120A
J107,J108	OPEN	11 - ENET TJA1120A
J51,J69	1-2	12 - CAN TJA1462A
J43,J46,J47,J52,J53	1-2	13 - CAN TJA1482 & TJA1483
J49	2-3	13 - CAN TJA1482 & TJA1483
J77,J78,J79,J80,J81,J82,J157, J158,J159	1-2	14 - S32K5 IO IND/CTRL
J151	1-2	16 - OpenSDA FOR S32K5
J54,J55	1-2	17 - LIN SJA1124
J72	OPEN	18 - FS2530 FOR S32J100
J56	OPEN	19 - S32J100 POWER & GND
J60	1-2	22 - S32J100 RESET & JTAG
J57	2-4, 5-6, 7-8	22 - S32J100 RESET & JTAG
J61,J63,J64,J111,J112,J113, J115,J117	1-2	26 - ENET TJA1230A
J114	2-3	26 - ENET TJA1230A
J116,J118	OPEN	26 - ENET TJA1230A
J30,J31,J32,J119,J120,J123, J125	1-2	27 - ENET TJA1230A
J121,J122	2-3	27 - ENET TJA1230A
J124,J126	OPEN	27 - ENET TJA1230A
J83,J84,J127,J128,J129,J132	1-2	28 - ENET TJA1120B
J85,J133	2-3	28 - ENET TJA1120B
J130,J131	OPEN	28 - ENET TJA1120B
J75,J76,J134,J135,J139	1-2	29 - ENET TJA1120B
J86,J136,J140	2-3	29 - ENET TJA1120B
J137,J138	OPEN	29 - ENET TJA1120B
J33,J34,J35,J36	1-2	30 - ENET TJA1410B
J27,J28	1-2	31 - ENET PHY RJ45
J90,J91	1-2	32 - CAN TJA1462A
J178	1-2, 3-4, 5-6	33 - OPNET KD7251
J167	1-2	33 - OPNET KD7251
J164,J165,J166	2-3	33 - OPNET KD7251
J179	1-2, 3-4, 5-6	34 - OPNET KD7251
J169,J170,J171	1-2	34 - OPNET KD7251
J168	2-3	34 - OPNET KD7251
J6,J9,J10,J13,J14,J175,J176, J177,R925,R927,R928	1-2	35 - PF5123 & PWR JUMPER

REF DES	SWITCH(DEFAULT)	PAGE NAME
SW1	OFF	03 - FS2530 FOR S32K566
SW25	0000_0000	20 - S32J100 IO
SW24	1100_0101	20 - S32J100 IO
SW3	0010_0010	23 - PCIE CLOCK

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ3	OPEN	03 - FS2530 FOR S32K566
SJ17	OPEN	18 - FS2530 FOR S32J100
SJ26	OPEN	35 - PF5123 & PWR JUMPER

REF DES	ASSY_OPT	PAGE NAME
C180,J38,L4,R441,R929,R930, R933,R972,R974	DNP	03 - FS2530 FOR S32K566
C27,C340,R224	DNP	05 - S32K566 B DOMAIN
C224	DNP	06 - S32K566 A DOMAIN
R982,R984	DNP	08 - S32K566 A DOMAIN
R203,R204,R205,R206,R207	DNP	09 - ENET 60PIN CON
R49,R51,R54,R56,R57,R58,R59, R60,R65,R66,R67,R75,R76,R286, R287,R504,R505,R506,R507, R508,R509,R657,R856,R857, R858,R863,R864,R865	DNP	10 - ENET TJA1410A
C317,C334,D78,D79,R70,R222, R872,R880	DNP	11 - ENET TJA1120A
R363,R364,R377,R381,R382	DNP	12 - CAN TJA1462A
R229,R230,R235,R240,R244, R290,R331,R332,R333,R334, R337,R352,R367,R368,R369, R370,R411,R412,R413,R414, R512,U9,U10	DNP	13 - CAN TJA1482 & TJA1483
R10,R79,R127	DNP	16 - OpenSDA FOR S32K5
R317	DNP	17 - LIN SJA1124
R534,R535,R543,R544	DNP	18 - FS2530 FOR S32J100
C609,C610,C611,C612,C613, C614,C718	DNP	19 - S32J100 POWER & GND
R553,R578,R745,R746,R750	DNP	20 - S32J100 IO
R648	DNP	21 - S32J100 SERDES
C735,C738,J23,R595,R596,R597, R609,R610,R658,R659,R662, R665,R679	DNP	22 - S32J100 RESET & JTAG
R436,R702	DNP	23 - PCIE CLOCK
R635,R636,R637,R638,R641, R642,R643,R644	DNP	24 - PCIE
C423,C426,C435,C439,C443, C447,C450,C453,C455,C459, C817,C818,C819,C820,C858, R451,R456,R458,R467,R468, R470,U23,Y10	DNP	26 - ENET TJA1230A
C554,C557,C566,C570,C576, C580,C583,C586,C588,C592, C593,C821,C822,C823,C824, R797,R800,R801,R802,R805, R806,U68,Y12	DNP	27 - ENET TJA1230A
C921,C922,R342,R343,R402	DNP	28 - ENET TJA1120B
C927,C928,R282,R346,R347	DNP	29 - ENET TJA1120B
R288,R289,R359,R360,R361, R362,R438,R439,R764,R765, R766,R767,R775,R776,R780,R781	DNP	30 - ENET TJA1410B
C803,R828,R830,R832,R834, R838,R842,R845	DNP	31 - ENET PHY RJ45
R371,R372,R785,R789,R790	DNP	32 - CAN TJA1462A
C1043,C1047,R1050,R1051	DNP	33 - OPNET KD7251
C527,C531,R373,R375	DNP	35 - PF5123 & PWR JUMPER

Classification: <Classification>

Drawing Title: **X-RING-OPTIC-ETH**

Page Title: **38 - SETTINGS**

Size	Document Number	Rev
C	SCH-97122 / PDF: SPF-97122	A

Date: Tuesday, August 25, 2026 | Sheet 38 of 38